AMENDMENT OF SOLICITATION/MODIFICATION OF CONTRACT

1. CONTRACT ID CODE	PAGE OF PAGES
J W912QR26RA016_Specs-0001	1 3

2. AMENDMENT/MODIFICATION NUMBER	3. EFFECTIVE DATE	4. REQUISITION/PURCHASE REQUISITION NUMBER	5. PROJECT NUMBER *(If applicable)*
0001	20 APR 2026		99248

6. ISSUED BY CODE W912QR	7. ADMINISTERED BY *(If other than Item 6)* CODE
W072 ENDIST LOUISVILLE KO CONTRACTING DIVISION, 600 DR MARTIN LUTHER KING JR PL LOUISVILLE, KY 40202-2230 UNITED STATES RANZEL MERIDETH, EMAIL: RANZEL.L.MERIDETH@USACE.AMY.MIL TELEPHONE: (502) 315-6590	SCD: PAS:

8. NAME AND ADDRESS OF CONTRACTOR *(Number, street, county, State and ZIP Code)*	(X)	
	☒	9A. AMENDMENT OF SOLICITATION NUMBER W912QR26RA016
		9B. DATED *(SEE ITEM 11)* 13 MAR 2026
	☐	10A. MODIFICATION OF CONTRACT/ORDER NUMBER
CODE / FACILITY CODE		10B. DATED *(SEE ITEM 13)*

11. THIS ITEM ONLY APPLIES TO AMENDMENTS OF SOLICITATIONS

☒ The above numbered solicitation is amended as set forth in Item 14. The hour and date specified for receipt of Offers ☒ is extended. ☐ is not extended.

Offers must acknowledge receipt of this amendment prior to the hour and date specified in the solicitation or as amended, by one of the following methods:
(a) By completing items 8 and 15, and returning 1 copies of the amendment; (b) By acknowledging receipt of this amendment on each copy of the offer submitted; or (c) By separate letter or electronic communication which includes a reference to the solicitation and amendment numbers. FAILURE OF YOUR ACKNOWLEDGMENT TO BE RECEIVED AT THE PLACE DESIGNATED FOR THE RECEIPT OF OFFERS PRIOR TO THE HOUR AND DATE SPECIFIED MAY RESULT IN REJECTION OF YOUR OFFER. If by virtue of this amendment you desire to change an offer already submitted, such change may be made by letter or electronic communication, provided each letter or electronic communication makes reference to the solicitation and this amendment, and is received prior to the opening hour and date specified.

12. ACCOUNTING AND APPROPRIATION DATA *(If required)*

SEE SECTION G - CONTRACT ADMINISTRATION DATA

13. THIS ITEM APPLIES ONLY TO MODIFICATIONS OF CONTRACTS/ORDERS. IT MODIFIES THE CONTRACT/ORDER NUMBER AS DESCRIBED IN ITEM 14.

CHECK ONE	
☐	A. THIS CHANGE ORDER IS ISSUED PURSUANT TO: *(Specify authority)* THE CHANGES SET FORTH IN ITEM 14 ARE MADE IN THE CONTRACT ORDER NUMBER IN ITEM 10A.
☐	B. THE ABOVE NUMBERED CONTRACT/ORDER IS MODIFIED TO REFLECT THE ADMINISTRATIVE CHANGES *(such as changes in paying office, appropriation data, etc.)* SET FORTH IN ITEM 14, PURSUANT TO THE AUTHORITY OF FAR 43.103(b).
☐	C. THIS SUPPLEMENTAL AGREEMENT IS ENTERED INTO PURSUANT TO AUTHORITY OF:
☐	D. OTHER *(Specify type of modification and authority)*

E. IMPORTANT: Contractor ☐ is not ☐ is required to sign this document and return ______ copies to the issuing office.

14. DESCRIPTION OF AMENDMENT/MODIFICATION (*Organized by UCF section headings, including solicitation/contract subject matter where feasible.)*

See Summary of Changes

Except as provided herein, all terms and conditions of the document referenced in Item 9A or 10A, as heretofore changed, remains unchanged and in full force and effect.

15A. NAME AND TITLE OF SIGNER *(Type or print)*		16A. NAME AND TITLE OF CONTRACTING OFFICER *(Type or print)*	
15B. CONTRACTOR/OFFEROR *(Signature of person authorized to sign)*	15C. DATE SIGNED	16B. UNITED STATES OF AMERICA *(Signature of Contracting Officer)*	16C. DATE SIGNED

AMENDMENT 0001 - P2 511686 UEPH FY 26 Barracks

Ft Campbell, KY
W912QR26RA016
Amendment 0001

SUMMARY OF CHANGES

1. The following SPECIFICATION SECTIONS have been revised and <u>replaced</u> in their entirety:

	TABLE OF CONTENTS
00 10 00	PRICE BRAKOUT SCHEDULE HAS BEEN REPLACED
00 22 00	CUI MARKINGS REMOVED FROM EVALUATION CRITERIA
05 12 00	STRUCTURAL STEEL
05 50 13	MISCELLANEOUS METAL FABRICATIONS
05 51 33	METAL LADDERS
05 52 00	METAL RAILINGS
06 41 16.00 10	PLASTIC-LAMINATE-CLAD ARCHTECTURAL CABINETS
10 26 00	WALL AND DOOR PROTECTION
10 28 13	TOILET ACCESSORIES
21 13 13	WET PIPE SPRINKLER SYSTEMS, FIRE PROTECTION
28 31 76	FIRE ALARM / MASS NOTIFICATION

2. The following SPECIFICATION SECTIONS have been <u>added</u> in their entirety:

11 31 13	ELECTRIC KITCHEN EQUIPMENT
07 13 53	ELASTOMERIC SHEET WATER PROOFING

3. The following SPECIFICATION SECTIONS have been <u>deleted</u> in their entirety:

DFARS 252.204-7055 HAS BEEN REMOVED

4. The following PLAN SHEETS have been revised and <u>replaced</u> in their entirety:

G1002	SHEET INDEX
G1003	SHEET INDEX
CS102	CIVIL SITE PLAN
CS103	CIVIL SITE PLAN
CS104	CIVIL SITE PLAN
CS105	CIVIL SITE PLAN
CU101	OVERALL UTILITY PLAN
CU102	UTILITY PLAN
CU103	UTILITY PLAN
CU104	UTILITY PLAN
CU105	UTILITY PLAN
CU106	UTILITY PLAN
CU107	UTILITY PLAN
A-201	EXTERIOR NORTH AND SOUTH ELEVATIONS
A-452	ENLARGED DUMPSTER PLANS AND SECTION AND DETAILS
A-691	MISC DETAILS
S-113	FOUNDATION PLAN – AREA 1C
S-502	TYPICAL FOUNDATION DETAILS
S-523	TYPICAL FRAMING DETAILS
S-601	COLUMN SCHEDULE

5. The following PLAN SHEETS have been <u>added</u> new:

A-445	ENLARGED BIKE RACK SECTIONS AND DETAILS

6. The following PLAN SHEETS have been deleted:

NONE for Amendment 0001

7. The following REFERENCE documents have been added new:

NONE for Amendment 0001

8. The proposal due date has been extended from 27-APR-2026 to 12-MAY-2026.

PRICE BREAKOUT

SCHEDULE PROJECT : UEPH Barracks FY26

LOCATION : Ft. Campbell, Kentucky

PROPOSER'S NAME :

____

BASE PROPOSAL

Item No.	Description	Unit	Amount
0001	Primary Facilities	JOB	$ ____________
0002	Project Site Work (excluding demo)	JOB	$ ____________
0003	Demolition	JOB	$ ____________
0004	Base Bid 30 Inch Drilled Shaft – No Socket	JOB	$ ____________
0005	Base Bid 42 Inch Drilled Shaft – With Socket	JOB	$ ____________
0006	City Light & Power Inc. (CLP) Construction and Demolition	JOB	$ 745,100.00
0007	Jacobs Engineering Water and Sewer construction and demolition	JOB	$ 1,060,467.00

BASE PROPOSAL SUBMITTAL $ ____________

BID OPTION ITEMS

			UNIT OF MEASURE	Est. QTY	TOTAL
0008	Optional Bid Item '001' 30" Drilled Shaft – No Socket	$ __________	PER FOOT	240 ft	$ __________
0009	Optional Bid Item '002' 42" Drilled Shaft – With Socket	$ __________	PER FOOT	60 ft	$ __________
0010	Optional Bid Item '003' Additional Proof Test Hole	$ __________	PER FOOT	120 ft	$ __________

TOTAL BID OPTION ITEMS $ ____________

TOTAL BASE PROPOSAL + BID OPTION ITEMS $ ____________

Description of Base Proposal Items

(a) **Item No. 0001** “Primary Facilities – Barracks Building” includes all Base Proposal construction work required within a line five feet outside of the building.

(b) **Item No. 0002** “Project Site Work”, includes all Base Proposal construction work required beyond a line five feet outside the building. The Project Site Work includes, but is not limited to, the construction of grading and drainage, utilities (outside of 5’ from the building), pavements for the parking lot, entrances, sidewalks, equipment pads, etc.

(c) **Item No. 0003** “Demolition’, includes the cost for demolition of all existing structures currently covering area of building site.

(d) **Item No. 0004 “**Base Bid 30 Inch Drilled Shaft – No Socket” includes the cost, material, proof hole inspection and labor to install 30 inch drilled shafts with no rock socket. This includes 4,080 feet of soil and rock drilling and 800 feet of proof hole drilling. Drilling is unclassified.

(e) **Item No. 0005 “**Base Bid 42 Inch Drilled Shaft – Socket” includes the cost, material, proof hole inspection and labor to install 42 inch drilled shafts with a rock socket. This includes 1,120 feet of soil and rock drilling, and 200 feet of proof hole drilling. Drilling is unclassified.

(f) **Item No. 0006 “**City Light & Power Inc. (CLP) Construction and Demolition” includes all existing site overhead distribution-level primary conductors, street lighting, associated poles, and underground wiring to be removed/demolished by CLP prior to construction. CLP (City Light and Power) will tie into existing overhead utility and install primary ug duct bank consisting of one (1) 6 inch conduit containing one (1) 15kv 4/0 al 133% 1/3 concentric neutral cable. Transformer sizes specified and supplied by CLP based on available load data. CLP utilizes a concrete pad of a sufficient size for three phase transformers. CLP will install bollards around transformers. For each transformer, CLP will install a grounding electrode system consisting of two ground rods and a ground ring. CLP subcontractor will install an empty elbow with caps and will mark through GIS for future (re)location. CLP will make secondary connections to transformers, then route cables through separate duct banks to line side of MCB in 1st floor electrical room for final connection by customer. CLP will bond neutral to ground at the transformer secondary and will not run a dedicated ground to service entrance equipment.

(g) **Item No. 0007 “**Jacobs Engineering Water and Sewer construction and demolition” includes the demolition of existing water and sewer lines that serve the two (2) existing facilities on site and any water lines serving hydrants on site. The demolition plan outlines existing lines and features to be demolished by Jacobs. Jacobs is responsible for the construction of the 8 inch sanitary sewer service line and connections to the existing pipe and junctions and cleanouts until 5 feet outside of the facility. Jacobs is responsible for the construction of the 12-inch water main loop that will serve the hydrants and facilities for the FY26 Barracks and tie into the existing water line infrastructure for the surrounding area. Jacobs will provide the water service lines to the building, including domestic and fire protection lines up to five (5) feet outside of the building. Jacobs provides the meter for the building. Jacobs provides the three (3) fire hydrants exterior to the building and post indicator valve and gate valves and bends of various sizes for lines outside of the facility. The contractor must appropriately phase construction activities to ensure water line and sewer line services are not impacted to existing facilities still in use.

Description of Option Items

(h) **Item No. 0008** "Option '001' 30 Inch Drilled Shaft – No Socket includes all cost, material and labor to install extra footage of 30 inch drilled shaft to account for variations in rock elevation that differ for the original lump sum. Drilling is unclassified.

Option Item No. 001 can be exercised 365 calendar days after Administrative NTP.

(i) **Item No. 0009** "Option '002' 42 Inch Drilled Shaft – With Socket includes all cost, material and labor to install extra footage of 42 inch drilled shaft to account for variations in rock elevation that differ from the original lump sum. Drilling is unclassified.

Option Item No. 002 can be exercised 365 calendar days after Administrative NTP.

(j) **Item No. 0010** "Option '003' Additional Proof Test Holes includes all cost, material, and labor to install additional proof test holes to account for variations in rock irregularities that differ from the original lump sum.

Option Item No. 003 can be exercised 365 calendar days after Administrative NTP.

00 22 00 Evaluation Criteria

1.1 The intent of this solicitation is to select one contractor for the construction of a Standard Design Barracks to accommodate 260 unaccompanied enlisted personnel. Primary facilities include living and sleeping quarters, baths, storage, service areas, building information systems, fire protection and alarm systems, antiterrorism measures, Intrusion Detection System (IDS) installation, and Energy Monitoring Control Systems (EMCS) connection. Supporting facilities include site development, utilities and utility connections, exterior lighting, paving, sidewalks, curbs and gutters, storm drainage, landscaping and signage. Heating and air conditioning will be provided by self-contained system for each living quarters unit along with a central system for common areas. Post Construction Award Services (PCAS) will be provided. Utility connections are required to a privatized electric distribution, natural gas, water, and wastewater systems. Measures in accordance with the Department of Defense (DoD) Minimum Antiterrorism for Buildings standards will be provided. Comprehensive building and furnishings related interior design services are required. Access for individuals with disabilities will be provided. Cyber Security Measures will be incorporated into this project. Sustainability/Energy measures will be provided. Facilities will be designed to a minimum life of 40 years in accordance with DoD's Unified Facilities Criteria (UFC 1-200-02) including energy efficiencies, building envelope and integrated building systems performance. Demolish 2 buildings at Fort Campbell, KY (6,389 Total SF). Air Conditioning (Estimated 266 Tons).

This project will be funded with Military (Army) Construction (MILCON) type Procurement dollars (Fiscal Year 26).

The design for this work is being completed in-house by USACE Louisville (CELRL).

There are no existing contract vehicles that have the capacity and scope to accomplish the scope of this project in the division or across the entire enterprise, as further discussed below.

There have not been any previous construction acquisitions for this project. The project will be competed as an unrestricted procurement. In-house effort during the procurement of this project will be for both the procurement activities and the related design activities of this project.

2. **Submittal of offers**.

2.1 Offerors submitting proposals for this project should limit submissions to data essential for evaluation of proposals so that a minimum of time and monies will have been expended in preparing information required herein. However, in order to be effectively and equitably evaluated, the proposals must include information sufficiently detailed to clearly describe the offeror's capability for successfully completing the solicited project. Requirements stated in this Request for Proposal (RFP) are minimums. Proposals should follow in the order of sequence set forth in the RFP. Failure to place the required submission information under the appropriate tab may result in a lower rating if the evaluators cannot readily find the appropriate information.

2.2 Offerors shall submit their proposals to the US Army Corps of Engineers via Procurement Integrated Enterprise Environment (PIEE) as detailed in the Electronic Proposal Submission Instructions, no later than the time and date specified in Block 13 of Standard Form 1442. Mark your proposal submission to the attention of Ranzel Merideth. NOTE: The Louisville District is in the **Eastern Time Zone**.

2.3 Offerors are required to submit a proposal consisting of the information identified in paragraphs 2.4 – 2.5 below. All proposal materials shall be submitted electronically with a table of contents and section dividers. The sections should parallel the submission requirements identified herein.

2.4 Volume I shall be submitted electronically in accordance with the Proposal Submission Instructions in Section 00 21 00 and include the following information:

- Volume I – Factor I: Past Performance
- Volume I – Factor II: Management Plan
- Volume I – Factor III: Small Business Participation Plan

***NOTE**: Failure to place the required submission information under the appropriate tab may result in a lower rating if the evaluators cannot readily find the appropriate information. Any specified page limits will be strictly adhered to and enforced. Information submitted that exceeds the specified limit will not be evaluated.

2.5 Volume II shall be submitted electronically in accordance with the Proposal Submission Instructions in Section 00 21 00 and include the following information:

- Volume II – Tab A: Standard Form 1442 and Price Breakout Schedule
- Volume II – Tab B: Joint Venture Agreements
- Volume II – Tab C: Bid Bond and Proof of Financial Ability
- Volume II – Tab D: Pre-Award Information
- Volume II – Tab E: Subcontracting Plan
- Volume II – Tab F: Project Labor Agreement (PLA)

***NOTE**: Failure to place the required submission information under the appropriate section may result in a lower rating if the evaluators cannot readily find the appropriate information. Any specified page limits will be strictly adhered to and enforced. Information submitted that exceeds the specified limit will not be evaluated.

2.6 Proposals: ALL SUBMISSIONS TO THIS PROPOSAL ANNOUNCEMENT SHALL BE SUBMITTED ELECTRONICALLY. No paper copies, CD-ROMs or facsimile submissions will be accepted. Electronic Proposal Submission is required through the Procurement Integrated Enterprise Environment (PIEE) Solicitation Module https://wawf.eb.mil/.

Instructions on how to create a vendor account will be posted to www.sam.gov and can be located at:

https://www.acq.osd.mil/asda/dpc/ce/cap/docs/piee/PIEE_Solicitation_Module_Vendor_Access_Instructions.pdf

Instructions on how to submit an offer will be posted to www.sam.gov and can be located at:

https://pieetraining.eb.mil/wbt/sol/Posting_Offer.pdf

NOTE: Do NOT enter Privacy Act Data (Personal Identification Information (PII)) in the File Description.)

File Naming Convention: To ensure your submission is received and processed appropriately, it is important that interested parties CAREFULLY ensure their electronic files adhere to the following naming convention:

- W912QR26RA016-FIRMNAME-VOLUME I
- W912QR26RA016-FIRMNAME-VOLUME II

Each file name shall begin with the solicitation number followed by the firm's name and a brief file description. Please see examples above.

File Organization: Although hard copies are not accepted, each file shall be clearly indexed, and logically assembled. Font size shall be 11 or larger. Pages shall be letter sized (larger page sizes (such as 11x17 foldouts, etc.) will be counted as two pages. Proposals shall be in a narrative format, organized and titled so that each section of the proposal follows the order and format of the factors. Information presented should be organized so as to pertain to only the evaluation factor in the section that the information is presented. Information pertaining to more than one evaluation factor should be repeated in each section for each factor.

Upload Completion & Deadline: Interested parties shall submit responses no later than the date specified on the solicitation document. The time & date of proposal receipt will be the upload completion / delivery time & date recorded within PIEE. Do not assume that electronic submission will occur instantaneously. Large files (e.g., 10MB or more) will take some time to upload. Offerors should time their upload effort with prudence by not waiting until the last few minutes—this will allow for unexpected delays in the transmittal process. Offerors are encouraged to keep a copy of the upload confirmation for their record. As noted above, offerors will not receive a confirmation email but instead will receive upload confirmation in the browser following successful upload. Submissions after the deadline will be considered late and will be processed in accordance with FAR 15.107.

Electronic Files: Files shall be in their native format (i.e., doc, xls, ppt, etc.), or if in pdf format, shall be in searchable text. Text and graphics portfolios of the electronic copies shall be in a format readable by Microsoft Office or Adobe applications. Data submitted in a spreadsheet format shall be readable by MS Excel (all cells and formulas should be unlocked).

Any information, presented in a proposal that the Offeror wants safeguarded from disclosure to other parties must be identified and labeled in accordance with the requirements of Provision "FAR 52.215-1, Instructions to Offerors – Competitive Acquisition," subparagraph (e), which is found in Section 00 10 00 of the RFP. The Government will endeavor to honor the restrictions against release requested by Offerors, to the extent permitted under United States law and regulations.

3. **Proposal Evaluation Process**.

3.1. A Source Selection Evaluation Board (SSEB) comprised of representatives of the Corps of Engineers, User/Customer, and other required personnel will evaluate the proposals. Offerors are advised that the technical evaluation and rating of proposals will be conducted in strict confidence. Technical proposals (Volume I) will be reviewed and rated without knowledge of the price offered. The number and identities of offerors are not revealed to anyone not involved in the evaluation and award process or to other offerors. Proposals will be evaluated based on the factors described herein, and the basis of award is a Best Value Trade-Off.

3.2 The evaluation process essentially consists of four parts: proposal compliance review and responsibility review, technical evaluation, price evaluation, and price/technical trade-off analysis.

3.2.1 Proposal Compliance/Responsibility Review: This is an initial review to ensure that all required forms and certifications are complete, that both a technical and price proposal were received, and that the offeror is financially capable of sustaining performance under the contract and is able to obtain the required level of performance and payment bonds from an acceptable surety.

3.2.2 Technical Evaluation: The SSEB will evaluate and rate the Volume I proposals against the RFP requirements. Factor I – Past Performance will be rated using Tables 1 and 2 below. The rating will be based on overall confidence in performance, with the final confidence assessment rating based on the extent of recent, relevant past performance and the quality of the offeror's performance. Factor II – Management Approach will be rated using Table 3 below. Factor III – Small Business Participation Plan will be rated using Table 4 below.

3.2.3 Price Evaluation: The SSEB and Contracting Officer/SSA will evaluate price proposals independent of, and subsequent to, the technical evaluation. The SSEB will not have access to price information until completion of the technical evaluation.

3.2.4 Price/Technical Trade-off Analysis: After all the above evaluations are complete, the Contracting Officer/SSA will compare the relative advantages and disadvantages of technical proposals and compare prices. The Source Selection Authority (SSA) will then consider all factors to select the proposal offering the best value to the Government.

4. **Proposal Information and Related Evaluation Factors.**

4.1 Proposals will be evaluated in accordance with the factors below, listed in relative order of importance. **All non-price factors, when combined, are considered approximately equal to price**. The Government intends to evaluate proposals and award a contract without discussions with offerors. Therefore, the offeror's initial proposal should contain the offeror's best terms from a price and technical standpoint. The Government reserves the right to conduct discussions if the Contracting Officer later determines them to be necessary. If the Contracting Officer determines that the number of proposals that would otherwise be in the competitive range exceeds the number at which an efficient competition can be conducted, the Contracting Officer may limit the number of proposals in the competitive range to the greatest number that will permit an efficient competition among the most highly rated proposals.

	Volume/Factor	**Relative Order of Importance**
4.2	Volume I – Factor I – Past Performance	1st
4.3	Volume I – Factor II – Management Plan	2nd
4.4	Volume I – Factor III – Small Business Participation Plan	3rd
4.5	Volume II - Price and Pro Forma Information (Separate file from Volume I)	
	Tab A - Standard Form 1442 and Price Breakout Schedule	Not Rated
	Tab B - Joint Venture Agreement	Not Rated
	Tab C – Bid Bond and Proof of Financial Ability	Not Rated
	Tab D - Pre-Award Information	Not Rated
	Tab E - Subcontracting Plan	Acceptable/Unacceptable
	Tab F – Project Labor Agreement (PLA)	Acceptable/Unacceptable

4.6 Ratings

Evaluators will apply the adjectival rating for the definition that most closely matches the evaluation.

TABLE 1

Past Performance Relevancy Ratings	
Rating	**Description**
Very Relevant	Present/past performance effort involved essentially the same scope and magnitude of effort and complexities this solicitation requires.
Relevant	Present/past performance effort involved similar scope and magnitude of effort and complexities this solicitation requires.
Somewhat Relevant	Present/past performance effort involved some of the scope and magnitude of effort and complexities this solicitation requires.
Not Relevant	Present/past performance effort involved little or none of the scope and magnitude of effort and complexities this solicitation requires.

TABLE 2

Performance Confidence Assessments	
Rating	**Description**
Substantial Confidence	Based on the offeror's recent/relevant performance record, the Government has a high expectation that the offeror will successfully perform the required effort.
Satisfactory Confidence	Based on the offeror's recent/relevant performance record, the Government has a reasonable expectation that the offeror will successfully perform the required effort.
Neutral Confidence	No recent/relevant performance record is available, or the offeror's performance record is so sparse that no meaningful confidence assessment rating can be reasonably assigned. The offeror may not be evaluated favorably or unfavorably on the factor of past performance.
Limited Confidence	Based on the offeror's recent/relevant performance record, the Government has a low expectation that the offeror will successfully perform the required effort.
No Confidence	Based on the offeror's recent/relevant performance record, the Government has no expectation that the offeror will be able to successfully perform the required effort.

TABLE 3

Combined Technical/Risk Rating Method	
Adjectival Rating	**Description**
Outstanding	Proposal demonstrates an exceptional approach and understanding of the requirements and contains multiple strengths and/or at least one significant strength, and risk of unsuccessful performance is low.
Good	Proposal indicates a thorough approach and understanding of the requirements and contains at least one strength or significant strength, and risk of unsuccessful performance is low to moderate.
Acceptable	Proposal meets requirements and indicates an adequate approach and understanding of the requirements, and risk of unsuccessful performance is no worse than moderate.
Marginal	Proposal has not demonstrated an adequate approach and understanding of the requirements, and/or risk of unsuccessful performance is high.
Unacceptable	Proposal does not meet requirements of the solicitation and, thus, contains one or more deficiencies and is unawardable, and/or risk of performance is unacceptably high.

TABLE 4

Small Business Participation Plan Ratings	
Rating	**Description**
Acceptable	Proposal indicates an adequate approach and understanding of small business objectives.
Unacceptable	Proposal does not meet small business objectives.

4.7 Definitions

1. Deficiency. A material failure of a proposal to meet a Government requirement or a combination of significant weaknesses in a proposal that increases the risk of unsuccessful contract performance to an unacceptable level. See FAR 15.001.

2. Strength. An aspect of an offeror's proposal that has merit or exceeds specified performance or capability requirements in a way that will be advantageous to the Government during contract performance.

3. Significant Strength. An aspect of an offeror's proposal that has appreciable merit or appreciably exceeds specified performance or capability requirements in a way that will be appreciably advantageous to the Government during contract performance.

4. Weakness. A flaw in the proposal that increases the risk of unsuccessful contract performance. See FAR 15.001.

5. Significant Weakness. A flaw in the proposal that appreciably increases the risk of unsuccessful contract performance. See FAR 15.001.

6. Uncertainty. Any aspect of a non-price factor proposal for which the intent of the offeror is unclear (e.g., more than one way to interpret the offer or inconsistencies in the proposal indicating that there may have been an error, omission, or mistake).

7. Clarifications. Limited exchanges between the Government and offerors that may occur when award without discussions is contemplated. See FAR 15.202(a)(2).

8. Adverse Past Performance. Past performance information that supports a less than satisfactory rating from sources where the information is from other than formal rating systems such as "PPIRS" or "FAPIIS."

9. Discussion. Exchanges (i.e., negotiations) in a competitive environment that are undertaken with the intent of allowing the offeror to revise its proposal. Discussions take place after establishment of the competitive range.

10. Best Value. The expected outcome of an acquisition that, in the Government's estimation, provides the greatest overall benefit in response to the requirement. See FAR 2.101.

11. Relevancy. As it pertains to past performance information, is a measure of the extent of similarity between the service/support effort, complexity, dollar value, contract type, and subcontract/teaming or other comparable attributes of past performance examples and the solicitation requirements; and a measure of the likelihood the past performance is an indicator of future performance.

12. Recency. As it pertains to past performance information, is a measure of the elapsed time since the past performance reference occurred. Recency is generally expressed as a time period during which past performance references are considered relevant.

13. Risk. As it pertains to source selection, is the potential for unsuccessful contract performance. The consideration of risk assesses the degree to which an offeror's proposed approach to achieving the technical factor or subfactor may involve risk of disruption of schedule, degradation of performance, the need for increased Government oversight, and the likelihood of unsuccessful contract performance.

5.0 Volume I – Factor I: Past Performance

5.1 Submission Requirements:

5.1.1 Provide descriptions of up to three (3) projects substantially complete or completed by the Prime Contractor within the last five (5) years (from the solicitation issue date) that are similar to this project in size and scope. Projects completed more than five (5) years before the solicitation issue date may be considered for evaluation purposes but may lessen the overall relevancy rating for that project. Projects are considered substantially complete if enough work has been performed such that it demonstrates the ability to complete all phases of the project (*i.e.*, only punchlist items remain) or at least 95% complete. An Indefinite Delivery Indefinite Quantity (IDIQ) contract may be submitted only if a single task order could be considered similar to this project. Task orders may not be combined in order for the project to be considered similar.

NOTE: Do not submit more than three (3) projects. Any projects submitted that exceed the three (3) project limit will not be evaluated.

NOTE: Renovation work is not considered similar in size or scope.

5.1.2 Projects considered similar in size would be a minimum of 75,000 SF of new construction, only primary residential and ancillary common areas and conference space square footage are included in the 75,000 SF.

5.1.3 Projects considered similar in scope to this project include new construction of multi-story barracks/dormitory building, hotels or apartment buildings.

5.1.4 The prime contractor must have self-performed at least 15 percent of the direct contract labor (including testing and layout personnel), exclusive of other general conditions or field overhead personnel, material, equipment, or subcontractors to be considered similar.

5.1.5 Additional consideration may be given for projects that contain the following features:

a) Project is built on a military installation for USACE.
b) Project includes deep foundations/drilled shaft piers
c) Project includes demolition work

5.1.6 The following information shall be provided for each project:

a) Project Title, Location, and Contract Number/Unique Identifier;
b) Current percentage of construction complete and the date it was or will be complete
    a. For projects that are not yet complete, provide description of work remaining to be completed;
c) Scope of the project, to include purpose/use of facility;
d) Size of the project, differentiate square footage of new construction versus renovation if the project includes both;
e) Portion and percentage of work that was self-performed;
f) Two references, including name, telephone number, and email address.

5.1.7 For this factor, also include any ratings, letters, awards, etc. that support past performance on these projects. Any of this information that is submitted shall clearly identify to which of the submitted projects it pertains. A sample Past Performance Questionnaire is attached for your convenience. If used, the Past Performance Questionnaire must be submitted by the offeror with the proposal submission and **not** sent directly to the agency from the reference. For each project, the offeror may provide information on problems encountered on the identified contracts and the offeror's respective corrective action.

NOTE: For purposes of evaluating past performance, the Prime Contractor is defined as the contractor identified in Block 14 of the Standard Form 1442. Projects performed by contractors other than the offeror, including, but not limited to, teaming partners, subcontractors, sister or parent companies, and ~~affiliates~~ will not be evaluated for past performance, unless those other contractors are part of a joint venture offeror as demonstrated by a signed joint venture agreement. If more than one contractor is listed in Block 14, then a signed joint venture must be submitted with the proposal and the joint venture shall be registered as such in the System for Award Management (SAM). However, each party of the Joint Venture (JV) must submit their own Unique Entity Identifier Number (formerly known as DUNS) with the JV proposal. If the offeror represents the combining of two or more companies as a JV for the purpose of this RFP, each company in the JV may submit project examples, but the total submitted by the JV will not exceed three (3).

Relevant past performance information provided for affiliates of the offeror will be considered. Affiliates (as defined by FAR 2.101) means associated business concerns or individuals if, directly or indirectly either one controls or can control the other; or third-party controls or can control both. Affiliation will be determined in accordance with 13 CFR 121.103. In determining whether affiliation exists, the Government will consider the totality of the circumstances and may find affiliation even though no single factor is sufficient to constitute affiliation. If an offeror is submitting the past performance of an affiliate, the offeror must submit proof of affiliation pursuant to 13 CFR 121.103.

5.2 Evaluation Criteria:

5.2.1 The SSEB will first evaluate the relevancy of recent past performance identified in the proposal in response to paragraph 5.1 above. By using the criteria identified above, the SSEB will determine how relevant a past project is when compared to the scope, size, and magnitude of effort and complexities of the solicited project. A relevancy rating will be assigned to each submitted project using the Past Performance Relevancy Ratings table above.

5.2.2 The SSEB will next review how well the offeror performed on those projects. The Government will, at a minimum, review past performance information retrieved through the Contractor Performance Assessment Reporting System (CPARS), using all Unique Entity Identifier (UEI) numbers identified in the offeror's proposal. The Government reserves the right to check any or all cited references to verify supplied information and to assess owner satisfaction. The Government also reserves the right to not contact the provided references. In addition to the information submitted by the offeror, the Government reserves the right to review any other sources of relevant information for evaluating past performance, including projects other than those submitted by the offeror. Other sources may include, but are not limited to, past performance information retrieved from inquiries of owner representative(s), Federal Awardee Performance and Integrity Information System (FAPIIS), Electronic Subcontract Reporting System (eSRS), and any other known sources not provided by the offeror.

5.2.3 The SSEB will review all past performance information collected and determine the quality of the offeror's performance, general trends, and usefulness of the information and incorporate this information into the performance confidence assessment. The SSEB will assign a final, overall Performance Confidence rating, using the ratings in the Performance Confidence Assessment table (Table 2) above, based on the SSEB's assessment of (1) the degree of the offeror's recent, relevant experience, and (2) how well the offeror performed that experience.

6.0 Volume I – Factor II: Management Plan

6.1 Submission Requirements:

Provide a management plan narrative for the project that describes how your labor, resources, designers, subcontractors, material suppliers, and procurement of long lead items will be managed, supervised, coordinated, and used to ensure successful completion of the project. At a minimum, the Management Plan shall also include the following information:

- Describe your process for managing, coordinating, and tracking changes that arise during construction.
- Describe how you will manage, supervise, and coordinate the subcontractors' work and who in the organization will be responsible for this management and coordination.
- Describe how the offeror will approach the construction of the Fort Campbell UEPH Barracks.
- Describe the teams' quality control approach and capabilities to maintain quality control of the construction.
- Describe, in a narrative format, the commissioning process and critical milestones of the commissioning process and how they relate to critical construction milestones. Describe, in narrative format, the key submittals for commissioning process. Describe, by role and qualifications, who will be in lead authority for the contractor (in-house or first tier sub) during the commissioning process. Lay out their roles and authority over all involved subcontractors.

NOTE: There is a page limit of ten (10) single-sided, 8.5" x 11" pages, using a minimum font size of 11 and a minimum margin of one-half inch on all sides, for the Management Plan.

6.2 Evaluation Criteria:

Plans will be evaluated based on the level of understanding of the work and the involvement the contractor will have in the management, oversight, control, and coordination of the work performed during construction of the project. Plans that demonstrate a clear understanding of the project requirements and provide a thorough approach for successfully managing the Fort Campbell UEPH Barracks project may be rated more favorably by the SSEB. In your proposal, please detail your proposed management approach for this project. We are particularly interested in your strategy for identifying and mitigating risks associated with critical solicitation tasks. Furthermore, demonstrate your qualifications by providing a summary of your past performance and experience on other USACE projects. A rating will be assigned using the Technical Assessment Rating table above.

7.0 Volume I – Factor III: Small Business Participation Plan

7.1 Submission Requirements

ALL OFFERORS ARE REQUIRED TO SUBMIT A SMALL BUSINESS PARTICIPATION PLAN.

The Small Business Participation Plan shall be based on the offeror's best effort and is required to address each of the following areas individually:

- The extent to which the small business programs listed in FAR 19 (small business, small disadvantaged business, woman-owned small business, HubZone, service disabled veteran owned small business, etc.) are specifically identified in the Small Business Participation Plan;
- The extent of participation of such firms in terms of the value of the total acquisition in %'s for the base year and for each individual option year; the extent of commitment to use such firms (for example, enforceable commitments, i.e., teaming agreements signed, are to be considered more heavily than non-enforceable ones);
- The complexity and variety of the work small firms are to perform on this acquisition;
- The practicality of the Small Business Participation Plan, i.e., aggressive goals.

The Small Business Participation Plan shall be organized as follows:

(1) Prime Contractor type of business (check all that apply):
( ) Large
( ) Small (also check type of small business)
( ) Small Non-Disadvantaged Business

( ) Small Disadvantaged Business
( ) Woman-Owned Small Business

( ) HUBZone Small Business
( ) Veteran Owned Small Business
( ) Service Disabled, Veteran Owned Small Business

(2) Percentage of your participation as a prime contractor: ___________________%

NOTE: Small Business primes' self-performance counts as Small Business Participation, and small business primes may achieve small business participation goals through their own performance/participation as a prime and/or through subcontracting to other small businesses.

(3) Percentage of total contract value of subcontracts planned for:

	% of Total Contract Value
Large	%
Total Small	%
Small Non-Disadvantaged	%
Small Disadvantaged	%
Small Woman Owned	%
Small HUB Zone	%
Small Veteran Owned	%
Small Service Disabled Veteran Owned	%

Each percentage above shall be accompanied by detailed supporting documentation regarding individual commitments.

NOTE: The sum of the percentages of Small Non-Disadvantaged and Small Disadvantaged should equal the entries for the Total Small; however, the sum of all of the percentages need not equal 100% since the prime is not included and individual subcontractors may be counted towards more than one category.

(4) List principal supplies/services (be specific) to be subcontracted to:

	Name of Company	Type of Service/Supply
Large		
Small Non-Disadvantaged		
Small Disadvantaged		
Small Woman Owned		
Small HUB Zone		
Small Veteran Owned		
Small Service Disabled Veteran Owned		

(5) Prior Performance Information: Provide any information substantiating the offeror's track record of utilizing small business on past contracts.

(6) For Large ***and*** Small Businesses provide descriptive information for all small business categories. Any information concerning long-term relationships with Small Business subcontractors, such as mentor-protégé relationships, should be provided.

(7) Extent of Commitment: Provide documentation regarding enforceable commitments to utilize any small business category as defined in FAR Part 19 as subcontractors.

(8) Small Business Subcontracting Plan: Each ***Large Business Offeror*** shall provide a Small Business Subcontracting Plan that contains all of the elements required by FAR Clause 52.219-9 ALT II. This Plan ***shall*** be submitted separately from the Small Business Participation Plan information required above which applies to both Large and Small Businesses. The Subcontracting Plan is not a requirement for evaluation in source selection but rather a requirement for award to a Large Business. The approved Small Business Subcontracting Plan will be incorporated into any resultant contract(s).

7.2 Evaluation Criteria:

ALL OFFERORS ARE REQUIRED TO SUBMIT A SMALL BUSINESS PARTICIPATION PLAN.
The Small Business Participation Plan will be evaluated based on the offeror's best efforts, the level of small business commitment that is being demonstrated for the proposed acquisition, and the prior level of commitment to utilizing small businesses in performance of prior contracts. The Small Business Participation Plan must meet the minimum Total Small Business Participation goal of 20% of the total contract value (through collective small business participation from any type of small business or sub-category small business).

Pursuant FAR 52.219-9 (d), Small Business Subcontracting Plan. the following elements will be considered in evaluating an offeror's Small Business Participation Plan:

- The extent to which such firms, as defined in FAR Part 19, are specifically identified in plans;
- The extent of commitment to use such firms (enforceable commitments will be weighted more heavily than non-enforceable ones);
- The complexity and variety of the work such firms are to perform;
- The realism of the plans;
- Past performance of offerors in complying with the requirements of the Subcontracting Plan Goals for such firms and monetary targets for participation;
- The extent of participation of such firms in terms of the proposed subcontracted value; and
- The extent to which the offeror provides detailed explanations/documentation supporting the proposed participation percentages, or lack thereof. The Department of Defense (DOD) has established small business goals to help ensure small business receives a fair proportion of DOD awards.

8.0 **Volume II - Price and Proforma Information**

8.1 **Tab A - Standard Form 1442 and Proposal Price Breakout Schedule**.

8.1.1 Submission Requirements:

The offeror shall complete and submit Standard Form 1442 and Section 00 10 00, Proposal Price Breakout Schedule.

8.1.2 Evaluation Criteria:

The price will be evaluated on the total price for the base proposal plus all options. The price will be evaluated by the Government for fairness and reasonableness through the use of a price analysis. Price will also be checked for unbalancing of line items. Unbalanced pricing exists when, despite an acceptable total evaluated price, the price of one or more line items is significantly over or understated as indicated by the application of price analysis techniques. An offer may be rejected if the contracting officer determines that the lack of balance poses an unacceptable risk to the Government. Offerors are cautioned to distribute pricing appropriately.

8.2 **Tab B – Joint Venture Agreements**

8.2.1 Submission Requirements:

If more than one contractor is listed in Block 14, or the offeror listed in Block 14 is a joint venture (JV), then a signed JV agreement must be submitted with the proposal and the offeror shall be registered in the System for Award Management (SAM) as a legal entity separate from the individual joint venture members. However, each member of the JV must submit its own Unique Entity Identifier (UEI) with the proposal. A copy of the offeror's signed joint venture agreement shall be submitted with the proposal.

In addition to the joint venture agreement, small business offerors submitting a proposal as a Mentor-Protégé JV under the SBA Mentor-Protégé program (MPP) shall submit a copy of their SBA-approved Mentor-Protégé agreement.

8.2.2 Evaluation Criteria:

This information will be used for the purpose of completing the Pre-Award Survey and will not be rated. Small business joint venture agreements must comply with the relevant regulations in Title 13 of the Code of Federal Regulations in order for an offeror to be eligible for any small business set-aside acquisitions and/or small-business price preference.

8.3 **Tab C – Evidence of Ability to Obtain Bonding and Proof of Financial Ability**

8.3.1 Submission Requirements:

A. Financial Capability. Submit Proof of Financial Ability (most recent financial statement covering assets and liabilities). Include the name, address, and telephone number of offeror's banking institution. If offeror is a joint venture, submit this information for all joint venture members.

B. Bonding Capability. Submit information showing offeror's ability to be bonded for this project (minimum amount of $50M single project bonding capacity required). Include the name, address, and telephone number of the offeror's bonding company.

8.3.2 Evaluation Criteria:

This information will be used for the purpose of completing the Pre-Award Survey and will not be rated. See RFO Part 28 for information related to bonds.

If the offeror is a joint venture, submit this information for all joint venture members.

8.4 **Tab D – Pre-Award Information**

8.4.1 Submission Requirements:

A. The offeror shall submit one completed copy of Section 00 45 00, Representations and Certification.

B. The offeror shall submit the following information:

a) Number of years the firm has been in business

b) Name, address, and telephone numbers of two credit/trade references

c) A list of present commitments, including the dollar value

Note: If the offeror is a joint venture, submit this information for all joint venture members.

8.4.2 Evaluation Criteria:
This information will be used for the purpose of completing the Pre-Award Survey and will not be rated.

8.5 Tab E - Subcontracting Plan

8.5.1 Submission Requirements:

Large business offerors shall submit a Subcontracting Plan in accordance with FAR Clauses 52.219-8, 52.219-9 ALT II, and RFO 19.109. The offeror may use the attached sample sub-contracting plan as a starting point. Percentage goals apply to the total amount being subcontracted.

8.5.2 Evaluation Criteria:
Submitted information will be evaluated for acceptability in accordance with FAR 52.219-9(d).

8.6 **Tab F – Project Labor Agreement (PLA)**

8.6.1 Submission Requirements

ALL OFFERORS ARE REQUIRED TO SUBMIT AN EXECUTED PROJECT LABOR AGREEMENT. All offerors shall submit a Project Labor Agreement in accordance with FAR Solicitation Provision 52.222-33 and FAR Clause 52.222-34. To be acceptable, the agreements must adequately address the required statutory elements and provide sufficient information to enable the Contracting Officer to answer affirmatively the requirements laid out in FAR 22.502-2(b).

8.6.2 Evaluation Criteria:

Submitted information will be evaluated for acceptability in accordance with FAR 22.502-2(b). To be acceptable, Project Labor Agreements must:

(a) Bind all contractors and subcontractors engaged in construction on the construction project to comply with the Project Labor Agreement;

(b) Allow all contractors and subcontractors to compete for contracts and subcontracts without regard to whether they are otherwise parties to collective bargaining agreements;

(c) Contain guarantees against strikes, lockouts, and similar job disruptions;

(d) Set forth effective, prompt, and mutually binding procedures for resolving labor disputes arising during the term of the project labor agreement; and

(e) Provide other mechanisms for labor-management cooperation on matters of mutual interest and concern, including productivity, quality of work, safety, and health.

NAVFAC/USACE PAST PERFORMANCE QUESTIONNAIRE (Form PPQ-0)

CONTRACT INFORMATION (Contractor to complete Blocks 1-4)

1. Contractor Information
Firm Name: CAGE Code:
Address: Unique Entity Identifier (UEI):
Phone Number:
Email Address:
Point of Contact: Contact Phone Number:

2. Work Performed as: ☐ Prime Contractor ☐ Sub Contractor ☐ Joint Venture ☐ Other (Explain)
Percent of project work performed:
If subcontractor, who was the prime (Name/Phone #):

3. Contract Information
Contract Number:
Delivery/Task Order Number (if applicable):
Contract Type: ☐ Firm Fixed Price ☐ Cost Reimbursement ☐ Other (Please specify):
Contract Title:
Contract Location:
Award Date (mm/dd/yy):
Contract Completion Date (mm/dd/yy):
Actual Completion Date (mm/dd/yy):
Explain Differences:

Original Contract Price (Award Amount):
Final Contract Price (*to include all modifications, if applicable*):
Explain Differences:

4. Project Description:
Complexity of Work ☐ High ☐ Med ☐ Routine
How is this project relevant to project of submission? (*Please provide details such as similar equipment, requirements, conditions, etc.*)

CLIENT INFORMATION (Client to complete Blocks 5-8)

5. Client Information
Name:
Title:
Phone Number:
Email Address:

6. Describe the client's role in the project:

7. Date Questionnaire was completed (mm/dd/yy)**:**

8. Client's Signature:

NOTE: NAVFAC/USACE REQUESTS THAT THE CLIENT COMPLETES THIS QUESTIONNAIRE AND SUBMITS DIRECTLY BACK TO THE OFFEROR. THE OFFEROR WILL SUBMIT THE COMPLETED QUESTIONNAIRE TO USACE WITH THEIR PROPOSAL, AND MAY DUPLICATE THIS QUESTIONNAIRE FOR FUTURE SUBMISSION ON USACE SOLICITATIONS. THE GOVERNMENT RESERVES THE RIGHT TO VERIFY ANY AND ALL INFORMATION ON THIS FORM.

ADJECTIVE RATINGS AND DEFINITIONS TO BE USED TO BEST REFLECT YOUR EVALUATION OF THE CONTRACTOR'S PERFORMANCE

RATING	DEFINITION	NOTE
(E) Exceptional	Performance meets contractual requirements and exceeds many to the Government/Owner's benefit. The contractual performance of the element or sub-element being assessed was accomplished with few minor problems for which corrective actions taken by the contractor was highly effective.	An Exceptional rating is appropriate when the Contractor successfully performed multiple significant events that were of benefit to the Government/Owner. A singular benefit, however, could be of such magnitude that it alone constitutes an Exceptional rating. Also, there should have been NO significant weaknesses identified.
(VG) Very Good	Performance meets contractual requirements and exceeds some to the Government's/Owner's benefit. The contractual performance of the element or sub-element being assessed was accomplished with some minor problems for which corrective actions taken by the Contractor were effective.	A Very Good rating is appropriate when the Contractor successfully performed a significant event that was a benefit to the Government/Owner. There should have been no significant weaknesses identified.
(S) Satisfactory	Performance meets minimum contractual requirements. The contractual performance of the element or sub-element contains some minor problems for which corrective actions taken by the Contractor appear or were satisfactory.	A Satisfactory rating is appropriate when there were only minor problems, or major problems that the Contractor recovered from without impact to the contract. There should have been NO significant weaknesses identified. Per DOD policy, a fundamental principle of assigning ratings is that Contractors will not be assessed a rating lower than Satisfactory solely for not performing beyond the requirements of the contract.
(M) Marginal	Performance does not meet some contractual requirements. The contractual performance of the element or sub-element being assessed reflects a serious problem for which the Contractor has not yet identified corrective actions. The Contractor's proposed actions appear only marginally effective or were not fully implemented.	A Marginal rating is appropriate when a significant event occurred from which the Contractor had trouble overcoming and that impacted the Government/Owner.
(U) Unsatisfactory	Performance does not meet most contractual requirements and recovery is not likely in a timely manner. The contractual performance of the element or sub-element contains serious problem(s) for which the Contractor's corrective actions appear or were ineffective.	An Unsatisfactory rating is appropriate when multiple significant events occurred from which the contractor had trouble overcoming and that impacted the Government/Owner. A singular problem, however, could be of such serious magnitude that it alone constitutes an Unsatisfactory rating.
(N) Not Applicable	No information or did not apply to your contract	Rating will be neither positive nor negative.

TO BE COMPLETED BY CLIENT

PLEASE CIRCLE THE ADJECTIVE RATING THAT BEST REFLECTS YOUR EVALUATION OF THE CONTRACTOR'S PERFORMANCE.

1. QUALITY:	
a) Quality of technical data/report preparation efforts	E VG S M U N
b) Ability to meet quality standards specified for technical performance	E VG S M U N
c) Timeliness/effectiveness of contract problem resolution without extensive customer guidance	E VG S M U N
d) Adequacy/effectiveness of quality control program and adherence to contract quality assurance requirements (without adverse effect on performance)	E VG S M U N
2. SCHEDULE/TIMELINESS OF PERFORMANCE:	
a) Compliance with contract delivery/completion schedules including any significant intermediate milestones. *(If liquidated damages were assessed or the schedule was not met, please address below)*	E VG S M U N
b) Rate the contractor's use of available resources to accomplish tasks identified in the contract	E VG S M U N
3. CUSTOMER SATISFACTION:	
a) To what extent were the end users satisfied with the project?	E VG S M U N
b) Contractor was reasonable and cooperative in dealing with your staff (including the ability to successfully resolve disagreements/disputes; responsiveness to administrative reports; efforts to keep lines of communication open)	E VG S M U N
c) To what extent was the contractor cooperative, businesslike, and concerned with the interests of the customer?	E VG S M U N
d) Overall customer satisfaction	E VG S M U N
4. MANAGEMENT/ PERSONNEL/LABOR	
a) Effectiveness of on-site management, including management of subcontractors, suppliers, materials, and/or labor force?	E VG S M U N
b) Ability to hire, apply, and retain a qualified workforce to this effort	E VG S M U N
c) Government Property Control	E VG S M U N
d) Knowledge/expertise demonstrated by contractor personnel	E VG S M U N
e) Utilization of Small Business concerns	E VG S M U N
f) Ability to simultaneously manage multiple projects with multiple disciplines	E VG S M U N
g) Ability to assimilate and incorporate changes in requirements and/or priority, including planning, execution, and response to Government changes	E VG S M U N
h) Effectiveness of overall management (including ability to effectively lead, manage, and control the program)	E VG S M U N
5. COST/FINANCIAL MANAGEMENT	
a) Ability to meet the terms and conditions within the contractually agreed price(s)?	E VG S M U N
b) Contractor proposed innovative alternative methods/processes that reduced cost, improved maintainability, or other factors that benefited the client	E VG S M U N
c) If this is/was a Government cost type contract, please rate the Contractor's timeliness and accuracy in submitting monthly invoices with appropriate back-	E VG S M U N

up documentation, monthly status reports/budget variance reports, compliance with established budgets, and avoidance of significant and/or unexplained variances (under runs or overruns)						
d) Is the Contractor's accounting system adequate for management and tracking of costs? *If no, please explain in Remarks section.*	Yes			No		
e) If a Government contract, has it been partially or completely terminated for default or convenience or are there any pending terminations? *Indicate if show cause or cure notices were issued, or any default action in comment section below.*	Yes			No		
f) Have there been any indications that the contractor has had any financial problems? *If yes, please explain below.*	Yes			No		
6. SAFETY/SECURITY						
a) To what extent was the contractor able to maintain an environment of safety, adhere to its approved safety plan, and respond to safety issues? (Includes: following the users rules, regulations, and requirements regarding housekeeping, safety, correction of noted deficiencies, etc.)	E	VG	S	M	U	N
b) Contractor complied with all security requirements for the project and personnel security requirements.	E	VG	S	M	U	N
7. GENERAL						
a) Ability to successfully respond to emergency and/or surge situations (including notifying the COR, PM, or Contracting Officer in a timely manner regarding urgent contractual issues).	E	VG	S	M	U	N
b) Compliance with contractual terms/provisions *(explain if specific issues)*	E	VG	S	M	U	N
c) Would you hire or work with this firm again? *(If no, please explain below)*	Yes			No		
d) In summary, provide an overall rating for the work performed by this Contractor.	E	VG	S	M	U	N

Please provide responses to the questions above (*if applicable*) and/or additional remarks. Furthermore, please provide a brief narrative addressing specific strengths, weaknesses, deficiencies, or other comments that may assist our office in evaluating performance risk *(please attach additional pages if necessary)***:**

AFARS -- Appendix DD Subcontracting Plan Evaluation Guide

DD-100 Purpose.
The guide provides a methodology for uniform and consistent evaluation of subcontracting plans within the Army. It is designed to facilitate compliance with the mandates of 15 U.S.C. § 637(d) to increase opportunities for small and small disadvantaged businesses.

DD-101 Applicability.
In accordance with requirements of RFO 19.206-3, the contracting officer shall use this guide to review all subcontracting plans (except those for commercial items). A copy of the completed evaluation shall be included in the contract file.

DD-102 Goals.
Contracting officers must place special emphasis on negotiating subcontracting goals that are realistic, challenging and attainable. The plan must express goals in terms of percentages of total planned subcontracting dollars and must be comparable to the dollar commitments in the small business participation plan. In accordance with RFO 19.206-3 the contracting officer must review enough evidence to determine that the:
1. Offeror can meet subcontracting plan goals;
2. Offeror’s goals are consistent with their cost or pricing data or information other than cost or pricing data;
3. Offeror will honor the terms of subcontract agreements (i.e., timely payments of amounts owed, use of firms cited in proposal, etc.); and
4. Offeror’s make or buy policy or program does not conflict with the proposed subcontracting plan and is in the Government’s best interest.
5. Plan includes the contractor’s commitment to adopt and comply with its requirements and goals for small business utilization.

DD-103 Evaluation Rating.
Either the contracting officer, the small business representative, or both, shall evaluate and rate the subcontracting plan as “acceptable” or “unacceptable,” in the context of the particular procurement. For instance, in smaller dollar value contracts, or contracts for uniquely manufactured items, it might be impracticable or not cost effective for offerors to take the type of actions that might be appropriate in contracts for larger dollar values or commercial components. To receive an “Acceptable” rating, the contractor must satisfy all objectives in Part 2 and meet each statutory subcontracting plan requirement outlined in Part 3. Failure to receive a subcontracting plan rating of acceptable could jeopardize the offeror’s selection for contract award. The contracting officer must document the decisions in the contract file.

DD-104 Modification of Guide.
Pursuant to AFARS 5101.403, only principal assistants responsible for contracting may approve individual deviations to this evaluation guide. This approval authority may not be further delegated.

DD-105 Use of Preaward Surveys.
For contracts administered by the Defense Contract Management Agency, obtain information needed to assess contractor compliance with subcontracting plans in current and previous contracts by requesting a preaward survey in accordance with FAR 9.106, DFARS 209.106 and DFARS PGI 209.106.

Part 2 – Rating System

DD-201 Acceptable Plans.
Objective: The subcontracting plan meets all of the requirements outlined in Part 3. The offeror has provided details that demonstrate an acceptable approach to assisting, promoting and utilizing small businesses, small disadvantaged businesses, women-owned small businesses, historically underutilized business zone small businesses, veteran-owned small businesses, service disabled veteran-owned small businesses and, for Defense Research Programs, historically black colleges and universities and minority serving institutions. The offeror has demonstrated an ability to meet prior subcontracting plan goals and honor the terms of subcontract agreements. Offeror has outlined an approach utilizing mentor protégé firms, joint venture teams, or other partners. The subcontracting goals are realistic, challenging, and attainable. Clarifications and minor rework of the submission may be required to correct slight omissions that do not prejudice other offers.

DD-202 Unacceptable Plans.
Objective: The subcontracting plan fails to meet a requirement outlined in Part 3. The offeror has not provided an acceptable approach to assisting, promoting, and utilizing small businesses. The offeror has a history of failing to honor subcontract

agreements. The offeror did not discuss the establishment of mentor protégé relationships, teaming, or joint venture agreements with other firms. Ensure the proposed subcontracting goals are attainable in light of the contractor's past performance in meeting subcontracting goals. Proposed subcontracting goals reflect less than a good faith effort. Substantial rework of the document is required to correct omissions and establish realistic, challenging, and attainable goals. Failure to receive a rating of acceptable may jeopardize offeror's eligibility for contract award. See FAR 19.702(a)(1).

Part 3 – Subcontracting Plan Requirements

DD-301 Requirements.

If any of the following are answered "NO", the plan is not acceptable, and the offeror must revise it before contract award. Does the plan:

1. Contain a policy statement or evidence of internal guidance to company buyers that commits to complying with the Small Business Act (Public Law 99-661, Section 1207 and Public Law 100-180)?

2. Identify separate percentage goals for utilizing small businesses (including Alaska Native Corporations (ANCs) and Indian tribes), veteran-owned small businesses (VOSB), service-disabled veteran-owned small businesses (SDVOSB), historically underutilized business zone small businesses (HUBZone), small disadvantaged businesses (SDB), women-owned small businesses (WOSB), and, for Defense Research Programs, historically black colleges and universities and minority serving institutions where applicable? Negotiated subcontracting goals must correlate with percentages of small business utilization identified in the contractor's small business participation plan, see FAR 15.304 and DFARS 215-304, and/or minimum targets identified in the solicitation or contract modification. FAR 19.704(a)(1)

3. Project the total dollars planned to be subcontracted and a separate statement of the total dollars planned to be subcontracted to small business (including ANCs and Indian tribes), VOSB, SDVOSB, HUBZone, SDB, and WOSB concerns? FAR 19.704(a)(2)

4. Describe the principal types of supplies and services to be subcontracted and identify the types planned for subcontracting to small business (including ANCs and Indian tribes), VOSB, SDVOSB, HUBZone, SDB and WOSB concerns?

5. Describe the method to be used to develop the subcontracting goals? FAR 19.704(a)(4)

6. Describe the method for identifying potential sources for solicitation purposes? FAR 19.704(a)(5)

7. State if the offeror included indirect costs in establishing subcontracting goals, and a description of the method used to determine the proportionate share of indirect costs to be incurred with small business, VOSB, SDVOSB, HUBZone, SDB (including ANCs and Indian tribes), and WOSB concerns? FAR 19.704(a)(6)

8. Identify the name of the employee who will administer the offeror's subcontracting program and describe that person's duties? FAR 19.704(a)(7)

9. Provide an approach for ensuring that small businesses, VOSB, SDVOSB, HUBZone, SDB, (including ANCs and Indian tribes) and WOSB concerns will have an equitable opportunity to compete for subcontracts?

10. Require the offeror to include the clause at FAR 52.219-8, Utilization of Small Business Concerns in all subcontracts that offer further subcontracting opportunities and require all subcontractors (except small business concerns) that receive subcontracts over $900,000 ($2,000,000 for construction) to adopt a plan that complies with the requirements of the clause at FAR 52.219-9, Small Business Subcontracting Plan?

11. Provide assurances that the offeror will:

   a. Cooperate in required studies or surveys;

   b. Submit periodic reports so that the Government can determine the extent of offeror's compliance with the subcontracting plan;

   c. Submit semi-annual Individual Subcontract Reports (ISRs) and/or Summary Subcontract Reports (SSR) in the Electronic Subcontracting Reporting System (eSRS) (http://www.esrs.gov) in accordance with FAR 52.219-9 ALT II or provide other ancillary reports as requested by the contracting officer or Army Small Business Office;

   d. Ensure that its subcontractors with subcontracting plans agree to submit the ISRs and/or SSRs using the eSRS;

e. Provide its prime contract number and its DUNS number and the e-mail address of the Government or contractor employee responsible for acknowledging or rejecting the reports, to all first-tier subcontractors with subcontracting plans so they can enter this information into the eSRS when submitting their reports; and

f. Require each subcontractor with a subcontracting plan to provide the prime contract number and its own DUNS number, and the e-mail address of the Government or contractor official responsible for acknowledging or rejecting the reports, to its subcontractors with subcontracting plans? FAR 19.704(10)

12. Describe the types of records that the contractor will maintain concerning procedures adopted to comply with the requirements and goals in the plan, including establishing source lists; and a description of the offeror's efforts to locate small business, VOSB, SDVOSB, HUBZone, SDB, and WOSB concerns and to award subcontracts to them? FAR 19.704(11)

13. Does plan, pursuant to FAR 19.704(11)(c), provide a separate goal for the basic contract and, if applicable, each option?

SMALL BUSINESS SUBCONTRACTING PLAN (SAMPLE)

The Revolutionary Far Overhaul, paragraph 19.109 prescribes the use of the clause at FAR 52.219-9 ALT II entitled "Small Business Subcontracting Plan." The following is a suggested model for use when formulating such subcontracting plan. While this model plan has been designed to be consistent with FAR 52.219-9, other formats of a subcontracting plan may be acceptable. However, failure to include the essential information as exemplified in this model may be cause for either a delay in acceptance or the rejection of a bid or offer where the clause is applicable. Further, the use of this model is not intended to waive other requirements that may be applicable under FAR 52.219-9. "Small Business Subcontracting Plan," as used in this clause, means any agreement (other than one involving an employer-employee relationship) entered into by a federal government prime contractor or subcontractor calling for supplies or services required for performance of the contract or subcontract.

I. IDENTIFICATION DATA:

Company Name: ____

Address: ____

Date Prepared: ______________________ Solicitation Number: ______________________

Description: ____

Estimated Contract Dollar Value: ____

II. TYPE OF PLAN (circle one)

A. Individual Plan (All elements developed specifically for this contract and applicable for the full term of this contract, including any option periods.)

B. Master Plan (Goals developed for this contract; all other elements standard; must be renewed every three years)

C. Commercial Plan Commercial products/service plan, including goals, covers the offeror's fiscal year and applies to the entire production of commercial items or delivery of services sold by either the entire company or a portion thereof (e.g., division, plant, or product line); this includes planned subcontracting for both commercial and Government business. In accordance with FAR 19.704(d), "A commercial plan (as defined in FAR 19.701) is the preferred type of subcontracting plan for contractors furnishing commercial items."(Contractor sells large quantities of off-the-shelf commodities to many Government agencies. Plans/goals negotiated by a lead agency on a company-wide basis rather than for individual contracts. Plan effective only during the year for which it is approved. The contractor must provide a copy of the lead agency approval.)

III. GOALS:

(For information purposes only: FAR 19.704(a)(1) requires separate percentage goals for using Small Business, Veteran-Owned Small Business, Service-Disabled Veteran-Owned Small Business, HUBZone Small Business, Small Disadvantaged Business, and Women-Owned Small Business concerns as subcontractors; and a statement of the total dollars planned to be subcontracted to Small Business, Veteran-Owned Small Business, Service-Disabled Veteran-Owned Small Business, HUBZone Small Business, Small Disadvantaged Business, and Women-Owned Small Business concerns. NOTE: The dollar amounts planned for subcontracting must be expressed as percentages of total subcontracting dollars as shown below.)

State separate dollar and percentage goals, expressed in terms of percentages of total subcontracting dollars, for the use of Large Business, Small Business, Veteran-Owned Small Business, Service Disabled Veteran-Owned Small Business, HUBZone Small Business, Small Disadvantaged Business, Woman-Owned Small Business, and Historically Black Colleges and Universities/Minority Institutions concerns as subcontractors. The offeror shall include all subcontracts that contribute to contract performance, and may include a proportionate share of products and services that are normally allocated as indirect costs in the following format. (For a contract with options, provide a separate statement for the basic contract and individual statements for each option year.)

A. **BASE BID ONLY:** The following percentage goals (expressed in terms of a percentage of total planned subcontracting dollars) and dollar amounts are applicable to the contract cited above or to the contract awarded under the solicitation cited. Total Base Bid is $______________________.

(i) Total estimated dollar value of all planned subcontracting for an individual contract plan; or the offerors total projected sales, expressed in dollars, and the total value of projected subcontracts to support the sales for a commercial plan; i.e., the sum of a and b above: $ (100 Percent) $__________________and ________________%

(ii) Total estimated dollar value and percent of planned subcontracting with Small Business (including Veteran-Owned Small Business, Service Disabled Veteran-Owned Small Business, HUBZone Small Business, Small Disadvantaged Business, Woman-Owned Small Business, and Historically Black Colleges and Universities/Minority Institutions concerns): (% of "(i)") $__________________and ______________%

(iii) Total estimated dollar value and percent of planned subcontracting with large businesses (all business concerns classified as "other than small"): (% of "(i)") $__________________and ________________%

(iv) Total estimated dollar value and percent of planned subcontracting with Small Disadvantaged Business concerns (SDB): $__________________and ________________% of total planned subcontracting dollars under this contract will be awarded to subcontractors who are small concerns owned and controlled by socially and economically disadvantaged individuals and appear on the Small Business Administration's list. (% of "(i)")

(v) Total estimated dollar value and percent of planned subcontracting with Women-Owned Small Business concerns (WOSB): $__________________and ________________% of total planned subcontracting dollars under this contract will be awarded to subcontractors who are WOSB. (% of "(i)")

(vi) Total estimated dollar value and percent of planned subcontracting with Veteran-Owned Small Business concerns (VOSB): $__________________and ________________% of total planned subcontracting dollars under this contract will be awarded to subcontractors who are VOSB. (% of "(i)")

(vii) Total estimated dollar value and percent of planned subcontracting with Service-Disabled Veteran-Owned Small Business concerns (SDVOSB): $__________________and ________________%
of total planned subcontracting dollars under this contract will be awarded to subcontractors who are SDVOSB. (% of "(i)")

(viii) Total estimated dollar value and percent of planned subcontracting with Historically Black Colleges and Universities/Minority Institutions (HBCU/MI): $__________________and ________________%
of total planned subcontracting dollars under this contract will go to HBCU's who are an institution determined by the Secretary of Education to meet the requirements of 34 CFR 608.2, the term also includes any nonprofit research institution that was an integral part of such a college or university before November 14, 1986; or MI's who are an institution of higher education meeting the requirements of Section 1046(3) of the Higher Education Act of 1965 (20 U.S.C. 1135d-5(3)) which, includes a Hispanic-serving institution of higher education as defined in Section 316(b)(1) of the Act (20 U.S.C. 1059c(b)(1)). (% of "(i)")

(ix) Total estimated dollar value and percent of planned subcontracting with HUBZone Small Business concerns: $__________________and ________________% of total planned subcontracting dollars under this contract will go to subcontractors who are small business concerns located in a historically underutilized business zone which is an area located within one or more qualified census tracts, qualified non-metropolitan counties, or lands within the external boundaries of an Indian reservation and appear on the Small Business Administration's HUBZONE web site at www.sba.gov/HUBZONE. (% of "(i)")

The following principal products and/or services will be subcontracted under the Base Bid of this contract, and the distribution among Large Business, Small Business, Veteran-Owned Small Business, Service Disabled Veteran-Owned Small Business, HUBZone Small Business, Small Disadvantaged Business, Woman-Owned Small Business, and Historically Black Colleges and Universities/Minority Institutions is as follows: **(Check all that apply)**

Subcontractor Name	Product or Service/ Description	Large Business	Small Business	VOSB	SDVOSB	HSB	SDB	WOSB	HBCU/ MI

(ATTACHMENT MAY BE USED IF ADDITIONAL SPACE IS REQUIRED)

B. **OPTIONS:** You must include a separate goal for each option. See the attached Continuation Sheet for Paragraph A for each option.

C. The following method was used in developing subcontract goals (i.e., Statement explaining how the product and service areas to be subcontracted were established, how the areas to be subcontracted to Small Business, Veteran-Owned Small Business, Service-Disabled Veteran-Owned Small Business, HUBZone Small Business, Small Disadvantaged Business, and Women-Owned Small Business concerns were determined, and how Small Business, Veteran-Owned Small Business, Service-Disabled Veteran-Owned Small Business, HUBZone Small Business, Small Disadvantaged Business, and Women-Owned Small Business concerns' capabilities were determined, to include identification of source lists utilized in making those determinations. Also a statement as to what efforts will be taken to improve on past goals and how SB and SDB firms will be included in areas without previous SB/SDB involvement).

____

____

D. A description of the method used to identify potential **SOURCES** for solicitation purposes (e.g., whether you used existing company source lists, the System for Award Management (SAM)) of the Small Business Administration (SBA), veterans service organizations, the National Minority Purchasing Council Vendor Information Service, the Research and Information Division of the Minority Business Development Agency in the Department of Commerce, or small, HUBZone, disadvantaged, and women-owned small business trade associations. A firm may rely on the information contained in SAM as an accurate representation of a concern's size and ownership characteristics for the purposes of maintaining a small, veteran-owned, service-disabled veteran-owned, HUBZone small, small disadvantaged and women-owned small business source list. Use of SAM as its source list does not relieve a firm of its responsibilities e.g., outreach, assistance, counseling, and publicizing subcontracting opportunities) in this clause.

____

____

E. Indirect and overhead costs (check one): _____HAVE _____HAVE NOT been included in the goals specified in Paragraph A and Paragraph B.

F. If "HAVE" was selected in Paragraph E, explain the method used in determining the proportionate share of indirect and overhead cost to be allocated as subcontracts to Small Business, Veteran-Owned Small Business, Service-Disabled Veteran-Owned Small Business, HUBZone Small Business, Small Disadvantaged Business, and Women-Owned Small Business concerns. (NOTE: Commercial Plans Must Include Indirect Costs).

____

____

IV. PROGRAM ADMINISTRATOR:

(For information purposes only: FAR 19.704(a)(7) requires information about the company employee who will administer the subcontracting program. Please provide the name, title, address, telephone number, fax machine number, email address, position within the corporate structure, and the duties of that employee.)

Name: ______________________________

Title: ______________________________

Position: ______________________________

Address: ______________________________

____

Telephone No: ______________________________

Fax No: ______________________________

Email Address: ______________________________

This individual's specific duties, as they relate to the firm's subcontracting program, are as follows:
General overall responsibility for this company's Small Business Program, the development, preparation and execution of individual subcontracting plans and for monitoring performance relative to contractual subcontracting requirements contained in this plan, including but not limited to:

A. Developing and maintaining offerors/bidders lists of small business, veteran-owned small business, service-disabled veteran-owned small business, HUBZone small business, small disadvantaged business, and women-owned small business concerns from all possible sources. Our firm may rely on the information contained in the SBA Small Business Source System, as an accurate representation of a concern's size and ownership characteristics for the purposes of maintaining a Small Business, Veteran-Owned Small Business, Service-Disabled Veteran-Owned Small Business, HUBZone Small Business, Small Disadvantaged Business, and Women-Owned Small Business source list. The Small Business Administration's (SBA's) list of Small Disadvantaged Businesses and small HUBZone businesses can be accessed through www.sam.gov. Select "Dynamic Small Business Search" to access the SBA small business source system.

B. Ensuring that procurement packages are structured to permit Small Business, Veteran-Owned Small Business, Service-Disabled Veteran-Owned Small Business, HUBZone Small Business, Small Disadvantaged Business, and Women-Owned Small Business concerns to participate to the maximum extent possible.

C. Assuring inclusion of Small Business, Veteran-Owned Small Business, Service-Disabled Veteran-Owned Small Business, HUBZone Small Business, Small Disadvantaged Business, and Women-Owned Small Business concerns in all solicitations for products or services that they are capable of providing.

D. Reviewing solicitations to remove statements, clauses, etc., which may tend to restrict or prohibit Small Business, Veteran-Owned Small Business, Service-Disabled Veteran-Owned Small Business, HUBZone Small Business, Small Disadvantaged Business, and Women-Owned Small Business participation, including recommendations to set aside competitions for SDB's

E. Ensuring periodic rotation of potential subcontractors on bidders' lists.

F. Ensuring that the bid proposal review board documents its reasons for not selecting low bids submitted by Small Business, Veteran-Owned Small Business, Service-Disabled Veteran-Owned Small Business, HUBZone Small Business, Small Disadvantaged Business, and Women-Owned Small Business concerns.

G. Ensuring the establishment and maintenance of records of solicitations and subcontract award activity.

H. Attending or arranging for attendance of company counselors at Business Opportunity Workshops, Minority Business Enterprise Seminars, Trade Fairs, etc.

I. Conducting or arranging for conduct of motivational training for purchasing personnel pursuant to the intent of Public Laws 95-507, 99-661, and 100-180.

J. Monitoring attainment of proposed goals.

K. Preparing and submitting timely, required subcontract reports

L. Coordinating contractor's activities during the conduct of compliance reviews by Federal agencies.

M. Coordinating the conduct of contractor's activities involving its Small Business, Veteran-Owned Small Business, Service-Disabled Veteran-Owned Small Business, HUBZone Small Business, Small Disadvantaged Business, and Women-Owned Small Business subcontracting program.

N. Ensuring Individual Subcontract Reports (ISRs) and Summary Subcontract Reports (SSRs) are submitted using eSRS (http://www.esrs.gov), following the instructions in the eSRS.

O. Notifying the Contracting Officer or his representative in writing of any substitutions of firms that are not Small Business, Veteran-Owned Small Business, Service-Disabled Veteran-Owned Small Business, HUBZone Small Business, Small Disadvantaged Business, and Women-Owned Small Business for the firms listed in the subcontracting plan.

P. Additions to (or deletions from) the duties specified above are as follows:

____
____
____

V. EQUITABLE OPPORTUNITY:

(For information purposes only: FAR 19-704(8) requires a description of the efforts the contractor will make to ensure that Small Business, Veteran-Owned Small Business, Service-Disabled Veteran-Owned Small Business, HUBZone Small Business, Small Disadvantaged Business, and Women-Owned Small Business concerns will have an equitable opportunity to compete for subcontracts.)

The following efforts will be taken to assure that Small Business, Veteran-Owned Small Business, Service-Disabled Veteran-Owned Small Business, HUBZone Small Business, Small Disadvantaged Business, and Women-Owned Small Business concerns will have an equitable opportunity to compete for subcontracts, including items not traditionally awarded to SB or SDB firms:

A. Outreach efforts will be made by:

(i) Contacts with minority and small business trade associations such as veterans service organizations, the National Minority Purchasing Council Vendor Information Service, the Research and Information Division of the Minority Business Development Agency in the Department of Commerce.

(ii) Contacts with business development organizations.

(iii) Attendance at small and minority business procurement conferences and trade fairs.

(iv) Sources will be requested from Small Business Administration's small business source system.

(v) Reviews to determine the competence, ability, experience and capacity available from SB and SDB firms and providing technical assistance to same.

(vi) Evaluations of our SB, SDB, WOSB, VOSB, SDVOSB and HUBZone award performance and program effectiveness against goals established company-wide.

B. The following internal efforts will be made to guide and encourage buyers:

(i) Workshops, seminars and training programs will be conducted.

(ii) Activities will be monitored to evaluate compliance with this subcontracting plan, evaluating SB, SDB, WOSB, VOSB, SDVOSB and HUBZone award performance and program effectiveness.

(iii) Small business, veteran-owned small business, service-disabled veteran-owned small business, HUBZone small business, small disadvantaged business, and women-owned small business concern source lists, guides and other data identifying small, small disadvantaged and women-owned small business concerns will be maintained and utilized by buyers in soliciting subcontracts.

(iv) Additions to (or deletion from) the above listed efforts are as follows:

____
____
____

VI. FLOW DOWN CLAUSE:
(For information purposes only: FAR 19-704(a)(9) requires that your company include FAR 52.219-8, "Utilization of Small Business Concerns," in all subcontracts that offer further subcontracting opportunities. Your company must require all subcontractors, except small business concerns, that receive subcontracts in excess of $900,000 ($2,000,000 for construction) to adopt a plan that complies with the requirements of FAR 52.219-9, "Small Business Subcontracting Plan.")

The offeror (contractor) agrees that the clause entitled "Utilization of Small Business Concerns" at FAR 52.219-8 will be included in all subcontracts that offer further subcontracting opportunities, and all subcontractors (except small business concerns) who receive subcontracts in excess of $900,000 ($2,000,000 for construction) will be required to adopt a subcontracting plan that complies with FAR 52.219-9. Such plans will be reviewed by comparing them with the provisions of Public Law 95-507, and assuring that all minimum requirements of an acceptable subcontracting plan have been satisfied. The acceptability of percentage goals shall be determined on a case-by-case basis depending on the supplies/services involved, the availability of potential Small Business, Veteran-Owned Small Business, Service-Disabled Veteran-Owned Small Business, HUBZone Small Business, Small Disadvantaged Business, and Women-Owned Small Business subcontractors, and prior experience. Once approved and implemented, plans will be monitored through the submission of periodic reports, and/or, as time and availability of funds permit, periodic visits to subcontractors facilities to review applicable records and subcontracting program progress.

VII. REPORTING AND COOPERATION:
(For information purposes only: FAR 19-704(a)(10) requires your company (i) cooperate in any studies or surveys as may be required, (ii) submit periodic reports which show compliance with the subcontracting plan; (iii) submit the Individual Subcontract Report (ISR), and the Summary Subcontract Report (SSR) using the Electronic Subcontracting Reporting System (eSRS);, (iv) ensure that subcontractors with subcontracting plans agree to submit the ISR and/or the SSR using eSRS, (v) provide the prime contract number, DUNS number, and the e-mail address of the offeror's official responsible for acknowledging receipt of or rejecting the ISRs, to all first-tier subcontractors with subcontracting plans so they can enter this information into the eSRS when submitting their ISRs, and (vi) require that each subcontractor with a subcontracting plan provide the prime contract number, its own DUNS number, and the e-mail address of the subcontractor's official responsible for acknowledging receipt of or rejecting the ISRs, to its subcontractors with subcontracting plans.)

The offeror/contractor agrees to submit such periodic reports and cooperate in any studies or surveys as may be required by the contracting agency or the Small Business Administration in order to determine the extent of compliance by the offeror/contractor with the subcontracting plan and with the clause entitled "Utilization of Small Business Concerns," contained in the contract. The above reports will include submission of its Individual Subcontracting Report (ISR) and Summary Subcontract Report (SSR)

The offeror/contractor further agrees to ensure that its subcontractors agree to submission of ISRs and SSRs.
ISRs and SSRs shall be submitted via the Electronic Subcontracting Reporting System (eSRS) website www.esrs.gov

Reporting Period	Report Due	Due Date
Oct 1 - Mar 31	ISR/SF294	4/30
Apr 1 - Sept 30	ISR/SF294	10/30
Oct 1 – Mar 31	SSR/SF295	4/30 (for contracts with the DOD)
Apr 1 – Sept 30	SSR/SF295	10/30 (for contracts with DOD)
Oct 1 - Sept 30	SSR/SF295	10/30 (for civilian agencies)
Contract Completion	SSR/SF295	30 days after close of contractor's fiscal year (Commercial Plan)

VIII. RECORDKEEPING:
(For information purpose only: FAR 19-704(a)(11) requires a list of the types of records your company will maintain to demonstrate the procedures adopted to comply with the requirements and goals in the subcontracting plan.)
The offeror/contractor agrees that he will maintain at least the following types of records to document compliance with this subcontracting plan:

A. Small Business, Veteran-Owned Small Business, Service-Disabled Veteran-Owned Small Business, HUBZone Small Business, Small Disadvantaged Business, and Women-Owned Small Business concern source lists, guides and other data identifying SB/SDB concerns.

B. Organizations contacted for Small Business, Veteran-Owned Small Business, Service-Disabled Veteran-Owned Small Business, HUBZone Small Business, Small Disadvantaged Business, and Women-Owned Small Business sources.

C. On a contract-by-contract basis, records on all subcontract solicitations over $150,000, indicating on each solicitation (i) whether small business concerns were solicited, and if not, why not; (ii) whether Veteran-Owned Small Business concerns were solicited, and if not, why not; (iii) whether Service-Disabled Veteran-Owned Small Business concerns were solicited, and if not, why not; (iv) whether HUBZone Small Business concerns were solicited, and if not, why not; (v) whether Small Disadvantaged business concerns were solicited, and if not, why not; (vi) whether Women-Owned Small Business concerns were solicited, and if not, why not; and (vii) reasons for the failure of solicited Small Business, Veteran-Owned Small Business, Service-Disabled Veteran-Owned Small Business, HUBZone Small Business, Small Disadvantaged Business, and Women-Owned Small Business concerns to receive the subcontract award.

D. Records to support other outreach efforts: Contacts with veteran service organizations, Minority and Small Business Trade Associations, etc., and attendance at small and minority business procurement conferences and trade fairs.

E. Records to support internal activities to guide and encourage buyers: Workshops, seminars, training programs, etc., monitoring activities to evaluate compliance.

F. On a contract-by-contract basis, records to support subcontract award data to include name and address and business size of each subcontractor. Contractors having commercial plans need not comply with this requirement.

G. Records to be maintained in addition to the above are as follows:

____
____
____

IX. ASSURANCES

(For information purpose only: FAR 19.704(a)(12-15) requires assurances from your company)

A. Assurances that the offeror will make a good faith effort to acquire articles, equipment, supplies, services, or materials, or obtain the performance of construction work from the small business concerns that the offeror used in preparing the bid or proposal, in the same or greater scope, amount, and quality used in preparing and submitting the bid or proposal. Responding to a request for a quote does not constitute use in preparing a bid or proposal. An offeror used a small business concern in preparing the bid or proposal if--

(i) The offeror identifies the small business concern as a subcontractor in the bid or proposal or associated small business subcontracting plan, to furnish certain supplies or perform a portion of the contract; or

(ii) The offeror used the small business concern's pricing or cost information or technical expertise in preparing the bid or proposal, where there is written evidence of an intent or understanding that the small business concern will be awarded a subcontract for the related work if the offeror is awarded the contract;

B. Assurances that the contractor will provide the contracting officer with a written explanation if the contractor fails to acquire articles, equipment, supplies, services or materials or obtain the performance of construction work as described in (a)(12) of this section. This written explanation will be submitted to the contracting officer within 30 days of contract completion; and

C. Assurances that the contractor will not prohibit a subcontractor from discussing with the contracting officer any material matter pertaining to payment to or utilization of a subcontractor.

D. Assurances that the offeror will pay its small business subcontractors on time and in accordance with the terms and conditions of the subcontract, and notify the contracting officer if the offeror pays a reduced or an untimely payment to a small business subcontractor (see 52.242-5).

X. SIGNATURES REQUIRED:

This subcontracting plan was SUBMITTED by:

Signature: ____ Date: _____________

Typed Name and Title: ____

Phone Number: ____

Contracting Officer Approval: _____________________________________ Date: _____________

CONTINUATION OF PARAGRAPH A

NAME OF OPTION: Option 1

	Dollars	**Percentage**
1. Total option contract price	____________	____________
2. Total to be subcontracted (to all types of businesses)	____________	____________
a. To Large Business	____________	____________
b. To Small Business	____________	____________
i. To Veteran-Owned Small Business	____________	____________
ii. To Service-Disabled Veteran-Owned Small Business	____________	____________
iii. To HUBZone Small Business	____________	____________
iv. To Small Disadvantaged Business	____________	____________
v. To Women-Owned Small Business	____________	____________

The following principal products and/or services will be subcontracted under Option A of this contract, and the distribution among LB, SB, SDB, WOSB, VOSB, SDVOSB, HBCU/MI, and HUBZone SB is as follows: **(Check all that apply)**

Subcontractor Name	**Product or Service/ Description**	**Large Business**	**Small Business**	**VOSB**	**SDVOSB**	**HSB**	**SDB**	**WOSB**	**HBCU/ MI**

NAME OF OPTION: Option 2

	Dollars	**Percentage**
1. Total option contract price	____________	____________
2. Total to be subcontracted (to all types of businesses)	____________	____________
a. To Large Business	____________	____________
b. To Small Business	____________	____________
i. To Veteran-Owned Small Business	____________	____________
ii. To Service-Disabled Veteran-Owned Small Business	____________	____________
iii. To HUBZone Small Business	____________	____________
iv. To Small Disadvantaged Business	____________	____________
v. To Women-Owned Small Business	____________	____________

The following principal products and/or services will be subcontracted under Option A of this contract, and the distribution among LB, SB, SDB, WOSB, VOSB, SDVOSB, HBCU/MI, and HUBZone SB is as follows: **(Check all that apply)**

Subcontractor Name	**Product or Service/ Description**	**Large Business**	**Small Business**	**VOSB**	**SDVOSB**	**HSB**	**SDB**	**WOSB**	**HBCU/ MI**

SECTION III

SOURCE SELECTION TEAM (SST) ROLES AND RESPONSIBILITIES

A. EVALUATION ORGANIZATION. The evaluation and selection of the successful offeror will be performed by the following organizational elements:

1. Source Selection Authority (SSA)
2. Source Selection Evaluation Board (SSEB)
3. Procuring Contracting Officer (PCO)

* The composition of the SSEB is shown in Appendix A and their responsibilities are discussed below.

B. RESPONSIBILITIES OF THE ORGANIZATIONAL ELEMENTS.

1. Source Selection Authority. The SSA is the individual designated to make the best-value decision. The SSA shall:

a. Ensure that the SSP and evaluation criteria are consistent with the requirements of the RFP and applicable regulations.

b. Be responsible for the proper and efficient conduct of the source selection process in accordance with this procedure and all applicable laws and regulations.

c. Appoint the chairperson for the SSEB.

d. Ensure that personnel appointed to the SST are knowledgeable of policy and procedures for properly and efficiently conducting the source selection. Ensure the SST members have the requisite acquisition experience, skills, and training necessary to execute the source selection, and ensure the highest level of team membership continuity for the duration of the selection process.

e. Ensure no senior leader is assigned to or performs dual leadership roles in the source selection in accordance with DFARS 203.170(a).

f. Ensure that realistic source selection schedules are established and source selection events are conducted efficiently and effectively in meeting overall program schedules. The schedules should support proper and full compliance with source selection procedures outlined in this document and the SSA-approved SSP for the acquisition.

g. Ensure all involved in the source selection are briefed and knowledgeable of applicable portions of 41 U.S.C. § 2102—Prohibitions on Disclosing and Obtaining Procurement Information; FAR 3.104 regarding unauthorized disclosure of contractor bid and proposal information and source selection information; and 5 Code of Federal Regulations Part 2635, Standards of Ethical Conduct for Employees of the Executive Branch, regarding applicable standards of conduct (including procedures to prevent the improper disclosure of information). To confirm statutory and regulatory compliance, ensure all persons receiving source selection information sign a Non-disclosure Agreement and a Conflict of Interest statement. Ensure Conflict of Interest Statements (from both Government members/advisors and nongovernment team advisors) are appropriately reviewed and actual or potential conflict of interest issues are resolved prior to granting access to any source selection information.

h. If the solicitation states the Government intends to award without discussions and it is later determined that discussions are necessary, review and approve the PCO's written rationale (see FAR 15.202(a)(2)). If discussions will be conducted, review and approve the PCO's written determination of the competitive range or elimination of an offeror previously determined to be in the competitive range.

i. Select the source whose proposal offers the best value to the Government in accordance with the evaluation criteria and basis for award stated in the solicitation.

j. Document the rationale in the Source Selection Decision Document (SSDD).

2. Source Selection Evaluation Board. Government personnel assigned to the SSEB shall consider this duty as their primary responsibility. Their source selection assignment shall take priority over other work assignments.

a. The SSEB Chairperson shall:

1. Review all aspects of all proposals and shall fully participate in all ratings and ensure preparation by the SSEB of narrative support substantiating evaluation ratings. The SSEB Chairperson is responsible for the conduct of a comprehensive and integrated evaluation of competitive proposals in an impartial and equitable manner.

2. Assure that the SSEB members understand the criteria for the evaluation of proposals so that there is a uniformity of approach in the rating effort.

3. Require the assigned members' attendance at the meetings and conferences of the board and assign work necessary for the accomplishment of its mission.

4. Relieve and replace members from assignment ***only*** in the event of a demonstrated emergency or other appropriate cause.

5. Arrange for members to work overtime, when necessary, allowed, and approved.

6. Assure the safeguarding of sensitive information used by the board.

7. Arrange for the needed administrative staff at the work site.

8. Plan the security requirements of the board and the work site and ensure their accomplishment through conclusion of the source selection, to include any protests.

9. Establish the agenda and the schedule for SSEB meetings.

10. Isolate policy issues and major questions requiring decision by the SSA.

11. Seek to build consensus among the SSEB members.

12. Be responsible for the overall management of the SSEB and act as the SSEB's interface to the SSA.

13. Establish functional evaluation teams, as appropriate, to support an efficient source selection evaluation. Identify chairpersons and members to the functional evaluation teams, subject to approval of the SSA.

14. Ensure the skills of the personnel, the available resources, and the time assigned are commensurate with the complexity of the acquisition.

15. Ensure members of the SSEB are trained and knowledgeable on how an evaluation is conducted prior to reviewing any proposals.

16. Ensure the evaluation process follows the evaluation criteria and ratings are being consistently applied.

17. Provide consolidated evaluation results to the SSA.

18. Support any post source selection activities such as debriefings and post-award reviews/meetings, as required.

b. The SSEB members shall:

1. Conduct a comprehensive review and evaluation of proposals based solely on the evaluation criteria outlined in the RFP.

2. Assist the SSEB Chairperson in documenting the SSEB evaluation results.

3. Support any post source selection activities such as debriefings and post-award reviews/meetings, as required.

4. Contemporaneously and thoroughly document evaluation of proposals in writing and provide the written evaluation narratives to the SSEB Chairperson to assist in documenting the results.

3. Procuring Contracting Officer. The PCO will serve as the primary business advisor and principal guidance source for the entire Source Selection. The PCO shall:

a. Manage all business aspects of the acquisition and work with the SSEB Chairperson to ensure the evaluation is conducted in accordance with the evaluation criteria specified in the solicitation.

b. Ensure that required approvals are obtained and the appropriate notification clause is included in the solicitation before non-Government personnel are allowed to provide source selection support (FAR 7.503)

c. In accordance with FAR 3.104 and DFARS 203.104, ensure that procedures exist to safeguard source selection information and contractor bid or proposal information (FAR 15.108(b)). Approve access to or release of source selection information and contractor bid or proposal information after consulting Legal Counsel before and after contract award.

d. Maintain, as a minimum, the documents and source selection evaluation records.

e. Release the final solicitation only after obtaining all required approvals, including the SSA approval of the SSP.

f. Serve as the single point of contact for all solicitation-related inquiries from actual or prospective offerors.

g. After receipt of proposals, control exchanges with offerors in accordance with FAR 15.202(a)(2).

h. With the approval of the SSA, establish the competitive range and enter into discussions.

i. Brief the SSA as requested.

j. Respond to special instructions from the SSA.

k. Award the contract.

l. Chair all required debriefings.

SECTION IV
EVALUATION PROCESS AND PROCEDURES

A. KEY MILESTONES. The source selection process will follow the sequence outlined below:

1. Issue RFP
2. Receive Proposals
3. Conduct Evaluation
4. Present Findings to SSA
5. SSA Decision
6. Contract Award
7. Debriefing of Unsuccessful Offerors

B. DEFINITIONS

1. Rating. The adjective descriptor assigned by the evaluators to the non-Price Factors. It represents their conclusions as to the quality of the proposal, supported by narrative write-ups identifying the associated strengths, weaknesses, deficiencies, risks, and uncertainties. The strengths, weaknesses, etc., are the findings that support the rating adjective.

2. Factor Rating Definitions. SSEB will be conducted using a common set of adjectival ratings to evaluate the Management Plan Factor and the Small Business Participation Factor. Keep in mind that mere promises to comply with contractual requirements are insufficient basis for a favorable rating; evidence is required in support of any statements relating to promised performance. Also, the chosen adjectival rating must be supported by the evaluator's narrative reason(s) for the choice. The components of each factor will be assessed for significant strengths, strengths, significant weaknesses, weaknesses, deficiencies, risks, and uncertainties. Each factor will be assigned an adjectival rating based upon these collective findings. The compilation of these factor adjectival ratings will form the basis of the factor rating. Each proposal will be evaluated in accordance with the information set forth in the RFP, utilizing adjectival ratings.

Evaluators will apply the adjectival rating for the definition that most closely matches the evaluation.

TABLE 1

Past Performance Relevancy Ratings	
Rating	**Description**
Very Relevant	Present/past performance effort involved essentially the same scope and magnitude of effort and complexities this solicitation requires.
Relevant	Present/past performance effort involved similar scope and magnitude of effort and complexities this solicitation requires.
Somewhat Relevant	Present/past performance effort involved some of the scope and magnitude of effort and complexities this solicitation requires.
Not Relevant	Present/past performance effort involved little or none of the scope and magnitude of effort and complexities this solicitation requires.

TABLE 2

Performance Confidence Assessments	
Rating	**Description**
Substantial Confidence	Based on the offeror's recent/relevant performance record, the Government has a high expectation that the offeror will successfully perform the required effort.
Satisfactory Confidence	Based on the offeror's recent/relevant performance record, the Government has a reasonable expectation that the offeror will successfully perform the required effort.

Neutral Confidence	No recent/relevant performance record is available, or the offeror's performance record is so sparse that no meaningful confidence assessment rating can be reasonably assigned. The offeror may not be evaluated favorably or unfavorably on the factor of past performance.
Limited Confidence	Based on the offeror's recent/relevant performance record, the Government has a low expectation that the offeror will successfully perform the required effort.
No Confidence	Based on the offeror's recent/relevant performance record, the Government has no expectation that the offeror will be able to successfully perform the required effort.

TABLE 3

Combined Technical/Risk Rating Method	
Adjectival Rating	**Description**
Outstanding	Proposal demonstrates an exceptional approach and understanding of the requirements and contains multiple strengths and/or at least one significant strength, and risk of unsuccessful performance is low.
Good	Proposal indicates a thorough approach and understanding of the requirements and contains at least one strength or significant strength, and risk of unsuccessful performance is low to moderate.
Acceptable	Proposal meets requirements and indicates an adequate approach and understanding of the requirements, and risk of unsuccessful performance is no worse than moderate.
Marginal	Proposal has not demonstrated an adequate approach and understanding of the requirements, and/or risk of unsuccessful performance is high.
Unacceptable	Proposal does not meet requirements of the solicitation and, thus, contains one or more deficiencies and is unawardable, and/or risk of performance is unacceptably high.

TABLE 4

Small Business Ratings	
Rating	**Description**
Acceptable	Proposal indicates an adequate approach and understanding of small business objectives.
Unacceptable	Proposal does not meet small business objectives.

3. Findings Definitions. The findings definitions will be the same definitions as shown in proposal evaluation criteria, paragraph 4.7 Definitions.

C. RATING METHOD

1. Rating Package. Each individual evaluating a factor of the offeror's proposal will receive a rating package containing the following:

a. Evaluation Policies and Procedures (Section IV and V of this Plan);

b. Basis for Award, Evaluation Factors and Evaluation;
(Section II of this Plan);

c. Proposal Instructions
(Section II of this Plan);

d. Past Performance Evaluation forms (Appendix D), as appropriate;

e. Guidelines for writing clarifications and discussion issues (Appendix E);

f. Evaluation Notice (EN) form (Appendix F); and,

g. Request for Proposal (RFP).

2. Rating Structure. The non-price factors will be evaluated and rated based upon the general and specific instructions supplied in Section IV of this Plan. The ratings will be used for the final evaluation. The Price factor, while evaluated for fairness and reasonableness (including findings), will not be assigned an adjectival rating.

D. PROPOSAL EVALUATION

1. All proposals will be received by the Procuring Contracting Officer (PCO) or designee not later than the hour and date given in the RFP. Upon receipt of proposals the PCO or designee should conduct an initial screening to ascertain that each offeror has submitted all of the required information, including electronic media, in the quantities and format specified in the RFP. The PCO will retain the original of the offerors proposals and associated data. The SSEB Chairperson will control all copies of the offerors' proposals and other associated data.

2. Upon receipt of proposals, evaluators will read each section to gain an understanding of the level of the information and determine if errors, omissions or a need for clarification exists. These will be reported to the SSEB Chairperson. The SSEB Chairperson will notify the Contracting Officer of such issues.

3. Upon completion of the evaluation, the SSEB team will assign the appropriate rating to each factor as set forth above upon ***reaching consensus*** as to the findings and the ratings. Care must be taken to ***cite the section, page and paragraph numbers*** generating these findings. If after discussion of the findings and ratings a consensus cannot be reached, a Minority Report will be completed for inclusion in the file. The SSEB Chairperson will prepare an overall narrative summary for each factor along with recommended factor ratings.

4. The SSEB Chairperson will review the narrative summary and recommended factor ratings and provide an overall summary report. This report will be forwarded to the SSA and shall contain the adjectival assessments for each factor as well as a Price report and the supporting rationale.

5. Any proposal(s) which the Contracting Officer determines, to not be among the most highly rated proposals will be considered outside the competitive range and eliminated from further consideration. The Contracting Officer will promptly inform the offeror(s) in writing.

6. The Contracting Officer may conduct discussions with each offeror retained in the competitive range. After the conclusion of discussions, any new information received from the offerors will be evaluated. This evaluation will be documented in a supplement to the Initial Evaluation Report (Interim Evaluation Report) and will consist of an update, which addresses each factor of the Initial Evaluation Report.

7. Using the Interim Evaluation Report, the Contracting Officer will amend the competitive range. The Contracting Officer will give all offerors within the revised competitive range an opportunity to submit final proposal revisions by a common cutoff date. Any final proposal revisions will be evaluated and the re-evaluation will be documented in another supplemental report (Final Evaluation Report), which addresses each factor of the Initial and Interim Evaluation Reports. The Final Evaluation Report will not contain a recommendation pertaining to which offeror should be selected for award.

E. COMMUNICATIONS AND SECURING SOURCE SELECTION MATERIALS. The PCO and SSEB Chair will ensure "information security", including information integrity, confidentiality and availability is maintained as described in FAR 2.101. Electronic source selection information shall be housed in an access controlled, password protected site. All source selection materials will be marked in accordance with FAR 2.101 (see definition of source selection information) and 3.104 and secured as prescribed by the PCO and SSEB Chair. Communications (internal and external) procedures and limitations (if any) will be provided by the PCO and SSEB Chair.

Written Comparative Analysis. The SSA must consider whether or not the benefits of the non-price strength warrant the additional price premium. This is accomplished by conducting a trade-off analysis among the competing proposals. While the award recommendation decision process may appear simple, it is far from it. The evaluation, proposal comparison, and tradeoff analysis process require a great deal of subjectivity and judgment**.** When performing the comparative analysis, consider each proposal's total evaluated price and the discriminators in the non-cost ratings as indicated by each proposal's strengths, weaknesses, and risks. Consider these differences in light of the relative importance of each evaluation factor. The comparative analysis does not preclude eventual selection of the lowest price offer as providing the best value. In fact, selection of a higher-priced offer always involves the necessity to state in the source selection decision document the rationale for concluding that payment of a higher price is justified by a proportionate superiority in non-cost factors. If the superior technical proposal is not selected, it is also imperative that the rationale for its non-selection be documented.

F. BEST VALUE DECISION. The SSA will compare the proposals to determine the offer(s) that represent(s) the best value to the Government, taking into consideration the stated evaluation factors and their respective weightings as specified in the RFP. The selection process is complex and depending upon the evaluation factors, the SSA may exercise a significant degree of judgment in selecting the successful offeror(s). The adjectival ratings assigned by the evaluation team are labels and not the sole basis for proposal comparison. ***The SSA must not base his/her decision merely on the adjectival ratings or the comparative analysis, which compares the strengths and weaknesses of the competing proposals, but shall exercise independent judgment.***

The SSA must document his/her rationale for selecting the successful offeror(s) in an independent, stand-alone document.

G. ANNOUNCEMENT OF SELECTION. The Contracting Officer will make the announcement of the selection of a successful contractor directly or through his/her designee.

H. DEBRIEFING OF UNSUCCESSFUL OFFERORS. The Contracting Officer will chair and conduct the debriefing. Debriefings will be given to offerors who submit a timely request.

SECTION V
POLICIES, INSTRUCTIONS AND STANDARDS OF CONDUCT

A. GENERAL. The impartial, equitable, and comprehensive evaluation of offerors' proposals is dependent on the capability of the Government to provide each offeror the same information and to evaluate each proposal independently.

B. SAFEGUARDING PROCUREMENT INFORMATION. The sensitivity of the proceedings and documentation require stringent and special safeguards throughout the evaluation process.

1. Inadvertent release of information could be a source of considerable misunderstanding and embarrassment to the Government. It is incumbent, therefore, upon all members of the team not to make any unauthorized disclosures of information pertaining to this evaluation. All evaluation participants will observe the following rules:

   a. Do not permit members of your organization to divulge your membership to casual callers.

   b. Refer all attempted communications by offerors' representatives to the Contracting Officer.

   c. Do not accept any invitation from personnel of an offeror for participation in any functions, regardless of how remote they may be from the evaluation process.

   d. Do not assume that a nonparticipating contractor can be told anything pertaining to the evaluation and source selection.

   e. Do not assume that it is safe to speak about the evaluation because you are among Government employees or in Government buildings.

   f. Do not discuss any aspect of the evaluation with other board members outside the area designated for deliberations.

   g. Do not discuss the substantive issues of the evaluation with any unauthorized individual even after the announcement of the winning offeror.

   h. Your supervisor does not have a "need-to-know" regarding any aspect of the SSO proceedings if they have not been appointed as a member of the SST.

2. Care must be exercised to ensure that copies of the evaluation records and information relating thereto are adequately marked and safeguarded throughout the entire proceedings.

3. Removal of proposals or evaluation documents from the evaluation work site is not authorized except as specifically approved by the SSEB Chairperson.

4. To the degree feasible, all proposals and working papers will be kept in a locked container or room except when being used in conjunction with evaluation and source selection.

5. All participating personnel will sign the Agreement set forth at Appendix B to the effect that they are familiar with the regulations and other guidance pertaining to security measures.

6. Proposals may contain information or data, which the offerors do not want disclosed to the public or to other contractors or used by the Government for any purpose other than the evaluation of this proposal. Data so marked shall not be disclosed outside the Government without the written permission of the offeror except under the conditions provided in the offerors' restrictions. Since it is anticipated that there will be group discussions concerning aspects of offerors' proposals, conferees shall advise other members when proprietary data is being discussed. **While special emphasis is given here to proprietary data, SST members should NEVER make any assumptions regarding release of any information relative to the source selection.**

C. EVALUATION POLICIES

1. The principal purpose of the evaluation procedure is to provide a sound basis for the SSA to make an informed judgment. The evaluation methodology and techniques employed should enhance the quality, credibility and confidence levels in

the adequacy of the evaluation results. The evaluation process, therefore, must be consistent, well thought out, adequately staffed, managed, and carried out in a professional, comprehensive and objective manner. It must frame the elements for the selection decision with sufficient clarity and visibility so that the SSA will be able to make a sound decision within a short time period.

2. Proposal evaluation requires a mixture of fact-finding and reporting and the application of professional judgment to provide a comprehensive picture of the adequacy of each proposal. This requires:

a. Examination and judgment of the merits of each proposal as compared to the criteria established for evaluation.

b. Validation of the information, estimates and projections of each offeror as presented in their proposal.

c. Successive summarization of the detailed evaluation results accompanied by analysis in sufficient depth to give visibility to any significant findings or reservations.

APPENDIX A
Source Selection Team

Unaccompanied Enlisted Personnel Housing (UEPH) Barracks - Fort Campbell, Kentucky

Source Selection Authority/ Contracting Officer – Stephanie M. Craig

Office of Counsel – Erin Konopa

Source Selection Evaluation Board (SSEB)
Ranzel L. Merideth – Chairperson (LRL Contracting Non-Voting)
Jack Oakley – LRL PM; ALT: Greg Moore– LRL PM
Michael Barry – LRL ED; ALT David Yankey – LRL ED
Marshal Biter – LRL CD
Tori Delasko – FTC DPW

Technical Advisor: Josh Bush (Structural)

APPENDIX B
SOURCE SELECTION PARTICIPATION AGREEMENT

Important! This Agreement concerns a matter within the jurisdiction of a United States government agency. This Agreement prohibits you from making false, fictitious, or fraudulent statements and/or certifications. If you do so, you may be subject to prosecution under 18 U.S.C §1001.

AGREEMENT

1. This Agreement applies to individuals involved in Solicitation W912QR26RA0016, also known as the construction of an Unaccompanied Enlisted Personnel Housing (UEPH) Barracks Fort Campbell, Kentucky

2. This Agreement contains the rules of conduct relating to this acquisition. It includes rules of conduct regarding conflicts of interest as well as rules of conduct regarding the safeguarding of confidential information.

3. Your signature on this Agreement indicates that you have read this Agreement and agree to be bound by its terms.

TERMS

4. I have read, understand and will abide by the requirements of the Federal Acquisition Regulation (FAR) §3.104. I understand that I may request a copy of FAR 3.104 from the Contracting Officer for this acquisition.

5. Except as set forth below, I do not presently hold, and will not obtain during my participation in this acquisition, any financial interest* or affiliation** in any reasonably likely offeror or subcontractor for this acquisition.

____

6. To the best of my knowledge, and except as set forth below, my spouse and dependent children do not have a financial interest* or affiliation** in any reasonably likely offeror or subcontractor for this acquisition.

____

7. To the best of my knowledge, and except as set forth below, none of the following is a reasonably likely offeror or proposed subcontractor for this acquisition, or represents a reasonably likely offeror or proposed subcontractor with regard to this acquisition:

- any person or company with whom I have or am seeking a business, contractual or other financial relationship that involves other than a routine consumer transaction;
- my spouse and dependent children;
- any person or company with whom I have been affiliated within the last year;
- any organization in which I am an active participant.

*Financial Interest - Any continuing financial interest (such as through a pension or retirement plan, shared income, continuing termination payments, or other arrangements as a result of any current or prior employment or business or professional association) or any financial interest through legal or beneficial ownership of stock, stock options, bonds, securities, or other arrangements including trusts.

**Affiliation - A relationship as an employee, officer, owner, director, member, trustee, partner, advisor, agent, representative, or consultant; or a person having any understanding, plans or pending contacts regarding such a relationship in the future. (This includes sending resumes, making telephone inquiries or any act that reasonably could be construed as an indication of interest in a future affiliation.)

8. I understand that I may request a statement from the Contracting Officer as to whether a person or company is considered to be a reasonably likely offeror or subcontractor.

9. I will not knowingly disclose any contractor bid or proposal information or source selection information regarding this acquisition directly or indirectly to any person other than a person authorized in accordance with FAR 3.104 to receive such information.

10. I will observe the following rules during the conduct of the acquisition:

a. I will not solicit or accept, directly or indirectly, any promise of future employment or business opportunity from, or engage, directly or indirectly, in any discussion of future employment or business opportunity with, any officer, employee, representative, agent, or consultant of any reasonably likely offeror or subcontractor for this acquisition.

b. I will not ask for, demand, exact, solicit, seek, accept, receive, or agree to receive, directly or indirectly, any gratuity, favor, discount, entertainment, hospitality, loan, forbearance or other thing of value from any officer, employee, representative, agent, or consultant of any reasonably likely offeror or subcontractor for this acquisition, unless permitted under Title 5 Code of Federal Regulations Part 2635, Subpart B.

c. I will instruct members of my parent or home organization not to divulge my participation in the evaluation and source selection process or my physical location while participating in the evaluation and source selection process to unauthorized persons.

d. I understand that all communications with offerors or their subcontractors concerning this acquisition must be made by/through the Contracting Officer, or the Contracting Officer's designee. I will divert all attempted communications by offerors' or subcontractors' representatives or any other unauthorized person to the Contracting Officer, and advise the Chairperson of the SSAC or the Chairperson of the SSEB and Legal Counsel.

e. I will not discuss evaluation or source selection matters, including proposal information, with any unauthorized individuals (including Government personnel), even after the announcement of the successful contractor, unless authorized by proper authority. All discussions of evaluation/source selection matters with other SSEB/SSAC members shall be conducted solely in those areas designated for deliberations.

11. I realize that my actions in connection with my participation in this evaluation and source selection are subject to intense scrutiny and I will conduct myself in a way that will not adversely affect the confidence of the public in the source selection process. I will avoid any action, whether or not prohibited, that could result in or create the appearance of my losing independence or impartiality. I will not use my public office for private gain, and I agree not to engage in any personal business or professional activity, or enter into any financial transaction, that involves or appears to involve, the direct or indirect use of "inside information" to further a private gain for myself or others.

12. I understand that my obligations under this certification are of a continuing nature, and if anything takes place which would cause a change to any statement or create a violation of any representation or rule of conduct herein, I will immediately bring such matter to the attention of the Chairperson of the SSAC or SSEB, or the Contracting Officer.

CERTIFICATION

13. I agree to the Terms of this Agreement and certify that I have read and understand the above Agreement. I further certify that the statements made herein are true and correct.

____
Signature

____
Name (Printed)

____
Organization

____
Date

APPENDIX C

INDIVIDUAL TECHNICAL EVALUATION FORM

<table>
<tr><td colspan="2">SUMMARY EVALUATION FORM
RFP No:
EVALUATOR'S NAME: OFFEROR:</td></tr>
<tr><td>RFP REFERENCES:
FACTOR:</td><td>PROPOSAL REFERENCES:
VOLUME/PARAGRAPH:</td></tr>
<tr><td colspan="2">Evaluation Rating: (Insert appropriate rating from applicable adjectival rating; e.g., Outstanding (O) Good (G) Acceptable (A) Marginal (M) Unacceptable (U))</td></tr>
<tr><td colspan="2">Evaluator's Rating: (e.g., G/M)</td></tr>
<tr><td colspan="2">STRENGTHS:
(Precede the strength with an (S) if it identifies a significant strength. Address any identified risks associated with the strength.)</td></tr>
<tr><td colspan="2">WEAKNESSES:
(Precede the weakness with an (S) if it identifies a significant weakness. Address the identified risks associated with the weakness.)</td></tr>
<tr><td colspan="2">DEFICIENCIES: (Address the identified risks associated with the deficiency.)</td></tr>
</table>

MINORITY OPINION REPORT

Evaluator Name:____

Offeror:____

FACTOR ____________________ **(To be attached to respective evaluation documentation)**	
FACTOR:	**PROPOSAL REFERENCES** **(Cite Volume/Section/Page(s)/Paragraph(s))**
Ratings (mark one): Outstanding (O) Good (G) Acceptable (A) Marginal (M) Unacceptable (U)	
Past Performance Ratings (mark one): Substantial Confidence Satisfactory Confidence Limited Confidence No Confidence Unknown Confidence (Neutral)	
Evaluation (circle one) Initial Interim Final Revised	
Date:	
I disagree with the Committee Consensus on the following issue for the reasons/rationale stated below:	
My recommended rating for the Factor in question is indicated above. My assessment produced the following findings and results:	

APPENDIX D

PERFORMANCE CONFIDENCE ASSESSMENT

INDIVIDUAL EVALUATOR REPORT
PERFORMANCE CONFIDENCE ASSESSMENT

PRIME CONTRACTOR

RFP No:

EVALUATOR'S NAME: **OFFEROR:**

Evaluation Rating: ***Substantial Confidence, Satisfactory Confidence, Neutral Confidence, Limited Confidence, No Confidence.***

Evaluator's Overall Confidence Rating:

Description	Completed within last 5 years (Y/N)	Similar Scope (Y/N)	Similar Size (Y/N)	Self-Performed Work (Y/N)	Project built on a military installation for USACE	Project includes deep foundations /drilled shaft piers	Project includes demolition work	Relevancy Rating

STRENGTHS:
(*Precede the strength with an (S) if it identifies a significant strength. Address any risks associated with the strength.)*

WEAKNESSES:
(*Precede the weakness with an (S) if it identifies a significant weakness. Address the risks associated with the weakness.)*

TELEPHONE INTERVIEW QUESTIONNAIRE FORM

Contractor:______________________ Contract No.:______________________________
Subcontract No. (if applicable):___________________________________
POC:__________________________ Title:__________________________

PART I. (To be completed by the Offeror)

A. CONTRACT IDENTIFICATION

Contractor/Company Name/Division:
Address:
Program Identification/Title:
Contract Number:
Contract Type:
Prime Contractor Name (if different from the contractor name cited above):
Contract Award Date:
Forecasted or Actual Contract Completion Date:
Nature of the Contractual Effort or Items Purchased:

B. IDENTIFICATION OF OFFEROR'S REPRESENTATIVE

Name:
Title:
Date:
Telephone Number:
Address:
E-mail Address:

PART II. EVALUATION
On a scale of 1-5 (with 5 being the most highly rated), please comment on the following.

A. Compliance of Products, Services, Documents, and Related Deliverables to Specification Requirements and Standards of Good Workmanship

Rating: ____________
Comments: ____

B. Effectiveness of Project Management (to include use and control of subcontractors).

Rating: ____________
Comments: ____

C. Timeliness of Performance

Rating: ____________
Comments: ____

D. Commitment to Customer Satisfaction and Business-like Concern for its Customers' Interest

Rating: ____________
Comments: ____

E. General Comments. Provide any other relevant performance information.

Comments: ____

APPENDIX E

GUIDELINES FOR WRITING EVALUATION NOTICES (ENs)

1. In the competitive procurement, your job is not to get the contractor to submit the best possible proposal. Your job is to evaluate the proposal submitted; obtain additional information (through the SSEB Chairperson/PCO) from the contractor when needed to perform this process; document your evaluation/assessment; and evaluate/assess the Final Revised Proposal (if any) including responses to the evaluation notices. The Government's obligation during discussions is to call the contractor's attention to all deficiencies, uncertainties, significant weaknesses and weaknesses at least once.

2. Always reference the RFP, proposal and prior related ENs when writing a notice. This provides an excellent method of tracking the finding.

3. Do not use adjectives (e.g. poor) which require some comparison to an unknown standard in order to establish the meaning of the objective, nor use the adjectival rating within the narrative (e.g. outstanding solution).

4. Remember to write the ENs so that, to the extent possible, contracting and legal can understand them as well as the offeror.

5. If the RFP establishes a firm requirement, the offeror's proposal must not deviate from the requirement. In writing a deficiency EN, reference the RFP requirement, offeror's proposal (if appropriate) and state that offeror has failed to comply with a firm requirement of the RFP.

6. If the RFP does not establish a firm requirement, you may not suggest or imply ways to improve the proposal. You should identify significant weaknesses and weaknesses and explain why the Government considers it to be a weakness. Include reference to the proposal and RFP (if appropriate).

7. If you believe the Government's requirement is in error or deficient, do not attempt to get the offerors to cure it in their proposal by an EN. Instead, bring it to the attention of the SSEB Chairperson or PCO immediately.

8. For ENs, do not discuss the impact to the Government should the proposal be accepted. Such qualifications are for internal Government evaluations. The purpose of the EN is to call the offeror's attention to the finding. The degree of impact should only be identified in your evaluation report.

9. Whenever possible write the EN in an unclassified form. Instead, include the classified information location in the RFP or proposal.

APPENDIX F

EVALUATION NOTICE (EN) FORM – DISCUSSION

Contracting Division DATE
Evaluation Notice

Offeror's Name
Address

Dear POC:

This letter is in response to your proposal submitted for RFP SOLICITATION NUMBER, PROJECT TITLE, PROJECT LOCATION.

The Source Selection Evaluation Board (SSEB) is opening discussions for this project.

The following weaknesses/deficiencies have been found in your submitted proposal:

Factor I- Past Performance

State RFP reference, Proposal Reference, Description of Issue

Factor II- Technical

State RFP reference, Proposal Reference, Description of Issue

Address the issues discussed above. The requested information (original and # of copies and electronic submission) must be received by this office no later than DATE and TIME. In accordance with RFP clause 52.215-1, any proposal revisions or modifications received after the above stated time is late, and will not be considered. Faxed and/or emailed responses will not be accepted. Revised proposal information shall be submitted to:

U.S. Army Corps of Engineers, Louisville District
ATTN: (Ranzel L. Merideth)
600 Dr. Martin Luther King, Jr. Place, Room 821
Louisville, KY 40202-2267

The revised proposal must include a completed Price Breakout Schedule. This information must be submitted in original only and placed in a separate envelope. Submit a revised Subcontracting Plan in accordance with the revised Price Breakout Schedule.

If you have any questions, please contact Ranzel Merideth at Ranzel.L.Merideth@usace.army.mil.

Sincerely,

Contracting Officer

APPENDIX G

MAJOR MILESTONE SCHEDULE

a.	Acquisition Planning Activities (e.g., Market Research, Review Previous Similar Acquisitions, Industry Days)	04/01/2025
b.	Appointment of Source Selection Authority (SSA)	04/01/2025
c.	Acquisition Plan Approved	3/11/2026
d.	Source Selection Plan Approved	3/13/2026
e.	Pre-Solicitation Peer Review	3/13/2026
f.	Issue RFP	3/13/2026
g.	Receive Proposals	4/30/2026
h.	SSEB Evaluation	5/1/2026
i.	SSA Brief	5/15/2026
j.	Source Selection Decision	5/17/2026
k.	Contract Award	5/25/2026
l.	Debriefing of Unsuccessful Offerors	5/25/2026

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08/18, CHG 2: 05/21

AMENDMENT 0001

-- End of Section Table of Contents --

SECTION 05 12 00

STRUCTURAL STEEL
08/18, CHG 2: 05/21
Amendment 0001

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 207	(2016; R 2017) Certification Standard for Steel Fabrication and Erection, and Manufacturing of Metal Components
AISC 303	(2016) Code of Standard Practice for Steel Buildings and Bridges
AISC 325	(2017) Steel Construction Manual
AISC 326	(2009) Detailing for Steel Construction
AISC 341	(2016) Seismic Provisions for Structural Steel Buildings
AISC 360	(2016) Specification for Structural Steel Buildings
AISC 420	(2010) Certification Standard for Shop Application of Complex Protective Coating Systems
AISC DESIGN GUIDE 10	(1997) Erection Bracing of Low-Rise Structural Steel Buildings

AMERICAN SOCIETY FOR NONDESTRUCTIVE TESTING (ASNT)

ANSI/ASNT CP-189	(2020) ASNT Standard for Qualification and Certification of Nondestructive Testing Personnel

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B46.1	(2020) Surface Texture, Surface Roughness, Waviness and Lay

AMERICAN WELDING SOCIETY (AWS)

AWS A2.4	(2012) Standard Symbols for Welding, Brazing and Nondestructive Examination
AWS D1.1/D1.1M	(2020; Errata 1 2021) Structural Welding

	Code - Steel
AWS D1.8/D1.8M	(2016) Structural Welding Code—Seismic Supplement
AWS QC1	(2016) Specification for AWS Certification of Welding Inspectors

ASTM INTERNATIONAL (ASTM)

ASTM A6/A6M	(2022) Standard Specification for General Requirements for Rolled Structural Steel Bars, Plates, Shapes, and Sheet Piling
ASTM A29/A29M	(2020) Standard Specification for General Requirements for Steel Bars, Carbon and Alloy, Hot-Wrought
ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
ASTM A53/A53M	(2022) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A123/A123M	(2017) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
ASTM A143/A143M	(2007; R 2020) Standard Practice for Safeguarding Against Embrittlement of Hot-Dip Galvanized Structural Steel Products and Procedure for Detecting Embrittlement
ASTM A307	(2021) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength
ASTM A500/A500M	(2021a) Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes
ASTM A563	(2021; E 2022a) Standard Specification for Carbon and Alloy Steel Nuts
ASTM A780/A780M	(2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings
ASTM A992/A992M	(2022) Standard Specification for Structural Steel Shapes
ASTM B695	(2021) Standard Specification for Coatings of Zinc Mechanically Deposited on Iron and Steel
ASTM C827/C827M	(2016) Standard Test Method for Change in

	Height at Early Ages of Cylindrical Specimens of Cementitious Mixtures
ASTM C1107/C1107M	(2020) Standard Specification for Packaged Dry, Hydraulic-Cement Grout (Nonshrink)
ASTM F436/F436M	(2019) Standard Specification for Hardened Steel Washers Inch and Metric Dimensions
ASTM F844	(2019) Standard Specification for Washers, Steel, Plain (Flat), Unhardened for General Use
ASTM F959/F959M	(2017a) Standard Specification for Compressible-Washer-Type Direct Tension Indicators for Use with Structural Fasteners, Inch and Metric Series
ASTM F1136/F1136M	(2011) Standard Specification for Zinc/Aluminum Corrosion Protective Coatings for Fasteners
ASTM F1554	(2020) Standard Specification for Anchor Bolts, Steel, 36, 55, and 105-ksi Yield Strength
ASTM F2329/F2329M	(2015) Standard Specification for Zinc Coating, Hot-Dip, Requirements for Application to Carbon and Alloy Steel Bolts, Screws, Washers, Nuts, and Special Threaded Fasteners
ASTM F2833	(2011; R 2017) Standard Specification for Corrosion Protective Fastener Coatings with Zinc Rich Base Coat and Aluminum Organic/Inorganic Type
ASTM F3125/F3125M	(2019) Standard Specification for High Strength Structural Bolts and Assemblies, Steel and Alloy Steel, Heat Treated, Inch Dimensions 120 ksi and 150 ksi Minimum Tensile Strength, and Metric Dimensions 830 MPa and 1040 MPa Minimum Tensile Strength

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC PA 1	(2016) Shop, Field, and Maintenance Coating of Metals
SSPC Paint 20	(2019) Zinc-Rich Primers (Type I, Inorganic, and Type II, Organic)
SSPC Paint 29	(2002; E 2004) Zinc Dust Sacrificial Primer, Performance-Based
SSPC SP 3	(2018) Power Tool Cleaning
SSPC SP 6/NACE No.3	(2007) Commercial Blast Cleaning

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-301-01	(2019, with Change 1, 2022) Structural Engineering

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR Part 1926, Subpart R	Steel Erection

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Erection and Erection Bracing Drawings; G, RO, AE

SD-02 Shop Drawings

Fabrication Drawings Including Details of Connections; G, AE

SD-03 Product Data

Shop Primer

Welding Electrodes and Rods

Direct Tension Indicator Washers

Non-Shrink Grout

Tension Control Bolts

Recycled Content for Structural Steel; S

Recycled Content for Structural Steel Tubing; S

Recycled Content for Steel Pipe; S

SD-05 Design Data

Design Calculations for Steel Connections; G, AE

Shoring and Temporary Bracing; G, RO

SD-06 Test Reports

Class B Coating

Bolts, Nuts, and Washers

Weld Inspection Reports

Direct Tension Indicator Washer Inspection Reports

Bolt Testing Reports

Embrittlement Test Reports

SD-07 Certificates

Steel

Bolts, Nuts, and Washers

Galvanizing

AISC Structural Steel Fabricator Quality Certification

AISC Structural Steel Erector Quality Certification

Welding Procedures and Qualifications

Welding Electrodes and Rods

Certified Welding Inspector

NDT Technician

Welding Procedure Specifications (WPS)

1.3 AISC QUALITY CERTIFICATION

Work must be fabricated by an AISC Certified Structural Steel Fabricator, in accordance with AISC 207, Category BU. Submit AISC Structural Steel Fabricator quality certification.

Work must be erected by an AISC Structural Steel Certified Erector, in accordance with AISC 207, Category CSE. Submit AISC Structural Steel erector quality certification.

1.4 SEISMIC PROVISIONS

Provide the structural steel system in accordance with AISC 341, Chapter J as amended by UFC 3-301-01.

1.5 QUALITY ASSURANCE

1.5.1 Preconstruction Submittals

1.5.1.1 Erection and Erection Bracing Drawings

Submit for record purposes. Indicate the sequence of erection, temporary shoring and bracing. The erection drawings must conform to AISC 303.

1.5.2 Fabrication Drawing Requirements

Submit fabrication drawings for approval prior to fabrication. Prepare in accordance with AISC 303, AISC 326 and AISC 325. Fabrication drawings must not be reproductions of contract drawings. Include complete information for the fabrication and erection of the structure's components, including the location, type, and size of bolts, welds, member sizes and lengths, connection details, blocks, copes, and cuts. Use AWS A2.4 standard welding symbols. Shoring and temporary bracing must be designed and sealed by a registered professional engineer and submitted for record purposes, with calculations, as part of the drawings. Clearly highlight any deviations from the details shown on the contract drawings highlighted on the fabrication drawings. Explain the reasons for any deviations from the contract drawings.

1.5.3 Delegated Connection Design

Design structural steel connection indicated in the contract documents per AISC 303, Option 3, using the connection loads indicated. Submit design calculations for steel connections signed and sealed by a registered professional engineer.

1.5.4 Certifications

1.5.4.1 Welding Procedures and Qualifications

Prior to welding, submit certification for each welder stating the type of welding and positions qualified for, the code and procedure qualified under, date qualified, and the firm and individual certifying the qualification tests. If the qualification date of the welder or welding operator is more than 6 months old, the welding operator's qualification certificate must be accompanied by a current certificate by the welder attesting to the fact that he has been engaged in welding since the date of certification, with no break in welding service greater than 6 months.

Conform to all requirements specified in AWS D1.1/D1.1M and AWS D1.8/D1.8M.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Provide the structural steel system, including finish, complete and ready for use. Provide structural steel systems including design, materials, installation, workmanship, fabrication, assembly, erection, inspection, quality control, and testing in accordance with AISC 303, AISC 360, AISC 341, and UFC 3-301-01 except as modified in this contract.

2.2 STEEL

2.2.1 Structural Steel

Wide flange and WT shapes, ASTM A992/A992M. Angles, Channels and Plates, ASTM A36/A36M. Submit data identifying percentage of recycled content for structural steel.

2.2.2 Structural Steel Tubing

ASTM A500/A500M, Grade C. Submit data identifying percentage of recycled content for structural steel tubing.

2.2.3 Steel Pipe

ASTM A53/A53M, Type E or S, Grade B, weight class STD (Standard) or as indicated. Submit data identifying percentage of recycled content for steel pipe.

2.3 BOLTS, NUTS, AND WASHERS

Submit the certified manufacturer's mill reports which clearly show the applicable ASTM mechanical and chemical requirements together with the actual test results for the supplied fasteners.

2.3.1 Common Grade Bolts

2.3.1.1 Bolts

ASTM A307, Grade A, plain finish hot dipped zinc coating or mechanically deposited zinc coating. The bolt heads and the nuts of the supplied fasteners must be marked with the manufacturer's identification mark, the strength grade and type specified by ASTM specifications.

2.3.1.2 Nuts

ASTM A563, Grade A, heavy hex style.

2.3.1.3 Self-Locking Nuts

Provide nuts with a locking pin set in the nut. The locking pin must slide along the bolt threads, and by reversing the direction of the locking pin, the nut can be removed without damaging the nut or bolt. Provide stainless steel locking pins.

2.3.1.4 Washers

ASTM F844.

2.3.2 High-Strength Bolts

High strength bolts and nuts must be shipped together in the same shipping container. Fasteners indicated to be galvanized shall be tested by the supplier to show that the galvanized nut with the supplied lubricant provided may be rotated from the snug tight condition well in excess of the rotation required for pretentioned installation without stripping. The supplier shall supply nuts that have been lubricated and tested with the supplied bolts.

2.3.2.1 Bolts

ASTM F3125/F3125M, Grade A325M A325 or A490M A490, Type 1 Heavy Hex Head Style, plain finish hot dipped zinc coating or mechanically deposited zinc coating.

2.3.2.2 Nuts

ASTM A563, Grade and Style as specified in the applicable ASTM bolt standard.

2.3.2.3 Direct Tension Indicator Washers

ASTM F959/F959M. Submit product data for direct tension indicator washers.

2.3.2.4 Washers

ASTM F436/F436M, plain carbon steel.

2.3.3 Tension Control Bolts

ASTM F3125/F3125M, Grade F1852 or matching the bolt, Type 1, twistoff style assemblies consisting of steel structural bolts with splined ends, heavy-hex carbon steel nuts, and hardened carbon steel washers. Assembly finish must be plain or mechanically deposited zinc coating. Submit product data for tension control bolts.

2.3.4 Foundation Anchorage

2.3.4.1 Anchor Rods

ASTM F1554 grade as indicated.

2.3.4.2 Anchor Nuts

ASTM A563, Grade A, hex style.

2.3.4.3 Anchor Washers

ASTM F844.

2.3.4.4 Anchor Plate Washers

ASTM A36/A36M.

2.4 STRUCTURAL STEEL ACCESSORIES

2.4.1 Welding Electrodes and Rods

AWS D1.1/D1.1M and AWS D1.8/D1.8M. Submit product data for welding electrodes and rods.

2.4.2 Non-Shrink Grout

ASTM C1107/C1107M, with no ASTM C827/C827M shrinkage. Grout must be nonmetallic. Submit product data for non-shrink grout.

2.4.3 Welded Shear Stud Connectors

ASTM A29/A29M, Grades 1010 through 1020. AWS D1.1/D1.1M, Table 7.1, Type B.

2.5 GALVANIZING

ASTM F2329/F2329M, ASTM F1136/F1136M, ASTM F2833 or ASTM B695 for threaded parts or ASTM A123/A123M for structural steel members, as applicable, unless specified otherwise galvanize after fabrication where practicable.

2.6 FABRICATION

Fabrication must be in accordance with the applicable provisions of AISC 325. Fabrication and assembly must be done in the shop to the greatest extent possible. Punch, subpunch and ream, or drill bolt and pin holes perpendicular to the surface of the member.

Compression joints depending on contact bearing must have a surface roughness not in excess of 500 micro inch as determined by ASME B46.1, and ends must be square within the tolerances for milled ends specified in

ASTM A6/A6M.

Shop splices of members between field splices will be permitted only where indicated on the Contract Drawings. Splices not indicated require the approval of the Contracting Officer.

2.6.1 Markings

Prior to erection, identify members by a painted erection mark. Connecting parts assembled in the shop for reaming holes in field connections must be match marked with scratch and notch marks. Do not locate erection markings on areas to be welded. Do not locate match markings in areas that will decrease member strength or cause stress concentrations. Affix embossed tags to hot-dipped galvanized members.

2.6.2 Shop Primer

SSPC Paint 20 or SSPC Paint 29, (zinc rich primer). Shop prime structural steel, except as modified herein, in accordance with SSPC PA 1. Do not prime steel surfaces embedded in concrete, galvanized surfaces, surfaces to receive sprayed-on fireproofing, surfaces to receive epoxy coatings, surfaces designed as part of a composite steel concrete section, or surfaces within 0.5 inch of the toe of the welds prior to welding (except surfaces on which metal decking and shear studs are to be welded). If flash rusting occurs, re-clean the surface prior to application of primer. Apply primer in accordance with endorsement "SPE-P1" of AISC 420 or approved equal NACE or SSPC certification to a minimum dry film thickness of 2.0 mil. Submit shop primer product data.

Prime slip critical surfaces with a Class B coating in accordance with AISC 325. Submit test report for Class B coating.

Prior to assembly, prime surfaces which will be concealed or inaccessible after assembly. Do not apply primer in foggy or rainy weather; when the ambient temperature is below 45 degrees F or over 95 degrees F; or when the primer may be exposed to temperatures below 40 degrees F within 48 hours after application, unless approved otherwise by the Contracting Officer. Repair damaged primed surfaces with an additional coat of primer.

2.6.2.1 Cleaning

SSPC SP 6/NACE No.3, except steel exposed in spaces above ceilings, attic spaces, furred spaces, and chases that will be hidden to view in finished construction may be cleaned to SSPC SP 3 when recommended by the shop primer manufacturer. Maintain steel surfaces free from rust, dirt, oil, grease, and other contaminants through final assembly.

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2.6.3 **Fireproofing and Epoxy Coated Surfaces**

DELETED

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2.6.4 Surface Finishes

ASME B46.1 maximum surface roughness of 125 for pin, pinholes, and sliding bearings, unless indicated otherwise.

2.7 DRAINAGE HOLES

Drill adequate drainage holes to eliminate water traps. Hole diameter must be 1/2 inch and location indicated on the detail drawings. Hole size and locations must not affect the structural integrity.

PART 3 EXECUTION

3.1 ERECTION

a. Erection of structural steel, except as indicated in item b. below, must be in accordance with the applicable provisions of AISC 325, AISC 303 and 29 CFR Part 1926, Subpart R.

b. For low-rise structural steel buildings (60 feet tall or less and a maximum of 2 stories), erect the structure in accordance with AISC DESIGN GUIDE 10.

After final positioning of steel members, provide full bearing under base plates and bearing plates using nonshrink grout. Place nonshrink grout in accordance with the manufacturer's instructions.

3.1.1 STORAGE

Store the material out of contact with the ground in such manner and location as to minimize deterioration.

3.2 CONNECTIONS

Except as modified in this section, design connections indicated in accordance with AISC 360. Build connections into existing work. Do not tighten anchor bolts set in concrete with impact torque wrenches. Holes must not be cut or enlarged by burning. Bolts, nuts, and washers must be clean of dirt and rust, and lubricated immediately prior to installation.

3.2.1 Common Grade Bolts

Tighten ASTM A307 bolts to a "snug tight" fit. "Snug tight" is the tightness that exists when plies in a joint are in firm contact. If firm contact of joint plies cannot be obtained with a few impacts of an impact wrench, or the full effort of a man using a spud wrench, contact the Contracting Officer for further instructions.

3.2.2 High-Strength Bolts

Provide direct tension indicator washers in all ASTM F3125/F3125M, Grade A325 and Grade A490 bolted connections. Bolts must be installed in connection holes and initially brought to a snug tight fit. After the initial tightening procedure, fully tension bolts, progressing from the most rigid part of a connection to the free edges.

Fastener components shall be protected from dirt and moisture in closed containers at the site of the installation. Fastener components that are not incorporated into the work shall be returned to protected storage at the end of the work shift.

3.2.2.1 Installation of Direct Tension Indicator Washers (DTIW)

Where possible, install the DTIW under the bolt head and tighten the nut.

If the DTIW is installed adjacent to the turned element, provide a flat washer between the DTIW and nut when the nut is turned for tightening, and between the DTIW and bolt head when the bolt head is turned for tightening. In addition to the LIW, provide flat washers under both the bolt head and nut when ASTM F3125/F3125M, Grade A490 bolts are used.

3.2.3 Tension Control Bolts

Bolts must be installed in connection holes and initially brought to a snug tight fit. After the initial tightening procedure, fully tension bolts, progressing from the most rigid part of a connection to the free edges.

3.3 GAS CUTTING

Use of gas-cutting torch in the field for correcting fabrication errors is not permitted on any major member in the structural framing. Use of a gas cutting torch will be permitted on minor members not under stress only after approval has been obtained from the Contracting Officer.

3.4 WELDING

Welding must be in accordance with AWS D1.1/D1.1M and AWS D1.8/D1.8M. Provide AWS D1.1/D1.1M qualified welders, welding operators, and tackers.

Develop and submit the Welding Procedure Specifications (WPS) for all welding, including welding done using prequalified procedures. Submit for approval all WPS, whether prequalified or qualified by testing.

3.4.1 Removal of Temporary Welds, Run-Off Plates, and Backing Strips

Remove only from finished areas.

3.5 SHOP PRIMER REPAIR

Repair shop primer in accordance with the paint manufacturer's recommendation for surfaces damaged by handling, transporting, cutting, welding, or bolting.

3.5.1 Field Priming

Field prime steel exposed to the weather, or located in building areas without HVAC for control of relative humidity. After erection, the field bolt heads and nuts, field welds, and any abrasions in the shop coat must be cleaned and primed with paint of the same quality as that used for the shop coat.

3.6 GALVANIZING REPAIR

Repair damage to galvanized coatings using ASTM A780/A780M zinc rich paint for galvanizing damaged by handling, transporting, cutting, welding, or bolting. Do not heat surfaces to which repair paint has been applied.

3.7 FIELD QUALITY CONTROL

Perform field tests, and provide labor, equipment, and incidentals required for testing, except that electric power for field tests will be furnished as set forth in Division 1. Notify the Contracting Officer in writing of defective welds, bolts, nuts, and washers within 7 working days of the date of the inspection.

3.7.1 Welds

3.7.1.1 Visual Inspection

AWS D1.1/D1.1M. Furnish the services of AWS-certified welding inspectors for fabrication and erection inspection and testing and verification inspections. A Certified Welding Inspector must perform visual inspection on 100 percent of all welds. Document this inspection in the Visual Weld Inspection Log. Submit certificates indicating that certified welding inspectors meet the requirements of AWS QC1.

Inspect proper preparation, size, gaging location, and acceptability of all welds; identification marking; operation and current characteristics of welding sets in use.

3.7.1.2 Nondestructive Testing

Nondestructive testing must be in accordance with AWS D1.1/D1.1M and AWS D1.8/D1.8M. Ultrasonic testing must be performed in accordance with Table 6.2 or 6.3 of AWS D1.1/D1.1M. Test locations must be selected by the Contracting Officer. All personnel performing NDT must be certified in accordance with ANSI/ASNT CP-189 in the method of testing being performed. Submit certificates showing compliance with ANSI/ASNT CP-189 for all NDT technicians. If more than 20 percent of welds made by a welder contain defects identified by testing, then all groove welds made by that welder must be tested by ultrasonic testing, and all fillet welds made by that welder must be inspected by magnetic particle testing (MT) or dye penetrant testing (PT) as approved by the Contracting Officer. When groove welds made by an individual welder are required to be tested, magnetic particle or dye penetrant testing may be used only in areas inaccessible to ultrasonic testing. Retest all repaired areas. Submit weld inspection reports.

Testing frequency: Provide the following types and number of tests:

Test Type	Number of Tests
Ultrasonic	50 percent of CJP Welds
Magnetic Particle	50 percent of PJP and Fillet Welds
Dye Penetrant	50 percent of PJP and Fillet Welds

3.7.2 Direct Tension Indicator Washers

3.7.2.1 Direct Tension Indicator Washer Compression

Test direct tension indicator washers in place to verify that they have been compressed sufficiently to provide the 0.015 inch gap, as required by ASTM F959/F959M. Submit direct tension indicator washer inspection reports.

3.7.2.2 Direct Tension Indicator Gaps

In addition to the above testing, an independent testing agency as approved by the Contracting Officer, must test in place the direct tension indicator gaps on 20 percent of the installed direct tension indicator washers to verify that the ASTM F959/F959M direct tension indicator gaps have been achieved. If more than 10 percent of the direct tension indicators tested have not been compressed sufficiently to provide the average gaps required by ASTM F959/F959M, test all in place direct tension indicator washers to verify that the ASTM F959/F959M direct tension indicator gaps have been achieved. Test locations must be selected by the Contracting Officer.

3.7.3 High-Strength Bolts

3.7.3.1 Testing Bolt, Nut, and Washer Assemblies

Test a minimum of 3 bolt, nut, and washer assemblies from each mill certificate batch in a tension measuring device at the job site prior to the beginning of bolting start-up. Demonstrate that the bolts and nuts, when used together, can develop tension not less than the provisions specified in AISC 360, depending on bolt size and grade. The bolt tension must be developed by tightening the nut. A representative of the manufacturer or supplier must be present to ensure that the fasteners are properly used, and to demonstrate that the fastener assemblies supplied satisfy the specified requirements. Submit bolt testing reports.

3.7.3.2 Inspection

Inspection procedures must be in accordance with AISC 360. Confirm and report to the Contracting Officer that the materials meet the project specification and that they are properly stored. Confirm that the faying surfaces have been properly prepared before the connections are assembled. Observe the specified job site testing and calibration, and confirm that the procedure to be used provides the required tension. Monitor the work to ensure the testing procedures are routinely followed on joints that are specified to be fully tensioned.

Inspect calibration of torque wrenches for high-strength bolts.

3.7.3.3 Testing

The Government has the option to perform nondestructive tests on 5 percent of the installed bolts to verify compliance with pre-load bolt tension requirements. Provide the required access for the Government to perform the tests. The nondestructive testing will be done in-place using an ultrasonic measuring device or any other device capable of determining in-place pre-load bolt tension. The test locations must be selected by the Contracting Officer. If more than 10 percent of the bolts tested contain defects identified by testing, then all bolts used from the batch from which the tested bolts were taken, must be tested at the Contractor's expense. Retest new bolts after installation at the Contractor's expense.

3.7.4 Testing for Embrittlement

ASTM A143/A143M for steel products hot-dip galvanized after fabrication. Submit embrittlement test reports.

3.7.5 Inspection and Testing of Steel Stud Welding

Perform verification inspection and testing of steel stud welding conforming to the requirements of AWS D1.1/D1.1M, Stud Welding Clause.

The Contracting Officer will serve as the verification inspector. Bend test studs that do not show a full 360 degree weld flash or have been repaired by welding as required by AWS D1.1/D1.1M, Stud Welding Clause. Studs that crack under testing in the weld, base metal or shank will be rejected and replaced by the Contractor at no additional cost.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 05 - METALS

SECTION 05 50 13

MISCELLANEOUS METAL FABRICATIONS

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-- End of Section Table of Contents --

SECTION 05 50 13

MISCELLANEOUS METAL FABRICATIONS
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PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ALUMINUM ASSOCIATION (AA)

AA DAF45	(2003; Reaffirmed 2009) Designation System for Aluminum Finishes

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 318	(2014; Errata 1-2 2014; Errata 3-5 2015; Errata 6 2016; Errata 7-9 2017) Building Code Requirements for Structural Concrete (ACI 318-14) and Commentary (ACI 318R-14)

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 303	(2016) Code of Standard Practice for Steel Buildings and Bridges

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B18.2.1	(2012; Errata 2013) Square and Hex Bolts and Screws (Inch Series)
ASME B18.2.2	(2022) Nuts for General Applications: Machine Screw Nuts, and Hex, Square, Hex Flange, and Coupling Nuts (Inch Series)
ASME B18.6.2	(2020) Square Head Set Screws and Slotted Headless Set Screws (Inch Series)
ASME B18.6.3	(2013; R 2017) Machine Screws, Tapping Screws, and Machine Drive Screws (Inch Series)
ASME B18.21.1	(2009; R 2016) Washers: Helical Spring-Lock, Tooth Lock, and Plain Washers (Inch Series)
ASME B18.21.2M	(1999; R 2014) Lock Washers (Metric Series)
ASME B18.22M	(1981; R 2017) Metric Plain Washers

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP A10.3	(2020) Safety Requirements for Powder-Actuated Fastening Systems American National Standard for Construction and Demolition Operations

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M	(2020; Errata 1 2021) Structural Welding Code - Steel

ASTM INTERNATIONAL (ASTM)

ASTM A29/A29M	(2020) Standard Specification for General Requirements for Steel Bars, Carbon and Alloy, Hot-Wrought
ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
ASTM A47/A47M	(1999; R 2022; E 2022) Standard Specification for Ferritic Malleable Iron Castings
ASTM A53/A53M	(2022) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A108	(2013) Standard Specification for Steel Bar, Carbon and Alloy, Cold-Finished
ASTM A123/A123M	(2017) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
ASTM A153/A153M	(2016a) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware
ASTM A307	(2021) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength
ASTM A500/A500M	(2021a) Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes
ASTM A653/A653M	(2022) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM A780/A780M	(2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings
ASTM A786/A786M	(2015; R 2021) Standard Specification for

	Hot-Rolled Carbon, Low-Alloy, High-Strength Low-Alloy, and Alloy Steel Floor Plates
ASTM A924/A924M	(2022a) Standard Specification for General Requirements for Steel Sheet, Metallic-Coated by the Hot-Dip Process
ASTM B26/B26M	(2018; E 2018) Standard Specification for Aluminum-Alloy Sand Castings
ASTM B108/B108M	(2019) Standard Specification for Aluminum-Alloy Permanent Mold Castings
ASTM B209	(2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate
ASTM B209M	(2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate (Metric)
ASTM B221	(2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes
ASTM B221M	(2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes (Metric)
ASTM C1513	(2018) Standard Specification for Steel Tapping Screws for Cold-Formed Steel Framing Connections
ASTM D1187/D1187M	(1997; E 2011; R 2011) Asphalt-Base Emulsions for Use as Protective Coatings for Metal
ASTM F1554	(2020) Standard Specification for Anchor Bolts, Steel, 36, 55, and 105-ksi Yield Strength

MASTER PAINTERS INSTITUTE (MPI)

MPI 79	(2016) Primer, Alkyd, Anti-Corrosive for Metal

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

NAAMM MBG 531	(2017) Metal Bar Grating Manual

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC SP 3	(2018) Power Tool Cleaning
SSPC SP 6/NACE No.3	(2007) Commercial Blast Cleaning

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2014) Safety -- Safety and Health Requirements Manual

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Expansion Joint Covers, Installation Drawings; G

Bollards/Pipe Guards; G

Roof Hatches, Installation Drawings; G

Bicycle Racks; G, AE

Mail Kiosk Canopy; G, AE

Bicycle Rack Canopy; G, AE

Building Entrance Canopy; G, AE

SD-03 Product Data

Expansion Joint Covers; G

Structural Steel Door Frames; G

Roof Hatches; G

Recycled Content; S

SD-04 Samples

Expansion Joint Covers

SD-05 Design Data

Building Entrance Canopy Calculations; G, AE

Bicycle Racks; G, AE

Mail Kiosk Canopy; G, AE

SD-07 Certificates

Certificates of Compliance; G

1.3 QUALIFICATION OF WELDERS

Qualify welders in accordance with AWS D1.1/D1.1M. Use procedures, materials, and equipment of the type required for the work.

1.4 DELIVERY, STORAGE, AND PROTECTION

Protect from corrosion, deformation, and other types of damage. Store items in an enclosed area free from contact with soil and weather. Remove and replace damaged items with new items.

1.5 MISCELLANEOUS REQUIREMENTS

1.5.1 Fabrication Drawings

Submit fabrication drawings showing layout(s), connections to structural system, and anchoring details as specified in AISC 303.

1.5.2 Installation Drawings

Submit templates, erection, and installation drawings indicating thickness, type, grade, class of metal, and dimensions. Show construction details, reinforcement, anchorage, and installation in relation to the building construction.

PART 2 PRODUCTS

2.1 RECYCLED CONTENT

Provide products with recycled content. Provide certificates of compliance for recycled content.

2.2 MATERIALS

Provide exposed fastenings of compatible materials (avoid contact of dissimilar metals). Coordinate color and finish with the material to which fastenings are applied.

2.2.1 Structural Carbon Steel

Provide in accordance with ASTM A36/A36M.

2.2.2 Structural Tubing

Provide in accordance with ASTM A500/A500M.

2.2.3 Steel Pipe

Provide in accordance with ASTM A53/A53M, Type E or S, Grade B.

2.2.4 Fittings for Steel Pipe

Provide standard malleable iron fittings in accordance with ASTM A47/A47M.

2.2.5 Floor Plates, Patterned

Provide floor plate in accordance with ASTM A786/A786M. Provide steel plate not less than 14 gage.

2.2.6 Anchor Bolts

Provide in accordance with ASTM F1554. Where exposed, provide anchor bolts of the same material, color, and finish as the metal to which they are applied.

2.2.6.1 Expansion Anchors and Adhesive Anchors

Provide expansion anchors and adhesive anchors as indicated.

a. Provide minimum ultimate pullout value indicated. Calculate pullout capacity according to ACI 318.

b. Provide minimum ultimate shear value as indicated. Calculate shear capacity according to ACI 318.

2.2.6.2 Lag Screws and Bolts

Provide in accordance with ASME B18.2.1, type and grade best suited for the purpose.

2.2.6.3 Toggle Bolts

Provide in accordance with ASME B18.2.1.

2.2.6.4 Bolts, Nuts, Studs and Rivets

Provide in accordance with ASME B18.2.2 or ASTM A307.

2.2.6.5 Powder Actuated Fasteners

Follow safety provisions in accordance with ASSP A10.3.

2.2.6.6 Screws

Provide in accordance with ASME B18.2.1, ASME B18.6.2, ASME B18.6.3 and ASTM C1513.

2.2.6.7 Washers

Provide plain washers in accordance with ASME B18.22M, ASME B18.21.1. Provide beveled washers for American Standard beams and channels, square or rectangular, tapered in thickness, and smooth. Provide lock washers in accordance with ASME B18.21.2M, ASME B18.21.1.

2.2.6.8 Welded Headed Shear Studs

Provide in accordance with ASTM A108 or ASTM A29/A29M-12.

2.2.7 Aluminum Alloy Products

Provide in accordance with ASTM B209M, ASTM B209 for sheet plate, ASTM B221M, ASTM B221M, ASTM B221 for extrusions and ASTM B26/B26M or ASTM B108/B108M for castings. Provide aluminum extrusions at least 1/8 inch thick and aluminum plate or sheet at least 0.050 inch thick.

2.3 FABRICATION FINISHES

2.3.1 Galvanizing

Hot-dip galvanize items specified to be zinc-coated, after fabrication where practicable. Provide galvanizing in accordance with ASTM A123/A123M, ASTM A153/A153M, ASTM A653/A653M or ASTM A924/A924M, Z275 G90.

2.3.2 Galvanize

Anchor bolts, grating fasteners, washers, and parts or devices necessary for proper installation, unless indicated otherwise.

2.3.3 Repair of Zinc-Coated Surfaces

Repair damaged surfaces with galvanizing repair method and paint in accordance with ASTM A780/A780M or by application of stick or thick paste material specifically designed for repair of galvanizing, as approved by Contracting Officer. Clean areas to be repaired and remove slag from welds. Heat, with a torch, surfaces to which stick or paste material will be applied. Heat to a temperature sufficient to melt the metals in the stick or paste. Spread molten material uniformly over surfaces to be coated and wipe off excess material.

2.3.4 Shop Cleaning and Painting

2.3.4.1 Surface Preparation

Blast clean surfaces in accordance with SSPC SP 6/NACE No.3. Surfaces that will be exposed in spaces above ceiling or in attic spaces, crawl spaces, furred spaces, and chases may be cleaned in accordance with SSPC SP 3 in lieu of being blast cleaned. Wash cleaned surfaces which become contaminated with rust, dirt, oil, grease, or other contaminants with solvents until thoroughly clean. Steel to be embedded in concrete must be free of dirt and grease prior to embed. Do not paint or galvanize bearing surfaces, including contact surfaces within slip critical joints. Shop coat these surfaces with rust prevention.

2.3.4.2 Pretreatment, Priming and Painting

Apply pre-treatment, primer, and paint in accordance with manufacturer's printed instructions. On surfaces concealed in the finished construction or not accessible for finish painting, apply an additional prime coat to a minimum dry film thickness of 1.0 mil. Tint additional prime coat with a small amount of tinting pigment.

2.3.5 Nonferrous Metal Surfaces

Protect by plating, anodic, or organic coatings.

2.3.6 Aluminum Surfaces

2.3.6.1 Surface Condition

Before finishes are applied, remove roll marks, scratches, rolled-in scratches, kinks, stains, pits, orange peel, die marks, structural streaks, and other defects which will affect uniform appearance of finished surfaces.

2.3.6.2 Aluminum Finishes

Unexposed sheet, plate and extrusions may have mill finish as fabricated. Sandblast castings' finish, medium, AA DAF45. Unless otherwise specified, provide all other aluminum items with anodized finish. Provide a coating thickness not less than that specified for protective and decorative type finishes for items used in interior locations or architectural Class I type finish for items used in exterior locations. Provide in accordance with AA DAF45. Provide a polished satin finish on items to be anodized.

2.4 EXPANSION JOINT COVERS

Provide expansion joint covers constructed of extruded aluminum with anodized satin aluminum finish for walls and ceilings and standard mill finish for floor covers and exterior covers. Furnish plates, backup angles, expansion filler strips and anchors as indicated.

2.5 FLOOR GRATINGS AND ROOF WALKWAYS

Design steel grating in accordance with NAAMM MBG 531 for bar type gratings, or in accordance with manufacturer's charts for plank grating. Galvanize steel floor gratings.

a. Design floor gratings to support a stress live load of 40 pounds per square foot for the spans indicated, with maximum deflection of L/240.

b. In accordance with NAAMM MBG 531, band edges of grating with bars of the same size as the bearing bars. Weld banding in accordance with the manufacturer's standard for trim unless otherwise indicated. Design tops of bearing bars, cross or intermediate bars to be in the same plane and to match grating finish.

d. Provide slip resistant surface finishes.

2.6 BOLLARDS/PIPE GUARDS

Provide 8 inch galvanized weight steel pipe in accordance with ASTM A53/A53M. Anchor posts as indicated and fill solidly with concrete with minimum compressive strength of 2500 psi.

2.7 DOWNSPOUT TERMINATIONS

Provide 4x3 inch, 5x4 inch or 4 inch diameter galvanized cast iron downspout boot with cleanout access and manufacturer's standard cast iron strap.

2.8 MISCELLANEOUS PLATES AND SHAPES

Provide items that do not form a part of the structural steel framework, such as lintels, sill angles, support framing for ceiling-mounted toilet partitions, miscellaneous mountings and frames. Provide lintels fabricated from structural steel shapes over openings in masonry walls and partitions as indicated and as required to support wall loads over openings. Provide with connections and fasteners. Construct to have at least 8 in bearing on masonry at each end.

Provide angles and plates in accordance with ASTM A36/A36M, for embedment as indicated. Galvanize embedded items exposed to the elements in accordance with ASTM A123/A123M.

Amendment 0001

2.9 **SAFETY CHAINS**

DELETED

Amendment 0001

2.10 ROOF HATCHES (SCUTTLES)

Provide zinc-coated steel sheets not less than 14 gauge with 3 inch beaded flange, welded and ground at corners. Provide a minimum clear opening of 30 by 36 inches. Insulate cover and curb with one inch thick rigid fiberboard insulation, covered and protected by zinc-coated steel liner of not less than 26 gage. Provide with 12 inches high curb, formed with 3 inch mounting flanges with holes for securing to the roof deck.

Amendment 0001*

2.11 **GUY CABLES**

DELETED

Amendment 0001

2.12 WINDOW SUB-SILL

Provide window sub-sill of extruded aluminum alloy, standard mill finish, of size(s) and design(s) indicated. Provide a minimum of two anchors per window section for securing to mortar joints of masonry sill course. Provide sills with protective coating for shipment, of two coats of a clear, colorless, methacrylate lacquer applied to all surfaces of the sills.

2.13 BICYCLE RACKS

The bicycle rack shall be 2 3/8", 9-gauge, schedule 40 steel pipe with thermoplastic powder coating. Rack loading and mounting shall be as indicated on the civil drawing details. Refer to civil drawings for pavement details.

2.14 MAIL KIOSK CANOPY

Provide (1) canopy over mail kiosks approximately 35'-4" wide x 10'-4" deep. Vertical head clearance must be 7-0" minimum. The mail kiosk canopy shall be constructed of a hot dip galvanized frame with galvanized or PVDF (polyvinylidene flouride) finish on frame. See architectural drawings for more information and a rendering. The structural frame shall include post connections to the foundation system. The structural frame and foundation shall be stamped by a licensed PE. Contractor shall coordinate canopy installation with concrete pad construction details and specifications. The post-supported canopy shall have a sloped roof at pitches indicated on the architectural drawings. The canopy shall have drainage components including roof drain, downspout and downspout boot as indicated on the drawings. Colors and finishes of canopy drainage components are indicated on the architectural drawings. The canopy roof shall meet a minimum of 41 lbs per sq ft snow load and canopy C&C wind loads on S-004.

Amendment 0001*

2.15 **BIKE STAND CANOPY**

Provide (1) canopy over bike stand approximately 23'-4" wide x 10'-4" deep. Vertical head clearance must be 7-0" minimum. The mail kiosk canopy shall be constructed of a hot dip galvanized frame with galvanized or PVDF (polyvinylidene flouride) finish on frame. See architectural drawings for more information and a rendering. The structural frame shall include post connections to the foundation system. The structural frame and foundation shall be stamped by a licensed PE. Contractor shall coordinate canopy installation with concrete pad construction details and specifications. The post-supported canopy shall have a sloped roof at pitches indicated on the architectural drawings. The canopy shall have drainage components including roof drain, downspout and downspout boot as indicated on the drawings. Colors and finishes of canopy drainage components are indicated on the architectural drawings. The canopy roof shall meet a minimum of 41 lbs per sq ft snow load and canopy C&C wind loads on S-004.

2.16 **MAIL CLUSTER BOXES**

Provide and install (10) USPS-STD-4C Type III, Front Loader 16-Compartment Cluster Box Units (CBU), constructed of welded aluminum in accordance with ASTM B209 and ASTM B221, with stainless teel components complying with ASTM A666, and including integral parcel lockers and one outgoing mail slot. Provide manufacturer’s standard locking hardware, stainless steel hinges, and powder-coated finish in standard “Sandstone” color. Units shall comply with USPS-B-1118 requirements and be suitable for exterior installation. Mount units on contractor designed and constructed masonry base in lieu of manufacturer pedestals, securely anchored, plumb, and level, in accordance with manufacturer’s written installation instructions. Locate units on the masonry constructed base as indicated on drawings.

2.17 **MECHANICAL YARD FENCING AND GATE**

Provide mechanical yard fence posts, panels, and gates as illustrated on drawings. Provide powder coated posts and panels of steel construction with posts of 12 gauage steel, 2.5" square (min), 2" x 6" mesh using no less than 6 gauge wire. Set posts in concrete to depth and diameter recommended by manufacturer in order to suppor height of fencing panels shown on drawings and in compliance with local ground frost depths. Provide matching powder coated surfaces on all fastening hardware, hinges, drop bolts (cane bolts), locks and hasps or galvanized finishes where necessary. Apply matching primer and finish coat upon final completion of construction of fence per manufacturers recommendation. Provide matching steel caps on all posts and 1/4" drain holes in posts at 1/2" above finished concretes slab only if posts were uncovered from weather during construction. Provide manufacturers recommended primer and paint on exposed metals resulting from drilling or scratches to pre-coated surfaces occuring during construction. Provide (2) top rail guiding track with roller assemblies per manufacturers specifications compatible with fencing system at the locations indicated on drawings. Provide galvanized wedge type expansion anchors no less than 1/2" dia x 4" long where necessary or per manufacturers recommendation.

Amendment 0001

2.18 BICYCLE RACK CANOPY AND FOUNDATION

Provide (1) canopy over bicycle rack approximately 23'-4" wide x 10'-4" deep. Vertical head clearance must be 7-0" minimum. The bicycle canopy shall be constructed of a hot dip galvanized frame with galvanized or PVDF (polyvinylidene flouride) finish on frame. See architectural drawings for more information and a rendering. The structural frame shall include post connections to the foundation system. The structural frame and foundation shall be stamped by a licensed PE. Contractor shall coordinate canopy installation with concrete pad construction details and specifications. The post-supported canopy shall have a sloped roof at pitches indicated on the architectural drawings. The canopy shall have drainage components including roof drain, downspout and downspout boot as indicated on the drawings. Colors and finishes of canopy drainage components are indicated on the architectural drawings. The canopy roof shall meet a minimum of 41 lbs per sq ft snow load and canopy C&C wind loads on S-004. Bike rack listed in subpart 2.13 above can be utilized or the unit purchased as an integrated canopy and rack unit together.

2.19 BUILDING ENTRANCE CANOPY AND FOUNDATION

Provide a canopy at the north and south entrance of the barracks. See Architectural drawings for canopy geometry,clear heights, renderings, and additional information. The canopy shall be constructed of a hot dip galvanized frame with galvanized. The structural frame shall include post connections to the foundation system. The structural frame and foundation shall be stamped by a licensed PE. Contractor shall coordinate canopy installation with concrete pad construction details and specifications. The post-supported canopy shall have a sloped roof at pitches indicated on the architectural drawings. The canopy shall have drainage components including roof drain, downspout and downspout boot as indicated on the Civil and Archtectural drawings. Colors and finishes of canopy drainage components are indicated on the architectural drawings. The canopy roof shall meet a minimum of 41 lbs per sq ft snow load

PART 3 EXECUTION

3.1 GENERAL INSTALLATION REQUIREMENTS

Install items at locations indicated in accordance with manufacturer's instructions. Verify all field dimensions prior to fabrication. Include materials and parts necessary to complete each assembly, whether indicated or not. Miss-alignment and miss-sizing of holes for fasteners is cause for rejection. Conceal fastenings where practicable. Joints exposed to weather must be watertight.

3.2 WORKMANSHIP

Provide miscellaneous metalwork that is true and accurate in shape, size, and profile. Make angles and lines continuous and straight. Make curves consistent, smooth and unfaceted. Provide continuous welding along the entire area of contact except where tack welding is permitted. Do not tack weld exposed connections. Unless otherwise indicated and approved, provide a smooth finish on exposed surfaces. Provide countersuck rivets where exposed. Provide coped and mitered corner joints aligned flush and without gaps.

3.3 ANCHORAGE, FASTENINGS, AND CONNECTIONS

Provide anchorage as necessary, whether indicated or not, for fastening miscellaneous metal items securely in place. Include slotted inserts, expansion shields, powder-driven fasteners, toggle bolts (when approved for concrete), through bolts for masonry, headed shear studs, machine and carriage bolts for steel, through bolts, lag bolts, and screws for wood. Do not use wood plugs. Provide non-ferrous attachments for non-ferrous metal. Provide exposed fastenings of compatible materials (avoid contact of dissimilar metals), that generally match in color and finish the surfaces to which they are applied. Conceal fastenings where practicable. Provide all fasteners flush with the surfaces they fasten, unless indicated otherwise.

3.4 BUILT-IN WORK

Where necessary and not otherwise indicated, form built-in metal work for anchorage with concrete or masonry. Provide built-in metal work in ample time for securing in place as the work progresses.

3.5 WELDING

Perform welding, welding inspection, and corrective welding in accordance with AWS D1.1/D1.1M. Use continuous welds on all exposed connections. Grind visible welds smooth in the finished installation. Provide welded headed shear studs in accordance with AWS D1.1/D1.1M, Clause 7, except as otherwise specified. Provide in accordance with the safety requirements of EM 385-1-1.

3.6 DISSIMILAR METALS

Where dissimilar metals are in contact, protect surfaces with a coating in accordance with MPI 79 to prevent galvanic or corrosive action. Where aluminum is in contact with concrete, plaster, mortar, masonry, wood, or absorptive materials subject to wetting, protect in accordance with ASTM D1187/D1187M, asphalt-base emulsion. Clean surfaces with metal shavings from installation at the end of each work day.

3.7 PREPARATION

3.7.1 Material Coatings and Surfaces

Remove rust preventive coating just prior to field erection, using a remover approved by the metal manufacturer. Surfaces, when assembled, must be free of rust, grease, dirt and other foreign matter.

3.7.2 Environmental Conditions

Do not clean or paint surfaces when damp or exposed to foggy or rainy weather, when metallic surface temperature is less than minus 5 degrees F above the dew point of the surrounding air, or when surface temperature is below 45 degrees F or over 95 degrees F, unless approved by the Contracting Officer. Metal surfaces to be painted must be dry for a minimum of 48 hours prior to the application of primer or paint.

3.8 EXPANSION JOINT COVERS

Provide in accordance with manufacturer's written instructions and with seismic requirements indicated. Verify installation allows specified movement prior to completion of work

3.9 ROOF HATCH (SCUTTLES)

Construction and accessories as follows:

a. Provide insulated cover and curb with mounting flanges for securing to roof deck. Provide curbs with integral metal cap flashing of the same gage and metal as the curb, fully welded and ground at corners for weather tightness.

b. Provide hatches completely assembled, with pintle hinges, compression spring operators enclosed in telescopic tubes, positive snap latches with turn handles on inside and outside, and neoprene draft seals. Provide fasteners for padlocking from the inside. Provide covers with automatic hold-open arms complete with grip handle to permit one hand release. Cover action must be smooth through its entire range of motion with an operating pressure of approximately 30 pounds.

3.10 INSTALLATION OF BOLLARDS/PIPE GUARDS

Set bollards/pipe guards vertically in concrete piers. Fill hollow cores with concrete having a compressive strength of 3000 psi.

3.11 INSTALLATION OF DOWNSPOUT TERMINATIONS

Secure downspouts terminations to downspouts and substrate per manufacturer's instructions.

3.12 MOUNTING OF SAFETY CHAINS

Provide safety chains where indicated. Mount the top chain 3 feet 6 inches ground and mount the lower chain 2 feet above the ground.

3.13 INSTALLATION OF WHEEL GUARDS

Fill wheel guards with concrete and anchor to slab in accordance with manufacturer's recommendations.

3.14 BAR-GRILLE WINDOW GUARDS

Securely anchor bar-grille window guards to masonry with 1/2 inch diameter prison-type screws or bolts and expansion shields, or other type of fastenings if the ends of such fastenings are welded to the adjoining metal grilles or otherwise made tamperproof in manner as approved by the Contracting Officer. Spanner-head screws or bolts are not considered prison-type fasteners.

3.15 INSTALLATION MISCELLANEOUS PLATES AND SHAPES

Provide lintels fabricated from structural steel shapes over openings in masonry walls and partitions as indicated and as required to support wall loads over openings. Provide with connections and fasteners or welds. Construct to have at least 8 inches bearing on masonry at each end.

-- End of Section --

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DIVISION 05 - METALS

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METAL LADDERS

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SECTION 05 51 33

METAL LADDERS
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PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ALUMINUM ASSOCIATION (AA)

AA DAF45	(2003; Reaffirmed 2009) Designation System for Aluminum Finishes

AMERICAN LADDER INSTITUTE (ALI)

ALI A14.3	(2008; R 2018) Ladders - Fixed - Safety Requirements

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP Z359.16	(2016) Safety Requirements for Climbing Ladder Fall Arrest Systems

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M	(2020; Errata 1 2021) Structural Welding Code - Steel

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
ASTM A47/A47M	(1999; R 2022; E 2022) Standard Specification for Ferritic Malleable Iron Castings
ASTM A53/A53M	(2022) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A123/A123M	(2017) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
ASTM A153/A153M	(2016a) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware
ASTM A500/A500M	(2021a) Standard Specification for Cold-Formed Welded and Seamless Carbon

	Steel Structural Tubing in Rounds and Shapes
ASTM A653/A653M	(2022) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM A780/A780M	(2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings
ASTM A924/A924M	(2022a) Standard Specification for General Requirements for Steel Sheet, Metallic-Coated by the Hot-Dip Process
ASTM B26/B26M	(2018; E 2018) Standard Specification for Aluminum-Alloy Sand Castings
ASTM B108/B108M	(2019) Standard Specification for Aluminum-Alloy Permanent Mold Castings
ASTM B209	(2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate
ASTM B221	(2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes
ASTM D1187/D1187M	(1997; E 2011; R 2011) Asphalt-Base Emulsions for Use as Protective Coatings for Metal

MASTER PAINTERS INSTITUTE (MPI)

MPI 79	(2016) Primer, Alkyd, Anti-Corrosive for Metal

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC SP 3	(2018) Power Tool Cleaning
SSPC SP 6/NACE No.3	(2007) Commercial Blast Cleaning

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.23	(Nov 2016) Ladders
29 CFR 1910.28	(Nov 2016)Duty to Have Fall Protection and Falling Object Protection
29 CFR 1910.29	(Nov 2016) Fall Protection System and Falling Object Protection - Criteria and Practices

Amendment 0001 ***

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Ladders, Installation Drawings

SD-03 Product Data

Ladders

Ladder Safety Devices (Climbing Ladder Fall Arrest Systems)

SD-07 Certificates

Fabricator Certification for Ladder Assembly

1.3 **CERTIFICATES**

Provide fabricator certification for ladder assembly stating that the ladder and associated components have been fabricated according to the requirements of 29 CFR 1910.23.

** **Amendment 0001**

1.4 QUALIFICATION OF WELDERS

Qualify welders in accordance with AWS D1.1/D1.1M. Use procedures, materials, and equipment of the type required for the work.

1.5 DELIVERY, STORAGE, AND PROTECTION

Protect from corrosion, deformation, and other types of damage. Store items in an enclosed area free from contact with soil and weather. Remove and replace damaged items with new items.

PART 2 PRODUCTS

2.1 MATERIALS

2.1.1 Structural Carbon Steel

ASTM A36/A36M.

2.1.2 Structural Tubing

ASTM A500/A500M.

2.1.3 Steel Pipe

ASTM A53/A53M, Type E or S, Grade B.

2.1.4 Fittings for Steel Pipe

Standard malleable iron fittings ASTM A47/A47M.

2.1.5 Aluminum Alloy Products

Conform to ASTM B209 for sheet plate, ASTM B221 for extrusions and ASTM B26/B26M or ASTM B108/B108M for castings, as applicable. Provide aluminum extrusions at least 1/8 inch thick and aluminum plate or sheet at least 0.050 inch thick.

2.2 FABRICATION FINISHES

2.2.1 Galvanizing

Hot-dip galvanize items specified to be zinc-coated, after fabrication where practicable. Galvanizing: ASTM A123/A123M, ASTM A153/A153M, ASTM A653/A653M or ASTM A924/A924M, G90, as applicable.

2.2.2 Galvanize

Anchor bolts, washers, and parts or devices necessary for proper installation, unless indicated otherwise.

2.2.3 Repair of Zinc-Coated Surfaces

Repair damaged surfaces with galvanizing repair method and paint conforming to ASTM A780/A780M or by application of stick or thick paste material specifically designed for repair of galvanizing, as approved by Contracting Officer. Clean areas to be repaired and remove slag from welds. Heat surfaces to which stick or paste material is applied, with a torch to a temperature sufficient to melt the metallics in stick or paste; spread molten material uniformly over surfaces to be coated and wipe off excess material.

2.2.4 Shop Cleaning and Painting

2.2.4.1 Surface Preparation

Blast clean surfaces in accordance with SSPC SP 6/NACE No.3. Surfaces that will be exposed in spaces above ceiling or in attic spaces, crawl spaces, furred spaces, and chases may be cleaned in accordance with SSPC SP 3 in lieu of being blast cleaned. Wash cleaned surfaces which become contaminated with rust, dirt, oil, grease, or other contaminants with solvents until thoroughly clean.

2.2.4.2 Pretreatment, Priming and Painting

Apply pretreatment, primer, and paint in accordance with manufacturer's printed instructions. On surfaces concealed in the finished construction or not accessible for finish painting, apply an additional prime coat to a minimum dry film thickness of 1.0 mil. Tint additional prime coat with a small amount of tinting pigment.

2.2.5 Nonferrous Metal Surfaces

Protect by plating, anodic, or organic coatings.

2.2.6 Aluminum Surfaces

2.2.6.1 Surface Condition

Before finishes are applied, remove roll marks, scratches, rolled-in scratches, kinks, stains, pits, orange peel, die marks, structural streaks, and other defects which will affect uniform appearance of finished surfaces.

2.2.6.2 Aluminum Finishes

Unexposed plate and extrusions may have mill finish as fabricated. Sandblast castings' finish, medium, AA DAF45. Unless otherwise specified, provide all other aluminum items with standard mill finish. Provide a coating thickness not less than that specified for protective and decorative type finishes for items used in interior locations or architectural Class I type finish for items used in exterior locations in AA DAF45.

2.3 LADDERS

Fabricate vertical ladders conforming to 29 CFR 1910.23 and Section 5 of ALI A14.3. Ladders shall be capable of supporting their maximum intended load. Use 2 1/2 by 3/8 inch steel flats for stringers and 3/4 inch diameter steel rods for rungs. Ladder rungs, step and cleats must be spaced not less than 10 inches and not more than 16 inches wide (measured before installation of ladder safety system), spaced no more than 14 inches apart, plug welded or shouldered and headed into stringers. Install ladders so that the maximum perpendicular distance from the centerline of the steps or rungs, or grab bars, or both, to the nearest permanent object in the back of the ladder or to the finished wall surface will not be less than 7 inches, except for the elevator pit ladders, which have a minimum perpendicular distance of 4.5 inches. Provide heavy clip angles riveted or bolted to the stringer and drilled as indicated. Provide intermediate clip angles not over 48 inches on centers. The top rung of the ladder must be level with the top of the access level, parapet or landing served by the ladder except for hatches or wells. Extend the side rails of through or side step ladders 42 inches above the access level. Provide ladder access protective swing gates at the top of access/egress level. The drawings must indicate ladder locations and details of critical dimensions and materials.

2.3.1 Phasing out of Ladder Cages and Wells (29 CFR 1910.28, Nov 2016)

Conform to 29 CFR 1910.28 (Nov 2016).

Each newly installed ladder over 20 feet in length shall only be equipped with a personal fall arrest system or climbing ladder fall arrest system (ladder safety device), cages and wells are prohibited. When a fixed ladder, cage, or well, or any portion of a section thereof, is replaced, a personal fall arrest system or climbing ladder fall arrest system (ladder safety device) is installed in at least that section of the fixed ladder, cage, or well where the replacement is located. On and after November 18, 2036, all fixed ladders shall only be equipped with a personal fall arrest system or a ladder safety device (climbing ladder Fall Arrest System).

2.3.2 Ladder Safety Devices (Climbing Ladder Fall Arrest Systems)

Conform to 29 CFR 1910.29, Section 7 of ALI A14.3 and ASSP Z359.16.

Install ladder safety devices on ladders over 20 feet long or more. The ladder safety systems must meet the design requirement of the ladders which they serve. The ladder safety system must be capable of sustaining a minimum static load of 1,000 pounds. The applied loads transferred to the climbing ladder mounting locations as a result of a fall shall be specified by the manufacturer of the climbing ladder fall arrest system. Each ladder safety system must allow the worker to climb up and down using both hands and does not require the employee continuously, hold, push, or pull any part of the system while climbing. The connection between the carrier or lifeline and the point of attachment to the body harness does not exceed 9 inches. The ladder safety system consists of a rigid or flexible carrier. Mountings for the rigid carries are attached at each end of the carrier, with intermediate mountings spaced as necessary, along the entire length of the carrier. Mountings for flexible carrier are attached at each end of the carrier and cable guides for flexible carriers are installed at least 25 feet apart but not more than 40 feet apart along the entire length of the carrier. The design and installation of mountings and cable guides does not reduce the design strength of the ladder.

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2.3.3 Ship's Ladder

DELETED

***Amendment 0001**

PART 3 EXECUTION

3.1 GENERAL INSTALLATION REQUIREMENTS

Install items at locations indicated, according to manufacturer's instructions. Verify all measurements and take all field measurements necessary before fabrication. Provide Exposed fastenings of compatible materials, generally matching in color and finish, and harmonize with the material to which fastenings are applied. Include materials and parts necessary to complete each item, even though such work is not definitely shown or specified. Poor matching of holes for fasteners will be cause for rejection. Conceal fastenings where practicable. Thickness of metal and details of assembly and supports must provide strength and stiffness. Formed joints exposed to the weather to exclude water. Items listed below require additional procedures.

3.2 WORKMANSHIP

Metalwork must be well formed to shape and size, with sharp lines and angles and true curves. Drilling and punching must produce clean true lines and surfaces. Continuously weld along the entire area of contact. Do not tack weld exposed connections of work in place. Grid smooth exposed welds. Provide smooth finish on exposed surfaces of work in place, unless otherwise approved. Where tight fits are required, mill joints. Cope or miter corner joints, well formed, and in true alignment. Install in accordance with manufacturer's installation instructions and approved drawings, cuts, and details.

3.3 ANCHORAGE, FASTENINGS, AND CONNECTIONS

Provide anchorage where necessary for fastening metal items securely in place. Include for anchorage not otherwise specified or indicated slotted inserts, expansion anchors, and powder-actuated fasteners, when approved for concrete; toggle bolts and through bolts for masonry; machine bolts, carriage bolts and powder-actuated threaded studs for steel; through bolts, lag bolts, and screws for wood. Do not use wood plugs in any material. Provide non-ferrous attachments for non-ferrous metal. Make exposed fastenings of compatible materials, generally matching in color and finish, to which fastenings are applied. Conceal fastenings where practicable.

3.4 WELDING

Perform welding, welding inspection, and corrective welding, in accordance with AWS D1.1/D1.1M. Use continuous welds on all exposed connections. Grind visible welds smooth in the finished installation.

3.5 FINISHES

3.5.1 Dissimilar Materials

Where dissimilar metals are in contact, protect surfaces with a coat conforming to MPI 79 to prevent galvanic or corrosive action. Where aluminum is in contact with concrete, plaster, mortar, masonry, wood, or absorptive materials subject to wetting, protect with ASTM D1187/D1187M, asphalt-base emulsion.

3.5.2 Field Preparation

Remove rust preventive coating just prior to field erection, using a remover approved by the rust preventive manufacturer. Surfaces, when assembled, must be free of rust, grease, dirt and other foreign matter.

3.5.3 Environmental Conditions

Do not clean or paint surface when damp or exposed to foggy or rainy weather, when metallic surface temperature is less than 5 degrees F above the dew point of the surrounding air, or when surface temperature is below 45 degrees F or over 95 degrees F, unless approved by the Contracting Officer.

3.6 LADDERS

Secure to the adjacent construction with the clip angles attached to the stringer. Install intermediate clip angles not over 48 inches on center. Install brackets as required for securing of ladders welded or bolted to structural steel or built into the masonry or concrete. Ends of ladders must not rest upon finished roof or floor.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 05 - METALS

SECTION 05 52 00

METAL RAILINGS

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-- End of Section Table of Contents --

SECTION 05 52 00

METAL RAILINGS
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DELEGATED DESIGN

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B18.2.1	(2012; Errata 2013) Square and Hex Bolts and Screws (Inch Series)
ASME B18.6.1	(2016) Wood Screws (Inch Series)
ASME B18.6.3	(2013; R 2017) Machine Screws, Tapping Screws, and Machine Drive Screws (Inch Series)
ASME B18.21.1	(2009; R 2016) Washers: Helical Spring-Lock, Tooth Lock, and Plain Washers (Inch Series)

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M	(2020; Errata 1 2021) Structural Welding Code - Steel

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
ASTM A53/A53M	(2022) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A108	(2013) Standard Specification for Steel Bar, Carbon and Alloy, Cold-Finished
ASTM A153/A153M	(2016a) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware
ASTM A283/A283M	(2013) Standard Specification for Low and Intermediate Tensile Strength Carbon Steel Plates
ASTM A307	(2021) Standard Specification for Carbon

	Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength
ASTM A467/A467M	(2020) Standard Specification for Machine Coil Chain
ASTM A500/A500M	(2021a) Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes
ASTM A512	(2006; R 2012) Standard Specification for Cold-Drawn Buttweld Carbon Steel Mechanical Tubing
ASTM A575	(2020) Standard Specification for Steel Bars, Carbon, Merchant Quality, M-Grades

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

NAAMM AMP 521	(2001; R 2012) Pipe Railing Systems Manual

1.2 ADMINISTRATIVE REQUIREMENTS

1.2.1 Preinstallation Meetings

Within 30 days of contract award, submit fabrication drawings to the Contracting Officer for the following items:

a. Steel shapes, plates, bars and strips

b. Steel railings and handrails

c. Anchorage and fastening systems

Submit manufacturer's catalog data, including two copies of manufacturers specifications, load tables, dimension diagrams, and anchor details for the following items:

a. Structural-steel plates, shapes, and bars

b. Structural-steel tubing

c. Cold-finished steel bars

d. Cold-drawn steel tubing

e. Masonry anchorage devices

f. Protective coating

g. Steel railings and handrails

h. Anchorage and fastening systems

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Fabrication Drawings; G

Iron and Steel Hardware; G

Steel Shapes, Plates, Bars and Strips; G

SD-03 Product Data

Structural-Steel Plates, Shapes, and Bars; G

Structural-Steel Tubing; G

Cold-Finished Steel Bars; G

Cold-Drawn Steel Tubing; G

Masonry Anchorage Devices; G

Protective Coating; G

Steel Railings and Handrails; G

Anchorage and Fastening Systems; G

SD-06 Test Reports

Mill Test Reports; G

Welding Inspections; G

SD-07 Certificates

Welding Procedures; G

Welder Qualification; G

SD-08 Manufacturer's Instructions

Installation Instructions

1.4 QUALITY CONTROL

1.4.1 Welding Procedures

See specification 05 05 23.16.

1.4.2 Welder Qualification

See specification 05 05 23.16.

PART 2 PRODUCTS

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2.1 FABRICATION

Preassemble items in the shop to the greatest extent possible. Disassemble units only to the extent necessary for shipping and handling. Clearly mark units for reassembly and coordinated installation.

For the fabrication of work exposed to view, use only materials that are smooth and free of surface blemishes, including pitting, seam marks, roller marks, rolled trade names, and roughness. Remove blemishes by grinding, or by welding and grinding, before cleaning, treating, and applying surface finishes, including zinc coatings.

Provide railing and handrail detail plans and elevations at not less than 1 inch to 1 foot. Provide details of sections and connections at not less than 3 inches to 1 foot. Also detail setting drawings, diagrams, templates for installation of anchorages, including concrete inserts, anchor bolts, and miscellaneous metal items having integral anchors.

Use materials of size and thicknesses indicated or, if not indicated, of the size and thickness necessary to produce adequate strength and durability in the finished product for its intended use. Work the materials to the dimensions indicated on approved detail drawings, using proven details of fabrication and support. Use the type of materials indicated or specified for the various components of work.

Form exposed work true to line and level, with accurate angles and surfaces and straight sharp edges. Ensure that all exposed edges are eased to a radius of approximately 1/32 inch. Bend metal corners to the smallest radius possible without causing grain separation or otherwise impairing the work.

Weld corners and seams continuously and in accordance with the recommendations of AWS D1.1/D1.1M. Grind exposed welds smooth and flush to match and blend with adjoining surfaces (similar to NOMMA #2).

Form the exposed connections with hairline joints that are flush and smooth, using concealed fasteners wherever possible. Use exposed fasteners of the type indicated or, if not indicated, use countersunk Phillips flathead screws or bolts.

Provide anchorage of the type indicated and coordinated with the supporting structure. Fabricate anchoring devices and space as indicated and as required to provide adequate support for the intended use of the work.

Use hot-rolled steel bars for work fabricated from bar stock unless work is indicated or specified to be fabricated from cold-finished or cold-rolled stock.

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2.1.1 Steel Handrails

Fabricate joint posts, rail, and corners by one of the following methods:

a. Flush-type rail fittings of commercial standard, welded and ground smooth, with railing splice locks secured with 3/8 inch hexagonal-recessed-head setscrews.

b. Mitered and welded joints made by fitting post to top rail and intermediate rail to post, mitering corners, groove-welding joints, and grinding smooth. Butt railing splices and reinforce them by a tight-fitting interior sleeve not less than 6 inches long.

c. Railings may be bent at corners in lieu of jointing, provided that bends are made in suitable jigs and the pipe is not crushed.

Provide removable sections as indicated.

2.1.2 Protective Coating

Shop-prime the steelwork as indicated in accordance with Section 09 90 00 PAINTS AND COATINGS except the following:

a. steel surfaces encased in concrete

b. steel surfaces for welding

c. high-strength bolt-connected contact surfaces

d. crane rail surfaces

2.2 COMPONENTS

2.2.1 Structural Steel Plates, Shapes And Bars

Provide structural-size shapes and plates, except plates to be bent or cold-formed, conforming to ASTM A36/A36M, unless otherwise noted.

Provide steel plates, to be bent or cold-formed, conforming to ASTM A283/A283M, Grade C.

Provide steel bars and bar-size shapes conforming to ASTM A36/A36M, unless otherwise noted.

2.2.2 Structural-Steel Tubing

Provide structural-steel tubing, hot-formed, welded or seamless, conforming to ASTM A500/A500M, Grade C, unless otherwise noted.

2.2.3 Hot-Rolled Carbon Steel Bars

Provide bars and bar-size shapes conforming to ASTM A575, grade as selected by the fabricator.

2.2.4 Cold-Finished Steel Bars

Provide cold-finished steel bars conforming to ASTM A108, grade as selected by the fabricator.

2.2.5 Cold-Drawn Steel Tubing

Provide tubing conforming to ASTM A512, sunk-drawn, butt-welded, cold-finished, and stress-relieved.

2.2.6 Steel Pipe

Provide pipe conforming to ASTM A53/A53M, type as selected, Grade B; primed finish, unless galvanizing is required; standard weight (Schedule 40).

2.2.7 Fasteners

Provide galvanized zinc-coated fasteners in accordance with ASTM A153/A153M used for exterior applications or where built into exterior walls or floor systems. Select fasteners for the type, grade, and class required for the installation of steel stair items.

Provide standard hexagon-head bolts, conforming to ASTM A307, Grade A.

Provide square-head lag bolts conforming to ASME B18.2.1.

Provide cadmium-plated steel machine screws conforming to ASME B18.6.3.

Provide flat-head carbon steel wood screws conforming to ASME B18.6.1.

Provide plain round, general-assembly-grade, carbon steel washers conforming to ASME B18.21.1.

Provide helical spring, carbon steel lockwashers conforming to ASME B18.2.1.

2.2.8 Steel Railings And Handrails

Design handrails to resist a concentrated load of 200 lb in any direction at any point of the top of the rail or 50 lb per foot applied horizontally to the top of the rail, whichever is more severe. NAAMM AMP 521, provide the same size rail and post. Provide pipe collars of the same material and finish as the handrail and posts.

2.2.8.1 Steel Handrails

Provide steel handrails, including inserts in concrete, steel pipe conforming to ASTM A53/A53M . Provide steel railings of 1 1/2 inch nominal size, shop-painted.

Provide kickplates between railing posts where indicated, and consisting of 1/8 inch steel flat bars not less than 6 inches high. Secure kickplates as indicated.

Galvanize exterior railings, including pipe, fittings, brackets, fasteners, and other ferrous metal components. Provide black steel pipe for interior railings.

Provide galvanized railings, including pipe, fittings, brackets, fasteners, and other ferrous metal components.

2.2.9 Safety Chains And Guardrails

Provide safety chains of galvanized steel, straight-link type, 3/16 inch diameter, with at least 12 links per foot, and with snap hooks on each end. Test safety chain in accordance with ASTM A467/A467M, Class CS. Provide snap hooks of boat type. Provide galvanized 3/8 inch bolt with 3/4 inch eye diameter for attachment of chain, anchored as indicated.

Supply two chains, 4 inches longer than the anchorage spacing, for each guarded area.

2.3 MILL TEST REPORTS

Provide mill test reports for structural steel plates, shapes, structural steel tubing, hot-rolled carbon steel bars, and cold-drawn steel tubing. Mill test reports will include, but are not limited to material heat number, material grade, edition year and type of specification met, material dimension, mechanical properties, chemical analysis, heat treatment (if applicable), and certified inspector signature.

PART 3 EXECUTION

3.1 PREPARATION

Adjust stair railings and handrails before securing in place in order to ensure proper matching at butting joints and correct alignment throughout their length. Space posts not more than 4 feet on center. Plumb posts in each direction. Secure posts and rail ends to building construction as follows:

a. Anchor posts in concrete by means of pipe sleeves set and anchored into concrete. Provide sleeves of galvanized, standard-weight, steel pipe, not less than 6 inches long, and having an inside diameter not less than 1/2 inch greater than the outside diameter of the inserted pipe post. Provide steel plate closure secured to the bottom of the sleeve, with closure width and length not less than 1 inch greater than the outside diameter of the sleeve. After posts have been inserted into sleeves, fill the annular space between the post and sleeve with nonshrink grout ora quick-setting hydraulic cement. Cover anchorage joint with a round steel flange welded to the post.

b. Anchor posts to steel with oval steel flanges, angle type or floor type as required by conditions, welded to posts and bolted to the steel supporting members.

c. Anchor rail ends into concrete and masonry with round steel flanges welded to rail ends and anchored into the wall construction with lead expansion shields and bolts.

d. Anchor rail ends to steel with oval or round steel flanges welded to tail ends and bolted to the structural-steel members.

Secure handrails to walls by means of wall brackets and wall return fitting at handrail ends. Provide brackets of malleable iron castings, with not less than 3 inch projection from the finished wall surface to the center of the pipe, drilled to receive one 3/8 inch bolt. Locate brackets not more than 60 inches on center. Provide wall return fittings of cast iron castings, flush type, with the same projection as that specified for wall brackets. Secure wall brackets and wall return fittings to building construction as follows:

a. For concrete and solid masonry anchorage, use bolt anchor expansion shields and lag bolts.

b. For hollow masonry and stud partition anchorage, use toggle bolts having square heads.

Install toe boards and brackets where indicated. Make splices, where required, at expansion joints. Install removable sections as indicated.

3.2 INSTALLATION

Submit manufacturer's installation instructions for the following products to be used in the fabrication of steel stair railing and hand rail work:

a. Structural-steel plates, shapes, and bars

b. Structural-steel tubing

c. Cold-finished steel bars

d. Hot-rolled carbon steel bars

e. Cold-drawn steel tubing

f. Protective coating

g. Masonry anchorage devices

h. Steel railings and handrails

Provide complete, detailed fabrication and installation drawings for all iron and steel hardware, and for all steel shapes, plates, bars, and strips used in accordance with the design specifications cited in this section.

3.2.1 Steel Handrail

Install handrail in pipe sleeves embedded in concrete and filled with nonshrink grout or quick-setting anchoring cement with anchorage covered with standard pipe collar pinned to post.by means of base plates bolted or welded to stringers or structural-steel frame work. Secure rail ends by steel pipe flanges anchored by expansion shields and bolts or through-bolted to a back plate or by 1/4 inch lag bolts to studs or solid backing.

3.2.2 Touchup Painting

Immediately after installation, clean field welds, bolted connections, abraded areas of the shop paint, and exposed areas painted with the paint used for shop painting. Apply paint by brush or spray to provide a minimum dry-film thickness of 2 mils.

3.3 FIELD QUALITY CONTROL

3.3.1 Field Welding

Ensure that procedures of manual shielded metal arc welding, appearance and quality of welds made, and methods used in correcting welding work comply with AWS D1.1/D1.1M.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 06 - WOOD, PLASTICS, AND COMPOSITES

SECTION 06 41 16.00 10

PLASTIC-LAMINATE-CLAD ARCHITECTURAL CABINETS

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SECTION 06 41 16.00 10

PLASTIC-LAMINATE-CLAD ARCHITECTURAL CABINETS
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PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI A161.2	(1998) Decorative Laminate Countertops, Performance Standards for Fabricated High Pressure

ARCHITECTURAL WOODWORK INSTITUTE (AWI)

AWI AWS	(2nd Edition) Architectural Woodwork Standards

ASTM INTERNATIONAL (ASTM)

ASTM D1037	(2012; R 2020) Standard Test Methods for Evaluating Properties of Wood-Base Fiber and Particle Panel Materials
ASTM F547	(2017) Standard Terminology of Nails for Use with Wood and Wood-Base Materials

BUILDERS HARDWARE MANUFACTURERS ASSOCIATION (BHMA)

ANSI/BHMA A156.9	(2020) Cabinet Hardware
ANSI/BHMA A156.16	(2023) Auxiliary Hardware
ANSI/BHMA A156.18	(2020) Materials and Finishes

COMPOSITE PANEL ASSOCIATION (CPA)

ANSI/CPA A208.1	(2022) Particleboard
CPA A208.2	(2016) Medium Density Fiberboard (MDF) for Interior Applications

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI/NEMA LD 3	(2005) Standard for High-Pressure Decorative Laminates

WINDOW AND DOOR MANUFACTURERS ASSOCIATION (WDMA)

ANSI/WDMA I.S.1A	(2013) Interior Architectural Wood Flush

1.2 SYSTEM DESCRIPTION

Work in this section includes laminate clad custom casework as indicated on the drawings and as described in this Section. This Section includes high-pressure laminate surfacing and cabinet hardware.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings

Installation

SD-03 Product Data

Recycled Content; G

Wood Materials; G

Cabinet Hardware; G

Thermally Fused Laminates; G

SD-04 Samples

Cabinet Hardware; G

Thermally Fused Laminates; G

Edge Banding; G

SD-07 Certificates

Quality Assurance

Fabricator and Installer Qualifications

Fabricator Shop Certification

SD-11 Closeout Submittals

Sustainable Design Requirements

1.4 QUALITY ASSURANCE

1.4.1 General Requirements

Unless otherwise noted on the drawings, all materials, construction methods, and fabrication must conform to and comply with the custom grade quality standards in accordance with AWI AWS, Section for laminate clad cabinets. These standards apply in lieu of omissions or specific requirements in this specification. Contractors and their personnel engaged in the work must be able to demonstrate successful experience with work of comparable extent, complexity and quality to that shown and specified. Submit a quality control statement which illustrates compliance with and understanding of AWI AWS requirements, in general, and the specific AWI AWS requirements provided in this specification. The quality control statement must also certify a minimum of ten years Contractor's experience in laminate clad casework fabrication, construction and installation. In the quality control statement, provide a list of a minimum of five successfully completed projects of a similar scope, size, and complexity.

1.4.2 Fabricator and Installer Qualifications

Shop that employs skilled workers and installers who custom fabricate and install products similar to those required for this Project.

1.4.3 Mock-ups

Prior to final approval of shop drawings, provide a full-size mock-up of a typical floor cabinet and wall cabinet , including all components and hardware necessary to illustrate a completed unit with a minimum of one door and one drawer assembly. Include countertops and back splashes where specified. Utilize specified finishes in the patterns and colors as indicated. Upon disapproval, rework or remake the mock-up until approval is secured. Remove rejected units from the jobsite. Approved mock-up may remain as part of the finished work subject to the Contracting Officer's approval.

1.4.4 Shop Drawings

Include plan views, elevations, sections and attachment details. Show large-scale details. Show dimensions and locations and sizes of furring, blocking, and hanging strips. Show locations and sizes of cutouts and holes for items installed in architectural cabinets.

1.5 DELIVERY, STORAGE, AND HANDLING

Casework may be delivered knockdown or fully assembled. Deliver all units to the site in undamaged condition, stored off the ground in fully enclosed areas, and protected from damage. Ventilate the storage area and do not subject to extreme changes in temperature or humidity.

1.6 ENVIRONMENTAL LIMITATIONS WITH HUMIDITY CONTROL

Do not deliver or install cabinets until building is enclosed, wet-work is complete, and HVAC system is operating and maintaining temperature between 60 and 90 deg F and relative humidity between 43 and 65 percent during the remainder of the construction period.

1.7 SEQUENCING AND SCHEDULING

Coordinate work with other trades. Do not install units in any room or space until painting, and ceiling installation are complete within the room where the units are located. Install floor cabinets before finished flooring materials are installed.

PART 2 PRODUCTS

2.1 SUSTAINABLE DESIGN REQUIREMENTS

See Section 01 33 29 SUSTAINABILITY REQUIREMENTS AND REPORTING for sustainable design requirements.

2.1.1 Recycled Content of Products

Provide products with post-consumer recycled content plus one-half of pre-consumer recycled content to the greatest extent possible.

Product data indicating percentage by weight of post-consumer and pre-consumer recycled content for products having recycled content included in this Section. Include statement of costs for recycled materials specified in this Section. Submit in accordance with Section 01 33 29 SUSTAINABILITY REQUIREMENTS AND REPORTING.

2.2 WOOD MATERIALS

Provide materials that comply with requirements of referenced quality standard for each type of architectural cabinet and quality grade specified unless otherwise indicated.

a. Wood Moisture Content: 5 to 10 percent

2.2.1 Lumber

a. Provide kiln-dried Grade III framing lumber to dimensions as shown on the drawings. Frame front, where indicated on the drawings, must be nominal 3/4 inch hardwood.

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2.3 **Room Closet Shelving and Clothes Hanger Rods**

Support 1 inch nominal thick wood shelf material or 3/4 or 23/32 inch thick plywood shelf material with end and intermediate supports arranged to prevent buckling and sagging. Wood shelf material shall be finished in plastic laminate. Provide hook strips 1 by 4 inches nominal and cleats 1 by 2 inches nominal. Provide cleats except where hook strips are specified or indicated. Anchor standards to wall at not more than 4 feet on center.

Provide clothes hanger rods where indicated and in closets having hook strips. Set rods parallel with front edges of shelves and support by sockets at each end and intermediate brackets spaced not more than 4 feet on center.

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2.4 PANEL CORE PRODUCTS

2.4.1 Particleboard

Particleboard must be industrial grade, medium density (40 to 50 pounds per cubic foot), Grade M-2 or better, in thickness and types meeting AWS Quality grade specified. A moisture-resistant particleboard in grade Type

2-M-2 or 2-M-3 must be used as the substrate for plastic laminate covered components as located on the drawings and other areas subjected to moisture. Particleboard must meet the minimum standards listed in ASTM D1037 and ANSI/CPA A208.1.

2.4.2 Medium Density Fiberboard

Medium density fiberboard (MDF), Grade 130 or better, in thickness and types meeting AWS Quality grade specified is an acceptable panel substrate where noted on the drawings. Medium density fiberboard must meet the minimum standards listed in CPA A208.2.

2.5 SOLID POLYMER MATERIAL

Solid surfacing casework components must be in accordance with Section 06 61 16 SOLID SURFACING FABRICATIONS.

2.6 HIGH PRESSURE DECORATIVE LAMINATE (HPDL)

Plastic laminates must meet the requirements of ANSI/NEMA LD 3 and ANSI A161.2 for high-pressure decorative laminates, HPDL. Provide design, colors, surface finish and texture, and locations as indicated on the drawings. Submit two samples of each plastic laminate pattern and color. Samples to be a minimum of 5 by 7 inches in size. Plastic laminate types and nominal minimum thicknesses for casework components must be as indicated in the following paragraphs.

2.6.1 Horizontal General Purpose Standard (HGS) Grade

Provide horizontal general purpose standard grade plastic laminate that is 0.048 inches (plus or minus 0.005 inches) in thickness. This laminate grade is intended for horizontal surfaces where postforming is not required.

2.6.2 Vertical General Purpose Standard (VGS) Grade

Provide vertical general purpose standard grade plastic laminate that is 0.028 inches (plus or minus 0.004 inches) in thickness. This laminate grade is intended for exposed exterior vertical surfaces of casework components where postforming is not required.

2.6.3 Horizontal General Purpose Postformable (HGP) Grade

Provide horizontal general purpose postformable grade plastic laminate that is 0.042 inches (plus or minus 0.005 inches) in thickness. This laminate grade is intended for horizontal surfaces where post forming is required.

2.6.4 Vertical General Purpose Postformable (VGP) Grade

Provide vertical general purpose postformable grade plastic laminate that is 0.028 inches (plus or minus 0.004 inches) in thickness. This laminate grade is intended for exposed exterior vertical surfaces of components where postforming is required for curved surfaces.

2.6.5 Cabinet Liner Standard (CLS) Grade

Provide cabinet liner standard grade plastic laminate of 0.020 inches in thickness. This laminate grade is intended for light duty semi-exposed interior surfaces of casework components.

2.6.6 Backing Sheet (BK) Grade

Undecorated backing sheet grade laminate is formulated specifically to be used on the backside of plastic laminated panel substrates to enhance dimensional stability of the substrate. Provide backing sheet thickness of 0.020 inches. Backing sheets must be provided for all laminated casework components where plastic laminate finish is applied to only one surface of the component substrate.

2.7 THERMALLY FUSED LAMINATES (MELAMINE)

Provide Thermally Fused Laminate decorative overlays (melamine panels) for casework cabinet interiordrawer interiorall semi-exposed surfaces.

2.8 EDGE BANDING

Edge banding for casework doors and drawer fronts: PVC tape, 0.020 inch minimum thickness, matching laminate in color, pattern and finish edge profile. Submit two samples of each edge band color. Edge banding types and nominal minimum thickness to be as indicated.

2.9 CABINET HARDWARE

Submit one sample of each cabinet hardware item specified to include hinges, pulls, drawer glides, and other related. Hardware must be in accordance with ANSI/BHMA A156.9, Grade 2 requirements at a minimum.

2.9.1 Door Hinges

a. Frameless Concealed Hinges (European Type): ANSI/BHMA A156.9, B01602, 170 degrees of opening, self-closing.

2.9.2 Back Mounted Pulls

ANSI/BHMA A156.9, B02011.

3-1/2" stainless steel wire pull type, BHMA No. B52011.

2.9.3 Drawer Slide

ANSI/BHMA A156.9, Side mounted, Full extension type with positive stop; BHMA No. B05051 minimum 75 pound load capacity, Include integral stop to avoid accidnetal drawer removal.

2.9.4 Adjustable Shelf Support System

Multiple holes with metal pin supports.

2.9.5 Door and Door Silencers

ANSI/BHMA A156.16, L03011.

2.9.6 Hardware Finishes

For concealed hardware, provide manufacturer's standard finish that complies with product class requirements in ANSI/BHMA A156.9. For exposed hardware, provide finish that complies with ANSI/BHMA A156.18 for BHMA finish number indicated.

a. Satin Stainless Steel: BHMA 630.

2.10 FASTENERS

Provide nails, screws, and other suitable fasteners that are the size and type best suited for the purpose and conforming to ASTM F547 where applicable.

2.11 ADHESIVES, CAULKS, AND SEALANTS

2.11.1 Adhesives

Provide adhesives of a formula and type recommended by AWI. Select adhesives for their ability to provide a durable, permanent bond and take into consideration such factors as materials to be bonded, expansion and contraction, bond strength, fire rating, and moisture resistance.

2.11.1.1 Wood Joinery

Adhesives used to bond wood members must be a Type II for interior use polyvinyl acetate resin emulsion. Adhesives must withstand a bond test as described in ANSI/WDMA I.S.1A.

2.11.1.2 Laminate Adhesive

Adhesive used to join high-pressure decorative laminate to wood must be adhesive consistent with AWI and laminate manufacturer's recommendations. PVC edgebanding must be adhered using a polymer-based hot melt glue.

2.11.2 Caulk

Caulk used to fill voids and joints between laminated components and between laminated components and adjacent surfaces must be clear, 100 percent silicone.

2.11.3 Sealant

Provide sealant of a type and composition recommended by the substrate manufacturer to provide a moisture barrier at sink cutouts and all other locations where unfinished substrate edges may be subjected to moisture.

2.12 ACCESSORIES

2.13 FABRICATION

Verify field measurements as indicated in the shop drawings before fabrication. Accomplish fabrication and assembly of components at the shop site to the maximum extent possible.

2.13.1 Plastic Laminate Casework

Unless otherwise indicated, comply with the "Architectural Woodwork Standards" for grades of cabinets indicated for construction, finishes, installation, and other requirements.

a. Grade: AWS Premium

b. Type of Construction: Frameless

c. Door and Drawer Front Style: Flush overlay

(1) Reveal dimension: 1/4 inch.

d. High Pressure Decorative Laminate (HPDL): ANSI/NEMA LD 3, grades as indicated or if not indicated, as required by AWS quality standard.

e. Exposed Surfaces meeting requirements of AWS grade specified: HPDL

(1) Horizontal Surfaces: Grade HGS

(2) Vertical Surfaces: Grade HGS

(3) Pattern Direction: Vertically for drawer fronts, doors, and fixed panels.

f. Exposed Interior Surfaces: LPDL (melamine) of a color and pattern compatible with exposed surfaces .

g. Semi-exposed Surfaces: LPDL (melamine)

h. Edges: Grade HGS .

i. Concealed Backs of Panels with Exposed Plastic-Laminate Surfaces: High-pressure decorative laminate (HPDL), ANSI/NEMA LD 3, Grade BK.

2.13.2 Drawers

a. Sides and Backs: Particle Board with melamine surfaces

b. Bottoms: Tempered Hardboard .

c. Joinery: Meeting requirements of AWS for Grade specified.

2.13.2.1 Wall Anchor Strips

Provide Wall Anchor Strips for cabinets with backs less than 1/2 inch thick. Strips must consist of minimum 1/2 inch thick lumber, minimum 2-1/2 inches width; securely attached to wall side of cabinet back - top and bottom for wall hung cabinets, top only for floor standing cabinets.

2.13.3 Cabinet Floor Base

Mount floor cabinets on a base constructed of nominal 2 inch thick lumber. Provide base assembly components that are treated lumber. Make finished height for each cabinet base as indicated on the drawings. Make bottom edge of the cabinet door or drawer face flush with top of base.

2.13.4 Shelving

2.13.4.1 General Requirements

Fabricate shelving from 3/4 inch medium density particleboard . Finish all shelving top and bottom surfaces with thermoset decorative overlay (melamine). Finish shelf edges in a thermoset decorative overlay (melamine) .

a. Shelving to be a thickness of 1 inch minimum when unsupported for 42 inches or more.

b. Panel cores to be subject to a 40 lb per sq ft load capacity with the exception of 50 lbs per sq ft at schools, hospitals and library bookshelves.

c. Shelving over 48 inches in length will be secured to back of cabinet and have a center support.

2.13.5 Laminate Application

Apply laminate to substrates following the recommended procedures and instructions of the laminate manufacturer and ANSI/NEMA LD 3, using tools and devices specifically designed for laminate fabrication and application. Provide a balanced backer sheet (Grade BK) wherever only one surface of the component substrate requires a plastic laminate finish. Apply required grade of laminate in full uninterrupted sheets consistent with manufactured sizes using one piece for full length only, using adhesives specified herein or as recommended by the manufacturer. Fit corners and joints hairline. Machined flush, file, sand, or buff all laminate edges to remove machine marks and ease (sharp corners removed). Clean up at easing must be such that no overlap of the member eased is visible. Perform fabrication in conformance to ANSI A161.2.

2.13.6 Finishing

2.13.6.1 Filling

Do not expose fasteners on laminated surfaces. Make all nails, screws, and other fasteners in non-laminated cabinet components countersunk and fill the holes with wood filler consistent in color with the wood species.

PART 3 EXECUTION

3.1 INSTALLATION

Before installation, condition cabinets to humidity conditions in installation areas for not less than 72 hours. Installation will comply with applicable requirements for AWI AWS premium quality standards. Install cabinets level, plumb and true in line to a tolerance of 1/8 inch in 96 inches using concealed shims. Scribe and cut cabinets to fit adjoining work, refinish cut surfaces, and repair damaged finish at cuts. Install cabinets without distortion so doors and drawers fit openings and to provide unencumbered operation. Complete installation of hardware and accessory items as indicated. Fasten wall cabinets through back, near top and bottom, and at ends not more than 16 inches on center with No. 10 wafer-head screws sized for not less than 1 1/2 inch penetration into wood framing, blocking or hanging strips .

3.1.1 Plumbing Fixtures

Install sinks, sink hardware, and other plumbing fixtures in locations as indicated on the drawings and in accordance with Section 22 00 00 PLUMBING, GENERAL PURPOSE .

3.1.2 Countertop

Install countertop in locations as indicated on the drawings and in accordance with Section 06 61 16 SOLID SURFACING FABRICATIONS .

3.1.3 Glass

Install glass and glazing in the casework using methods and materials specified in Section 08 81 00 GLAZING in locations as indicated on the drawings.

3.2 ADJUSTING AND CLEANING

Repair damaged and defective cabinets, where possible, to eliminate function and visual defects. Where not possible to repair, replace cabinets. Adjust joinery for uniform appearance. Clean, lubricate, and adjust hardware. Clean cabinets on exposed and semi-exposed surfaces.

3.3 CONSTRUCTION WASTE MANAGEMENT

Comply with requirements of the Waste Management Plan specified in Section 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL.

-- End of Section --

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ELASTOMERIC SHEET WATERPROOFING

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SECTION 07 13 53

ELASTOMERIC SHEET WATERPROOFING
02/16, CHG 1: 08/17
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PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C1305	(2008) Standard Test Method for Crack Bridging Ability of Liquid-Applied Waterproofing Membrane
ASTM D41/D41M	(2011; R 2016) Standard Specification for Asphalt Primer Used in Roofing, Dampproofing, and Waterproofing
ASTM D146/D146M	(2004; E 2012; R 2012) Sampling and Testing Bitumen-Saturated Felts and Woven Fabrics for Roofing and Waterproofing
ASTM D297	(2015; R 2019) Rubber Products - Chemical Analysis
ASTM D412	(2016) Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers - Tension
ASTM D429	(2014) Rubber Property-Adhesion to Rigid Substrates
ASTM D471	(2016a) Standard Test Method for Rubber Property - Effect of Liquids
ASTM D570	(1998; E 2010; R 2010) Standard Test Method for Water Absorption of Plastics
ASTM D573	(2004; R 2019) Standard Test Method for Rubber - Deterioration in an Air Oven
ASTM D624	(2000; R 2020) Standard Test Method for Tear Strength of Conventional Vulcanized Rubber and Thermoplastic Elastomers
ASTM D638	(2014) Standard Test Method for Tensile Properties of Plastics
ASTM D746	(2014) Standard Test Method for Brittleness Temperature of Plastics and Elastomers by Impact

ASTM D751	(2006; R 2011) Coated Fabrics
ASTM D903	(1998; R 2017) Standard Test Method for Peel or Stripping Strength of Adhesive Bonds
ASTM D1004	(2013) Initial Tear Resistance of Plastic Film and Sheeting
ASTM D1149	(2007; R 2012) Standard Test Method for Rubber Deterioration - Surface Ozone Cracking in a Chamber
ASTM D1204	(2014; R 2020) Linear Dimensional Changes of Nonrigid Thermoplastic Sheeting or Film at Elevated Temperature
ASTM D1876	(2008; R 2015; E 2015) Standard Test Method for Peel Resistance of Adhesives (T-Peel Test)
ASTM D2136	(2002; R 2012) Coated Fabrics - Low-Temperature Bend Test
ASTM D2240	(2015; E 2017) Standard Test Method for Rubber Property - Durometer Hardness
ASTM D3045	(1992; R 2010) Practice for Heat Aging of Plastics Without Load
ASTM D5385/D5385M	(1993; R 2014; E 2014) Standard Test Method for Hydrostatic Pressure Resistance of Waterproofing Membranes
ASTM E96/E96M	(2016) Standard Test Methods for Water Vapor Transmission of Materials
ASTM E154/E154M	(2008a; R 2013; E 2013) Water Vapor Retarders Used in Contact with Earth Under Concrete Slabs, on Walls, or as Ground Cover

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC	(2018) International Building Code

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Manufacturer's Standard Details; G,

Elastomeric Waterproofing Sheet Material; G,

Protection Board; G,

Primers, Adhesives, and Mastics; G

SD-06 Test Reports

Elastomeric Waterproofing Sheet Material; G

Field Quality Control documentation; G

Protective Covering; G

SD-07 Certificates

Elastomeric Waterproofing Sheet Material; G

Primers, Adhesives, and Mastics; GDraft Special Warranties; G

Final Special Warranties; G

Certificates Of Compliance; G

SD-08 Manufacturer's Instructions

Primers, Adhesives, and Mastics; G

SD-11 Closeout Submittals

Certificates Of Compliance with sustainable requirements for items listed in SD-07; S

1.3 MANUFACTURER'S DETAILS

Submit manufacturer's standard details indicating methods of attachment and spacing, transition and termination details, and installation details. Include verification of existing conditions.

1.4 PRODUCT DATA

Include data for material descriptions, recommendations for product shelf life, requirements for protective coverings, and manufacturer's Safety Data Sheets (SDS) for primers, adhesives, and mastics.

1.5 CODE REQUIREMENTS

Provide membrane waterproofing system in accordance with ICC IBC Section 1805 Dampproofing and Waterproofing.

1.6 DELIVERY, STORAGE, HANDLING, IDENTIFICATION

Deliver and store materials in accordance with manufacturer's printed instructions, out of the weather, in manufacturer's original packaging with brand name and product identification clearly marked. Keep materials wrapped and separated from off-gassing materials (such as drying paints and adhesives). Do not use materials that have visible moisture or biological growth. Do not permit unidentified materials in the work area or in the project.

1.7 ENVIRONMENTAL CONDITIONS

Do not apply waterproofing during inclement weather or when there is ice, frost, surface moisture, or visible dampness on the surface to receive waterproofing for when ambient and surface temperatures are 40 degrees F or below. The restriction on the application of waterproofing materials when ambient and surface temperatures are below 40 degrees F will be waived if the Contractor devises a means, approved by the Contracting Officer in writing, of maintaining the surface and ambient temperatures above 40 degrees F.

1.8 SPECIAL WARRANTIES

1.8.1 Guarantee

Guarantee waterproofing membrane installation against failure due to leaks for a period of two years from the date of Contract Completion Date. Submit draft and final guarantees in accordance with Sections 01 33 00 CLOSEOUT SUBMITTALS .

1.8.2 Warranty

Provide manufacturer's material warranty for all system components for a period of ten years from the date of Contract Completion Date. Submit draft and final warranties in accordance with Sections 01 33 00 CLOSEOUT SUBMITTALS .

PART 2 PRODUCTS

2.1 SUSTAINABILITY CRITERIA

Where allowed by performance criteria:

2.1.1 Reduced Volatile Organic Compound (VOC) Content

Provide products with reduced VOC content and provide certificates of compliance in accordance with Section 01 33 29 SUSTAINABILITY REQUIREMENTS AND REPORTING paragraph REDUCE VOLATILE ORGANIC COMPOUNDS.

2.1.2 Recycled Content

Provide products with recycled content and provide certificates of compliance in accordance with Section 01 33 29 SUSTAINABILITY REQUIREMENTS AND REPORTING paragraph RECYCLED CONTENT.

2.2 MATERIALS

Provide one of the types of elastomeric waterproofing sheet material and related primers, adhesives, and mastics as specified herein. Ensure compatibility of waterproofing materials with each other, and with materials on which they are applied. Provide materials that comply with applicable requirements cited below when tested in accordance with the referenced ASTM publications.

2.3 BUTYL RUBBER SHEETING

Not less than 60 mils minimum thickness.

2.3.1 Butyl Rubber Sheeting Performance Requirements

a. Thickness Tolerance, ASTM D412: Plus or minus 10 percent.

b. Specific Gravity, ASTM D297: 1.20, plus or minus 0.05.

c. Tensile Strength, ASTM D412: 1200 psi minimum.

d. Tensile Stress at 300 percent elongation, ASTM D412: 600 psi minimum.

e. Elongation, ASTM D412: 300 percent minimum.

f. Tear Resistance, Die C, ASTM D624: 150 pound force per inch (lbf/inch) minimum.

g. Shore A Hardness, ASTM D2240: 5-second interval before reading; 60 plus or minus 10.

h. Ozone Resistance, ASTM D1149: No cracks, 7 days - 50 pphm - 100 degrees F, 20 percent elongation.

i. Heating Aging-Accelerated, ASTM D573: Tensile retention, 60 percent of minimum original elongation retention; 60 percent of minimum original requirement; 7 days, 240 degrees F.

j. Butyl Identification, ASTM D471 REV A, Tricresyl Phosphate Immersion: Maximum volume swell 10 percent, 70 hrs, 212 degrees F.

k. Low Temperature Flexibility, ASTM D746: No failure at minus 40 degrees F.

l. Water Absorption, ASTM D471 REV A: Plus 1 percent maximum. 7 days, 158 degrees F.

m. Exposure to Fungi and Bacteria in Soil, ASTM E154/E154M REV A, Minimum 16 Weeks: Unaffected.

n. Water Vapor Transmission, 80 degrees F Permeance, ASTM E96/E96M, Procedure B or BW: 0.15 perms maximum.

2.3.2 Adhesive, Cement, and Tape for Use with Butyl Rubber

As recommended by the butyl rubber waterproofing membrane manufacturer.

2.4 THERMOPLASTIC MEMBRANE: POLYVINYL CHLORIDE (PVC)

Polyvinyl chloride (PVC) flexible sheets with non-woven fiberglass reinforcing not less than 60 mils minimum thickness.

2.4.1 Thermoplastic Membrane Performance Requirements

a. Overall thickness, ASTM D751: .059 inches minimum

b. Tensile strength, ASTM D638: 1600 psi minimum

c. Elongation at break, ASTM D638: 250 percent minimum

d. Seam strength, ASTM D638: 90 percent minimum of tensile strength

e. Retention of properties after heat aging, ASTM D3045

f. Tensile strength, ASTM D638: 95 percent of original

g. Elongation, ASTM D638: 95 percent of original

h. Tear resistance, ASTM D1004: 17 lbf

i. Low Temperature Bend, ASTM D2136: minus 40 degrees F

j. Liner Dimensional Change, ASTM D1204: 0.002 percent

k. Weight Change After Immersion in Water, ASTM D570: 2.0 percent maximum

2.4.2 Adhesives

a. Adhesive for thermoplastic flashings as recommended by manufacturer.

b. Adhesive for Sub-Membrane Grid: 100 percent solids, two part urethane, with minimum tensile strength of 150 psi, in accordance with ASTM D412 and adhesion to concrete of 12 ply in accordance with ASTM D429 as recommended by manufacturer.

2.4.3 Accessories

Securement Strip: 14 gauge stainless steel metal bar 1 inch wide, pre-punched 1 inch on center for securement.

2.5 COMPOSITE, SELF-ADHERING MEMBRANE SHEETING

Cold applied composite sheet consisting of rubberized asphalt and cross laminated, high density polyethylene film. Not less than 60 mils minimum thickness is required.

2.5.1 Composite, Self-Adhering Sheeting Performance Requirements

a. Tensile Strength ASTM D412, Die C: 250 psi minimum.

b. Ultimate Elongation, ASTM D412, Die C: 200 percent minimum.

c. Water Vapor Transmission, ASTM E96/E96M 80 degrees F Permeance, Procedure B: 0.1 perm maximum.

d. Pliability degrees, ASTM D146/D146M: (180 degrees Bend Over 1 Inch Mandrel): No cracks at minus minus 25 degrees F.

e. Provide test report data for crack bridging ability: Either in accordance with ASTM C1305 as modified for a dry film thickness specified by the manufacturer and conducted at low temperature; or in accordance with a cycling over crack test also conducted for the specified dry film thickness at low temperature. Using either test, verify crack bridging up to 1/4 inch without damage to the membrane system.

f. Puncture Resistance, ASTM E154/E154M REV A: 40 lb minimum.

g. Lap Adhesion at Minimum Application Temperature, ASTM D1876 Modified, 5 lbs/in..

h. Peel Strength, ASTM D903: Modified 9 lbs/in.

i. Resistance to Hydrostatic Head, ASTM D5385/D5385M: 231 ft of water.

j. Water Absorption, ASTM D570; 0.1 percent maximum.

2.5.2 Primers

Asphalt composition, ASTM D41/D41M, or synthetic polymer in solvent as recommended by the membrane manufacturer.

2.5.3 Mastics

Polymer modified asphalt in suitable solvent of trowel grade consistency and as recommended by the membrane manufacturer.

PART 3 EXECUTION

3.1 VERIFICATION OF CONDITIONS

Before starting the work, verify surfaces that must be waterproofed are in satisfactory condition. Notify the Contracting Officer of defects or conditions anticipated to prevent a satisfactory application. Do not start application until defects and conditions have been corrected.

3.2 SURFACE PREPARATION

Ensure surfaces to receive treatment are clean, dry, smooth, and free from deleterious materials and projections. Cut off high spots or grind smooth. Finish top surfaces of projecting masonry or concrete ledges below grade, except footings, to a steep bevel with Portland cement mortar. Sweep surfaces to receive covering before applying waterproofing to remove dust and foreign matter. Cure concrete by a method compatible with the waterproofing system.

3.3 APPLICATION

3.3.1 Building Envelope Requirements

Provide a continuous waterproofing system at all material and building transitions. Lap, wrap, fasten and seal products in accordance with manufacturer's printed instructions. Envelope assembly variations are not permitted without written approval from the Contracting Officer's Representative.

3.3.2 General Installation Requirements

Provide sheet waterproofing in accordance with manufacturer's printed installation instructions. Ensure the surface to receive membrane is clean, smooth and dry without surface irregularities; correct deficiencies prior to installation. When using solvent welding liquid, avoid prolonged contact with skin and breathing of vapor and provide adequate ventilation. Carry waterproofing of horizontal surfaces up abutting vertical surfaces and adhere solid to the substrate. Avoid wrinkles and buckles in applying membrane and joint reinforcement.

3.3.2.1 Self-Adhering Membrane

Apply composite, self-adhering membrane on surfaces primed at a uniform coverage rate in accordance with membrane manufacturer's printed instructions. Remove release sheet and apply with tacky surface in contact with dried primer.

3.3.2.2 Protection

Protect membrane over horizontal surfaces from traffic during installation. Use only equipment with rubber tires. Provide walkway protection where heavy traffic from other trades is expected. Do not store material on membrane.

3.3.3 Butyl Rubber

Lap sheets at sides and ends a minimum of 6 inches over the preceding sheet. Apply lap splicing cement over entire 6 inches splice area prior to application of sealant. Make sealant continuous along the entire length of the splice. Maintain a continuous bead of sealant at all membrane splices or as required by the manufacturer. Provide a tongue and groove cemented splice a minimum of 6 inches wide with factory made heat vulcanized seam of not less than 2 inches or as required by the manufacturer, when membrane is below water table.

3.3.4 Thermoplastic Membrane (PVC)

Consult with membrane manufacturer prior to grid application. Install 12 inches wide sub-membrane containment grid as required by manufacturer. Provide the containment grid at intervals across the width and length of the substrate, at the base of all transitions, walls, curbs, penetrations, and at the perimeter of each deck/substrate section. Fully adhere strips to the deck in a full bedding of two-part urethane adhesive. Weld adjacent sheets in accordance with manufacturer's instructions. Hot-air weld all side and end lap joints. Provide lap area a minimum of 3 inch wide when machine welding, and a minimum of 4 inch wide when hand welding but not less than recommended by the manufacturer. Orient overlaps with the direction of flow of water.

3.4 COMPOSITE, SELF-ADHERING MEMBRANE

Lap sheets at edges and ends a minimum of 2-1/2 inches over the preceding sheet. Provide all side laps a minimum 2-1/2 inches and end laps 5 inches. Provide self-adhesive, mastic laps in accordance with manufacturer's recommendation. Roll or firmly press to adhere membrane to substrate. Cover corners and joints with two layers of reinforcement by first applying a 12 inch width of membrane centered along the axis. Flash drains and projections with a second ply of membrane for a distance of 6 inches from the drain or projection. Finish exposed, terminated edges of membrane on horizontal or vertical surfaces with a toweled bead of mastic. Apply mastic around edges of membrane, and drains and projections. Apply mastic at end of each work day.

3.5 FLASHING

Flash penetrations through membrane. Seal all penetrations where reinforcing bars penetrate a waterproofing membrane with the appropriate sealant or mastic flashing component. Embed elastomeric membrane in a heavy coat of adhesive, except for self-adhering membrane. Position continuous metal reglets horizontally on footing and vertically on intersecting and connecting walls, and as specified in Section 07 60 00 FLASHING AND SHEET METAL. Metal reglets are to receive exposed edges of membrane waterproofing. Secure membrane into reglets by lead wedges and fill with cement as recommended in writing by manufacturer of waterproofing materials. Counterflash upper edge of membrane waterproofing and protective covering as specified in Section 07 60 00 FLASHING AND SHEET METAL.

3.6 FIELD QUALITY CONTROL

Notify the Contracting Officer 5 working days prior to date of performing tests. Before concealment, cover elastomeric waterproofing on horizontal surfaces over finished spaces with 4 inches of ponded water for 24 hours. Do not add water after start of 24 hour period. Accurately measure water level at beginning and end of 24 hour period. If water level falls, remove water and inspect waterproofing membrane. Make repairs or replacement as directed, and repeat test. Do not proceed with work that conceals membrane waterproofing before receiving approval and acceptance of the Contracting Officer.

3.7 PROTECTIVE COVERING

After installation has been inspected and approved by the Contracting Officer, apply a protective covering to the membrane waterproofing prior to backfilling. Protect vertical membrane waterproofing with a 1/2 inch minimum thickness of asphalt plank; 1/2 inch minimum thickness of fiberboard; or 1/8 inch minimum thickness of compatible water resistant bitumen type protection board with edges abutting adjacent edges and exposed surfaces covered by a taping system recommended by manufacturer of protection board. Cover horizontal membrane waterproofing with similar protection board and Portland cement mortar not less than 3/4 inch thick; place uniformly and allow to set before installing subsequent construction.

-- End of Section --

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WALL AND DOOR PROTECTION

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SECTION 10 26 00

WALL AND DOOR PROTECTION
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PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM B221	(2020) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes
ASTM D256	(2010; R 2018) Standard Test Methods for Determining the Izod Pendulum Impact Resistance of Plastics
ASTM D543	(2020) Standard Practices for Evaluating the Resistance of Plastics to Chemical Reagents
ASTM D635	(2018) Standard Test Method for Rate of Burning and/or Extent and Time of Burning of Plastics in a Horizontal Position
ASTM E84	(2020) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM G21	(2015; R 2021; E 2021) Standard Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi

CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

CDPH SECTION 01350	(2010; Version 1.1) Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources using Environmental Chambers

GREEN SEAL (GS)

GS-36	(2013) Adhesives for Commercial Use

SCIENTIFIC CERTIFICATION SYSTEMS (SCS)

SCS	SCS Global Services (SCS) Indoor Advantage

SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)

SAE J1545	(2005; R 2014) Instrumental Color Difference Measurement for Exterior Finishes, Textiles and Colored Trim

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

SCAQMD Rule 1168	(2017) Adhesive and Sealant Applications

UNDERWRITERS LABORATORIES (UL)

UL 2818	(2013) GREENGUARD Certification Program For Chemical Emissions For Building Materials, Finishes And Furnishings

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Corner Guards; G, RO

Wall Guards; G, RO

Door Protectors; G, RO

Wall Covering and Panels; G, RO

SD-03 Product Data

Corner Guards; G, RO

Wall Guards; G, RO

Door Protectors; G, RO

Wall Covering and Panels; G, ROSD-04 Samples

Corner Guards; G, RO

Wall Guards; G, RO

Door Protectors; G, RO

Wall Covering and Panels; G, RO

SD-06 Test Reports

Fire Resistance Rating

SD-07 Certificates

Indoor air quality for wall covering/panels; S

Indoor air quality for adhesives; S

SD-10 Operation and Maintenance Data

Corner Guards, Data Package 1; G, RO

Wall Guards, Data Package 1; G, RO

Door Protectors, Data Package 1; G, RO

Wall Covering and Panels, Data Package 1; G, RO

1.3 CERTIFICATIONS

1.3.1 Indoor Air Quality

1.3.1.1 Wall Covering and Panels

Provide sheet and high impact resistant resilient materials certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party program that products meet the requirements of this section. Provide current product certification documentation from certification body.

1.3.1.2 Adhesives and Sealants

Provide products certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party program that products meet the requirements of this section. Provide current product certification documentation from certification body. When product does not have certification, provide validation that product meets the indoor air quality product requirements cited herein.

1.4 DELIVERY, STORAGE, AND HANDLING

Deliver materials to the project site in manufacturer's original unopened containers with seals unbroken and labels and trademarks intact. Keep materials dry, protected from weather and damage, and stored under cover. Store materials at approximately 70 degrees F for at least 48 hours prior to installation.

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1.5 **WARRANTY**

Provide manufacturer's warranty to repair or replace defective materials and workmanship for a 5 year period of 5 years from date of final acceptance of the work.

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PART 2 PRODUCTS

2.1 STANDARD PRODUCTS

To the maximum extent possible, provide wall and door protection items that are standard products of a single manufacturer and furnished as detailed. Drawings show general configuration of products required, and items differing in minor details from those shown are acceptable.

Submit detailed shop drawings of each wall and door protection item indicated. Include elevations, dimensions, clearances, details of construction and anchorage, and details of joints and connections.

Submit manufacturers' descriptive product data for each wall and door protection item indicated. Include manufacturers' literature, finishes, profiles and thicknesses of materials.

Submit manufacturers' operations and maintenance data for each wall and door protection item indicated in accordance with Section 01 78 23 OPERATIONS AND MAINTENANCE DATA.

2.1.1 Resilient Material

Provide resilient material consisting of high impact resistant extruded acrylic vinyl conforming to the following:

2.1.1.1 Minimum Impact Resistance

Minimum impact resistance must be 18 ft-lbs/sq. inch when tested in accordance with ASTM D256, (Izod impact, ft-lbs per sq inch notched).

2.1.1.2 Fire Resistance Rating

Provide the following surface burning characteristics when tested and labeled in accordance with ASTM E84 by a qualified testing agency: maximum flame spread of 25 and a smoke developed rating of 450 or less. Provide material rated as self extinguishing when tested in accordance with ASTM D635. Provide resilient material used for protection on fire rated doors and frames listed by the qualified testing agency performing the tests. Provide resilient material installed on fire rated wood/steel door and frame assemblies tested on similar type assemblies. Test results of material tested on any other combination of door/frame assembly are not acceptable.

2.1.1.3 Integral Color

Provide colored components having integral color and matched in accordance with SAE J1545 to within plus or minus 1.0 on the CIE-LCH scales.

2.1.1.4 Chemical and Stain Resistance

Provide materials resistant to chemicals and stains reagents in accordance with ASTM D543.

2.1.1.5 Fungal and Bacterial Resistance

Provide materials resistant to fungi and bacteria in accordance with ASTM G21, as applicable.

2.2 CORNER GUARDS

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2.2.1 Resilient Corner Guards

Provide flush mounted corner guards, radius formed to profile shown **at all exposed outside corners subject to damage, including but not limited to dwelling units and common areas such as corridors and lobbies.** Provide corner guards that extend from floor to ceiling. Furnish mounting hardware, cushions, and base plates. Provide assembly consisting of a snap-on corner guard formed from high impact resistant resilient material, mounted on a continuous aluminum retainer. Extruded aluminum retainer conforms to ASTM B221, alloy 6063, temper T5 or T6. Provide aluminum components that contain a minimum of 35 percent recycled content. Provide data identifying percentage of recycled content for aluminum component of corner guards. Flush mounted type guards act as a stop for adjacent wall finish material. Furnish factory fabricated end closure caps for top and bottom of surface mounted corner guards. Provide flush mounted corner guards installed in fire rated wall that maintain the rating of the wall. Manufacturer to provide insulating materials that are an integral part of the corner guard system. Provide exposed metal portions of fire rated assemblies with a paintable surface.

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2.3 WALL COVERING AND PANELS

Provide wall covering and panels consisting of high impact rigid acrylic vinyl or polyvinyl chloride resilient material. Panel sizes are 4 by 8 feet. Provide wall covering material used on the interior of the building (defined as inside of the weatherproofing system) that meets either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) the VOC content requirements of SCAQMD Rule 1168, or VOC content requirements of GS-36. Provide certification of indoor air quality for wall covering/panels.

2.3.1 High Impact Wall Panels

Provide wall panel face and edge thickness that are 0.040 inch. Factory bond panel face to a 0.375 inch thick fiberboard core. Laminate the backside of the panel with a moisture resistant vapor barrier.

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2.3.1.1 Stainless-Steel Wall Panel

Provide 48-inch-high stainless steel wall panels at mop sink locations for wall protection, fabricated from minimum 18-gauge Type 304 stainless steel with smooth, satin finish, complying with ASTM A240. Panel shall be sized as indicated on drawings and installed over gypsum board substrate with concealed faseners or manufacturer's recommended adhesive system, providing a tight, flush fit with sealed edges. Coordinate installation with plumbing fixtures to ensure full coverage of wall surfaces subject to moisture exposure. Seal all perimeter joints watertight with mildew-resistant sealant to prevent water intrusion behind panel and protect adjacent construction.

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2.4 TRIM, FASTENERS AND ANCHORS

Provide vinyl trim, fasteners and anchors for each specific installation as indicated.

2.5 FINISH

Submit samples indicating color and texture of materials requiring color and finish.

2.5.1 Resilient Material Finish

Provide resilient material finish of embossed velour texture with colors in accordance with SAE J1545.

2.6 ADHESIVES

Provide adhesive for resilient material in accordance with manufacturers recommendations. Provide sealants and non-aerosol adhesive products used on the interior of the building (defined as inside of the weatherproofing system) that meet either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) the VOC content requirements of SCAQMD Rule 1168, or VOC content requirements of GS-36. Provide certification of indoor air quality for adhesives.

2.7 COLOR

Provide color as indicated; colors listed are not intended to limit the selection of equal colors from other manufacturers.

PART 3 EXECUTION

3.1 INSTALLATION

Do not install items that show visual evidence of biological growth. Install items on surfaces that are clean, smooth, and free of obstructions.

3.1.1 Corner Guards and Wall Guards

a. Mount guards as indicated and in accordance with manufacturer's written installation instructions.

b. For wall guards, space brackets at no more than 3 feet on centers and anchor to the wall in accordance with the manufacturer's written installation instructions.

3.1.2 Wall Coverings and Panels

Install as indicated in accordance with manufacturer's written instructions.

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DIVISION 10 - SPECIALTIES

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TOILET ACCESSORIES

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PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-STD-1691	**(1994; Rev F) Construction and Material Schedule for Military Medical and Dental Facilities**

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Product Schedule; G

Submit product Schedule indicating types, quantities, sizes, and installation locations by room for each toilet accessory item required. Identify locations using room designations indicated on the drawings.

SD-03 Product Data

Recycled content for stainless steel toilet accessories; S

Item A5047 Cabinet, Medicine; G

Item A5080 Dispenser, Paper Towel, SS, Surface Mounted; G

Item A5109 Grab Bar, 1-1/4 inch Dia., SS, 2 Wall, W/C Accessible; G

Item A5145 Hook, Garment, Double, SS, Surface Mounted; G

Item A5170 Rod, Shower Curtain, 1 inch Diameter, W/Curtain & Hooks; G

Item A5195 Dispenser, Toilet Tissue, SS, 1-Roll, Surface Mntd; G

Item A5202 Dispenser, Toilet Paper w/Utility Shelf, SS, 2-Roll; G

Submit catalog numbers, literature, data sheets, construction details, profiles, anchoring and mounting requirements and other pertinent data for each toilet accessory item to evaluate function, materials, dimensions and appearance.

Coat Rod; G

Soap Dispenser; G

Stainless Steel Framed Mirror; G

Vanity Mirror; G

Full Body Mirror; G

Fitness Mirror; G

SD-10 Operation and Maintenance Data

Item A5047 Cabinet, Medicine; G

Item A5080 Dispenser, Paper Towel, SS, Surface Mounted; G

Item A5109 Grab Bar, 1-1/4 inch Dia., SS, 2 Wall, W/C Accessible; G

Item A5145 Hook, Garment, Double, SS, Surface Mounted; G

Item A5160 Shelf, 8 inch Depth, SS, Surface Mounted; G

Item A5166 Shelf, 12 inch Depth, SS, Surface Mounted; G

Item A5170 Rod, Shower Curtain, 1 inch Diameter, W/Curtain & Hooks; G

Item A5195 Dispenser, Toilet Tissue, SS, 1-Roll, Surface Mntd; G

Item A5202 Dispenser, Toilet Paper w/Utility Shelf, SS, 2-Roll; G

1.3 DELIVERY, STORAGE, AND HANDLING

Wrap toilet accessories for shipment and storage, then deliver to the jobsite in manufacturer's original packaging, and store in a clean, dry area protected from construction damage and vandalism.

1.4 WARRANTY

Provide manufacturer's warranty to repair or replace defective materials and workmanship for a period of one year from date of final acceptance of the work.

PART 2 PRODUCTS

2.1 ACCESSORY ITEMS

Provide toilet accessories where indicated in accordance with Contractor-provided product schedule. Conform to the requirements for accessory items specified herein which are based on MIL-STD-1691 Joint Schedule Numbers (JSN). Provide each accessory item complete with the necessary mounting plates of sturdy construction with corrosion resistant surface.

Provide stainless steel products listed herein manufactured from materials containing a minimum of 50 percent recycled content. Provide data identifying percentage of recycled content for stainless steel toilet accessories.

2.1.1 Anchors and Fasteners

Provide corrosion-resistant anchors and fasteners capable of developing a restraining force commensurate with the strength of the accessory to be mounted and suited for use with the supporting construction. Accessories shall be mounted using methods that conceal fasteners where possible. Provide tamperproof design exposed fasteners with finish to match the accessory. Provide fasteners proposed for use for each type of wall construction and mounting.

2.1.2 Finishes

Except where noted otherwise, provide the following finishes on metal:

Metal	Finish
Stainless steel	No. 4 satin finish
Carbon steel, copper alloy, and brass	Chromium plated, bright

2.1.3 Coat Rod

Unit shall be of type 304 stainless steel, 1.006mm (20-gage), with satin finish. Outside diameter shall be 25.4mm (1-inch). Flanges shall be 80mm (3 3/16-inch) maximum diameter with satin finish. Unit shall include concealed mounting brackets.

2.1.4 Stainless Steel Framed Mirror

Mirror (T-MG1), (24 x 36 inches) shall be framed with one-piece, type 304, stainless steel angle, 19 x 19mm with continuous integral stiffener on all sides and beveled edge to hold frame tightly against mirror. Corners shall be heliarc welded, ground and polished smooth. All exposed surfaces shall be matte black finish. Mirror shall be No. 1 quality 6mm (1/4-inch) tempered glass. All edges shall be protected by high impact plastic filler strips. Back of mirror shall be protected by shock- absorbing, waterproof, non-abrasive 6.4mm thick polystyrene padding. Galvanized steel back shall have integral hanging brackets for mounting on concealed rectangular wall hanger(s) and be secured with concealed Phillips-head locking screws in lower frame.

Size: 24 inches wide by 36 inches height by 3/4 inch thick.

2.1.5 Vanity Mirror

Mirror (D-MG2), (36 x 48 inches) shall be framed with one-piece, type 304, stainless steel angle, 19 x 19mm with continuous integral stiffener on all sides and beveled edge to hold frame tightly against mirror. Corners shall be heliarc welded, ground and polished smooth. All exposed surfaces shall be matte black finish. Mirror shall be No. 1 quality 6mm (1/4-inch) tempered glass. All edges shall be protected by high impact plastic filler strips. Back of mirror shall be protected by shock- absorbing, waterproof, non-abrasive 6.4mm thick polystyrene padding. Galvanized steel back shall have integral hanging brackets for mounting on concealed rectangular wall hanger(s) and be secured with concealed Phillips-head locking screws in lower frame.

Size: 36 inches wide by 48 inches height by 3/4 inch thick.

2.1.6 Full Body Mirror

Mirror (D-MG3, T-MG3), (24 x 60 inches) shall be framed with one-piece, type 304, stainless steel angle, 19 x 19mm with continuous integral stiffener on all sides and beveled edge to hold frame tightly against mirror. Corners shall be heliarc welded, ground and polished smooth. All exposed surfaces shall be matte black finish. Mirror shall be No. 1 quality 6mm (1/4-inch) tempered glass. All edges shall be protected by high impact plastic filler strips. Back of mirror shall be protected by shock- absorbing, waterproof, non-abrasive 6.4mm thick polystyrene padding. Galvanized steel back shall have integral hanging brackets for mounting on concealed rectangular wall hanger(s) and be secured with concealed Phillips-head locking screws in lower frame.

Size: 24 inches wide by 60 inches height by 3/4 inch thick.

2.1.7 Fitness Mirror

Mirror (F-MG4), (14 x 6 feet) shall be bevealed edge frame with one-piece, type 304, stainless steel angle, 19 x 19mm with continuous integral stiffener on all sides and beveled edge to hold frame tightly against mirror. Corners shall be heliarc welded, ground and polished smooth. All exposed surfaces shall be matte black finish. Mirror shall be No. 1 quality 6mm (1/4-inch) tempered glass. All edges shall be protected by high impact plastic filler strips. Back of mirror shall be protected by shock- absorbing, waterproof, non-abrasive 6.4mm thick polystyrene padding. Galvanized steel back shall have integral hanging brackets for mounting on concealed rectangular wall hanger(s) and be secured with concealed Phillips-head locking screws in lower frame.

Size: 36 inches wide by 48 inches height by 3/4 inch thick.

2.1.8 Item A5047 Cabinet, Medicine

Medicine cabinet (D-RMC) constructed of heavy gauge stainless steel and have a mirror mounted in swinging door, a minimum of three shelves, magnetic catch and full length piano hinge. Mirror is 1/4 inch thick first quality float glass electrolytically copper-plated and guaranteed against silver spoilage for 15 years. Cabinet door can be inverted to change from left to right hand swing. Unit has concealed mounting holes. Mounting hardware included.

Approximate size: 17 inches wide by 25 inches high by 4 inches deep.

2.1.9 Soap Dispenser

Liquid soap vessel (T-SD) are satin-finish stainless steel. Valve shall be corrosion resistant design. Soap container shall be removable and have a capacity of 2.4L (80 fl oz) with unbreakable refill window. Adequate adjustments/accessories shall be provided to allow dispenser to work in location indicated in the drawings.

2.1.10 Item A5080 Dispenser, Paper Towel, SS, Surface Mounted

Surface mounted unit (T-PTD) constructed of stainless steel with satin finish, welded construction, and have full length piano hinge, tumbler lock, refill indicator. Unit has smooth corners, free of burrs and sharp edges. Unit has a capacity of 400 single fold paper towels.

Approximate size: 11 inches wide by 8 inches high by 6 inches deep.

2.1.11 Item A5109 Grab Bar, 1-1/4 Inch Diameter, SS, 2 Wall, W/C Accessible

Grab bar (T-GB) of 1-1/4 inch diameter satin finish stainless steel with peened gripping surface for use in toilet stall/room. Snap-on flange covers for concealed mounting are stainless steel and equipped with two screw holes for attachment to wall. Grab bars designed to meet and exceed ADA requirements for structural strength. Grab bars designed to withstand loads of 900 pounds when properly installed. Clearance from wall to grab bar is 1-1/2 inches to meet ADA and ANSI codes.

2.1.12 Item A5135 Shelf, Utility W/ Mop/Broom Holders, SS, Surf Mounted

Surface mounted mop/broom holder with shelf made (J-MH) of 18 gauge stainless steel with all exposed surfaces in satin finish. Unit has shelf 8 inches deep with shelf support brackets of satin finish stainless steel welded to mounting base, and a minimum of 3 hooks/3 holders. Mop holders have spring-loaded rubber cams and hold mop or broom handle with a diameter between 5/8 inch and 1 inch.

Approximate size: 36 inches wide by 8 inches deep.

2.1.13 Item A5145 Hook, Garment, Double, SS, Surface Mounted

Surface mounted double garment hook (D-TP1, T-CH1, T-CH2) made of stainless steel with satin finish. For use on door back or wall. Hook comes with concealed mounting bracket secured to concealed wall plate. Mounting hardware included. Flange size is approximately 2 inches by 2 inches.

2.1.14 Item A5170 Rod, Shower Curtain, 1 Inch Diameter, W/Curtain & Hooks

Shower Curtain Rod with concealed mounting (S11). Shower curtain rod made of satin finish stainless steel, 1 inch diameter, with flanges included, and have white vinyl shower curtain, 72 inches high, and stainless steel curtain hooks. Shower curtain has corrosion resistant grommets, reinforced heading, and treated with antibacterial and flame retardant agents. Shower hooks are stainless steel. Length as indicated on drawings.

2.1.15 Item A5195 Dispenser, Toilet Tissue, SS, 1-Roll, Surface Mounted

Concealed surface mounted single roll toilet tissue dispenser (D-TTD) of satin finish stainless steel. Spindle to be free-spinning for non-controlled delivery, chrome-plated high impact resistant plastic and equipped with heavy-duty internal spring. Unit accommodates standard core toilet paper roll up to 5-1/2 inches diameter. Mounting hardware included.

Approximate size: 7-3/4 inches wide by 2 inches high by 4 inches deep.

2.1.16 Item A5202 Dispenser, Toilet Paper w/Utility Shelf, SS, 2-Roll

Concealed surface mounted, double roll, toilet tissue dispenser (T-TTD) and utility shelf of satin finish stainless steel. Mounting brackets to be welded to shelf. Unit holds two standard 5-1/4 inch diameter rolls of toilet tissue. Spindles are free-spinning for non-controlled delivery, high impact-resistant plastic equipped with internal springs. Edges of shelf is 1/2 inch, with hemmed lip on front edge for safety.

Approximate size of shelf: 18 inches wide by 5 inches deep.

PART 3 EXECUTION

3.1 INSTALLATION

Do not install items that show visual evidence of biological growth. Provide the same finish for the surfaces of fastening devices exposed after installation as the attached accessory. Provide oval exposed screw heads. Install accessories at the location and height indicated. Protect exposed surfaces of accessories with strippable plastic or by other means until the installation is accepted. After acceptance of accessories, remove and dispose of strippable plastic protection. Coordinate accessory manufacturer's mounting details with other trades as their work progresses. Use sealants for brackets, plates, anchoring devices and similar items in showers (a silicone sealantsealant specified in Section 07 92 00 JOINT SEALANTS) as they are set to provide a watertight installation. After installation, thoroughly clean exposed surfaces and restore damaged work to its original condition or replace with new work.

3.1.1 Recessed Accessories

Fasten accessories with wood screws to studs, blocking or rough frame in wood construction. Set anchors in mortar in masonry construction. Fasten to metal studs or framing with sheet metal screws in metal construction.

3.1.2 Surface Mounted Accessories

Mount on concealed backplates, unless specified otherwise. Conceal fasteners on accessories without backplates. Install accessories with corrosion-resistant fasteners as required by the construction. Install backplates in the same manner, or provide with lugs or anchors set in mortar, as required by the construction. Fasten accessories mounted on gypsum board and plaster walls without solid backing into the metal or wood studs, or to backplates secured to metal studs.

3.2 **CLEANING**

Clean material in accordance with manufacturer's recommendations. Do not use alkaline or abrasive agents. Take precautions to avoid scratching or

marring exposed surfaces.

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DIVISION 11 - EQUIPMENT

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ELECTRIC KITCHEN EQUIPMENT

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PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2026; TIA 26-1; ERTA 26-1; TIA 26-2; TIA 26-3; TIA 26-4; TIA 26-5; TIA 26-6; TIA 26-7; ERTA 26-2; ERTA 26-3) National Electrical Code

UL SOLUTIONS (UL)

UL 858 (2014; Reprint Jun 2025) UL Standard for Safety Household Electric Ranges

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Kitchen Equipment

SD-08 Manufacturer's Instructions

Kitchen Equipment

Exhaust Hood

SD-10 Operation and Maintenance Data

Kitchen Equipment, Data Package 2; G,

Submit in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

SD-11 Closeout Submittals

Manufacturer's Warranty

Contractor's Warranty for Installation

PART 2 PRODUCTS

2.1 KITCHEN EQUIPMENT

2.1.1 Materials

Except as modified herein, provide manufacturer's standard materials for kitchen equipment. Provide quantities, physical dimensions, colors, and electrical characteristics as indicated.

2.1.2 Cooking Top

UL 858, GFGI, spill catching, seamless, with tubular plug-in surface elements. Provide indicating "on" lights. Cook top shall be stainless steel, 30-inch nomial width, with four (4) burners. Install in accordance with manufacturer's written instructions, coordinated with adjacent countertop and cabinetry for secure, flush fit and proper clearances.

2.1.3 Range Hood

UL 858, GFGI, nonvented, with two-speed fan, permanent washable filter, top exhaust, and eye level controls. Unit shall be stainless steel, 30-inch nominal width and include factory-installed lighting with incandescent or LED lamp. Install in accordance with manufacturer's written instructions, securely fastened and coordinated with electric cook top.

PART 3 EXECUTION

3.1 INSTALLATION

Do not install items that show visual evidence of biological growth.

NFPA 70, Section 22 00 00 PLUMBING, GENERAL PURPOSE and Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Install kitchen equipment in accordance with manufacturers' instructions.

3.2 FIELD QUALITY CONTROL

Conduct inspection and testing in the presence of the Contracting Officer.

3.2.1 Field Inspection

Before and after installation, inspect each piece of kitchen equipment for compliance with specified requirements.

3.2.2 Operation Tests

Upon completion, but before final acceptance, perform operation tests on each piece of equipment to determine that components, including controls, safety devices, and attachments, operate properly and in accordance with specified requirements.

3.3 MANUFACTURER'S WARRANTY

Submit all manufacturers' signed warranties to Contracting Officer prior to final commissioning and acceptance.

3.4 CONTRACTOR'S WARRANTY FOR INSTALLATION

Submit contractor's warranty for installation to the Contracting Officer prior to final commissioning and acceptance.

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DIVISION 21 - FIRE SUPPRESSION

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WET PIPE SPRINKLER SYSTEMS, FIRE PROTECTION

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SECTION 21 13 13

WET PIPE SPRINKLER SYSTEMS, FIRE PROTECTION
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DELEGATED DESIGN

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.1	(2020) Gray Iron Pipe Flanges and Flanged Fittings Classes 25, 125, and 250
ASME B16.3	(2021) Malleable Iron Threaded Fittings, Classes 150 and 300
ASME B16.4	(2021) Gray Iron Threaded Fittings; Classes 125 and 250
ASME B16.21	(2021) Nonmetallic Flat Gaskets for Pipe Flanges

AMERICAN SOCIETY OF SANITARY ENGINEERING (ASSE)

ASSE 1013	(2021) Performance Requirements for Reduced Pressure Principle Backflow Prevention Assemblies
ASSE 1015	(2021) Performance Requirements for Double Check Backflow Prevention Assemblies

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C104/A21.4	(2022) Cement-Mortar Lining for Ductile-Iron Pipe and Fittings for Water
AWWA C110/A21.10	(2021) Ductile-Iron and Gray-Iron Fittings
AWWA C111/A21.11	(2017) Rubber-Gasket Joints for Ductile-Iron Pressure Pipe and Fittings
AWWA C203	(2020) Coal-Tar Protective Coatings and Linings for Steel Water Pipelines - Enamel and Tape - Hot-Applied
AWWA M14	(2024) Manual: Recommended Practice for Backflow Prevention and Cross-Connection Control

ASTM INTERNATIONAL (ASTM)

ASTM A47/A47M	(1999; R 2022; E 2022) Standard Specification for Ferritic Malleable Iron Castings
ASTM A53/A53M	(2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A135/A135M	(2021) Standard Specification for Electric-Resistance-Welded Steel Pipe
ASTM A153/A153M	(2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware
ASTM A183	(2014; R 2020) Standard Specification for Carbon Steel Track Bolts and Nuts
ASTM A536	(1984; R 2019; E 2019) Standard Specification for Ductile Iron Castings

FM GLOBAL (FM)

FM APP GUIDE	(updated on-line) Approval Guide http://www.approvalguide.com/

INTELLIGENCE COMMUNITY STANDARD (ICS)

ICS 705-1	(2010) Physical and Technical Security Standard for Sensitive Compartmented Information Facilities

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

MSS SP-71	(2018) Gray Iron Swing Check Valves, Flanged and Threaded Ends

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 13	(2022; TIA 23-4) Standard for the Installation of Sprinkler Systems
NFPA 13R	(2022; ERTA 1 2022) Standard for the Installation of Sprinkler Systems in Low-Rise Residential Occupancies
NFPA 24	(2022) Standard for the Installation of Private Fire Service Mains and Their Appurtenances
NFPA 101	(2021; TIA 21-1) Life Safety Code
NFPA 291	(2022) Recommended Practice for Fire Flow Testing and Marking of Hydrants
NFPA 1963	(2019) Standard for Fire Hose Connections

NATIONAL INSTITUTE FOR CERTIFICATION IN ENGINEERING TECHNOLOGIES (NICET)

NICET 1014-7	(2012) Program Detail Manual for Certification in the Field of Fire Protection Engineering Technology (Field Code 003) Subfield of Automatic Sprinkler System Layout

UNDERWRITERS LABORATORIES (UL)

UL 199	(2020) UL Standard for Safety Automatic Sprinklers for Fire-Protection Service
UL 262	(2004; Reprint Jul 2023) Gate Valves for Fire-Protection Service
UL 312	(2022) UL Standard for Safety Check Valves for Fire-Protection Service
UL 405	(2013; Bul. 2020) UL Standard for Safety Fire Department Connection Devices
UL 668	(2004; Reprint Oct 2021) UL Standard for Safety Hose Valves for Fire-Protection Service
UL 789	(2004; Reprint May 2017) UL Standard for Safety Indicator Posts for Fire-Protection Service
UL 1626	(2008; Bul. 2018) UL Standard for Safety Residential Sprinklers for Fire-Protection Service
UL Fire Prot Dir	UL Product IQ (updated online) at https://productiq.ulpropsector.com/en

1.2 SYSTEM DESCRIPTION

Provide wet pipe sprinkler system(s) in all areas of the building. Except as modified herein, the system must meet the requirements of NFPA 13. Pipe sizes which are not indicated on the Contract drawings must be determined by hydraulic calculations.

1.2.1 Hydraulic Design

1.2.1.1 Basis for Calculations

A waterflow test was performed on 26 MAR 2024 at Fort Campbell and resulted in a static pressure of 54 psi with a residual pressure of 49 psi while flowing 1030 gpm. Perform a fire hydrant flow test prior to shop drawing submittal in accordance with NFPA 291. Results must include hydrant elevations relative to the building and hydrant number/identifiers for the tested hydrants, including which were flowed, which had a gauge. This information must be presented in a tabular form if multiple hydrants were flowed. The results must be included with the hydraulic calculations. Hydraulic calculations must be based on flow test noted in this paragraph, unless approved by Contracting Officer. Hydraulic calculations must be based upon the Hazen-Williams formula with a "C" value noted in NFPA 13 for piping. Hydraulic calculations must be based on operation of the fire pump(s) provided in Section 21 30 00 FIRE PUMPS. The minimum residual pressure in a service lateral (lead-in) at the 150 percent of the fire pump rated flow must be 20 psi at the inlet to the backflow preventer.

1.2.1.2 Hydraulic Calculations

a. Water supply curves and system requirements must be plotted on semi-logarithmic graph (N^1.85) paper so as to present a summary of the complete hydraulic calculation.

b. Provide a summary sheet listing sprinklers in the design area and their respective hydraulic reference points, elevations, minimum discharge pressures and minimum flows. Elevations of hydraulic reference points (nodes) must be indicated.

c. Documentation must identify each pipe individually and the nodes connected thereto. Indicate the diameter, length, flow, velocity, friction loss, number and type fittings, total friction loss in the pipe, equivalent pipe length and Hazen-Williams coefficient for each pipe.

d. Where the sprinkler system is supplied by interconnected risers, the sprinkler system must be hydraulically calculated using the hydraulically most demanding single riser. The calculations must not assume the simultaneous use of more than one riser.

e. All calculations must include the backflow preventer manufacturer's stated friction loss at the design flow or 8 psi for double check backflow preventer, whichever is greater.

f. All calculations must be performed back to the actual location of the flow test, taking into account the direction of flow in the service main at the test location.

g. For gridded systems, calculations must show peaking of demand area friction loss to verify that the hydraulically most demanding area is being used. A flow diagram indicating the quantity and direction of flows must be included.

1.2.1.3 Design Criteria

Hydraulically design the system to discharge a minimum density as indicated on the drawings. Hydraulic calculations must be in accordance with the Area/Density Method of NFPA 13. Add an allowance for exterior hose streams as shown on the drawings to the sprinkler system demand at the fire hydrant shown on the drawings closest to the point where the water service enters the building.

1.2.2 Sprinkler Coverage

Sprinklers must be uniformly spaced on branch lines. Provide coverage throughout 100 percent of the building. This includes, but is not limited to, telephone rooms, electrical equipment rooms (regardless of the fire resistance rating of the enclosure), boiler rooms, switchgear rooms, transformer rooms, attached electrical vaults and other electrical and mechanical spaces. Coverage per sprinkler must be in accordance with NFPA 13. Provide sprinklers below all obstructions in accordance with NFPA 13. Exceptions are as follows:

a. Sprinklers may be omitted from small rooms which are exempted for specific occupancies in accordance with NFPA 101.

b. Facilities that are designed in accordance with NFPA 13R.

1.2.3 Qualified Fire Protection Engineer (QFPE)

An individual who is a licensed professional engineer (P.E.) who has passed the fire protection engineering written examination administered by the National Council of Examiners for Engineering and Surveying (NCEES) and has relevant fire protection engineering experience. Services of the QFPE must include:

a. Reviewing SD-02, SD-03, and SD-05 submittal packages for completeness and compliance with the provisions of this specification. Working (shop) drawings and calculations must be prepared by, or prepared under the immediate supervision of, the QFPE. The QFPE must affix their professional engineering stamp with signature to the shop drawings, calculations, and material data sheets, indicating approval prior to submitting the shop drawings to the DFPE.

b. Provide a letter documenting that the SD-02, SD-03, and SD-05 submittal package has been reviewed and noting all outstanding comments.

c. Performing in-progress construction surveillance prior to installation of ceilings (rough-in inspection).

d. Witnessing pre-Government and final Government functional performance testing and performing a final installation review.

e. Signing applicable certificates under SD-07.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Partial submittals and submittals not fully complying with NFPA 13 and this specification section must be returned disapproved without review. SD-02, SD-03 and SD-05 must be submitted simultaneously.

Shop drawings (SD-02), product data (SD-03) and calculations (SD-05) must be prepared by the designer and combined and submitted as one complete package. The QFPE must review the SD-02/SD-03/SD-05 submittal package for completeness and compliance with the Contract provisions prior to submission to the Government. The QFPE must provide a Letter of Confirmation that they have reviewed the submittal package for compliance with the contract provisions. This letter must include their professional engineer stamp and signature. Partial submittals and submittals not reviewed by the QFPE must be returned disapproved without review.

Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualified Fire Protection Engineer (QFPE); G, AE

Sprinkler System Designer; G, AE

Sprinkler System Installer; G, AE

SD-02 Shop Drawings

Shop Drawing; G, AE

SD-03 Product Data

Pipe; G, AE

Fittings; G, AE

Valves, including gate, check, butterfly, and globe; G, AE

Relief Valves; G, AE

Sprinklers; G, AE

Pipe Hangers and Supports; G, AE

Sprinkler Alarm Switch; G, AE

Valve Supervisory (Tamper) Switch; G, AE

Fire Department Connection; G, AE

Backflow Prevention Assembly; G,

Air Vent; G,

Hose Valve; G, AE

Seismic Bracing; G, AE

Nameplates; G, AE

SD-05 Design Data

Seismic Bracing; G, AE

Load calculations for sizing of seismic bracing

Hydraulic Calculations; G, AE

SD-06 Test Reports

Test Procedures; G, AE

SD-07 Certificates

Verification of Compliant Installation; G, AE

Request for Government Final Test; G, AE

SD-10 Operation and Maintenance Data

Operating and Maintenance (O&M) Instructions; G

Spare Parts Data; G

SD-11 Closeout Submittals

As-built drawings

1.4 QUALITY ASSURANCE

1.4.1 Preconstruction Submittals

Within 36 days of contract award but no less than 14 days prior to commencing work on site, the prime Contractor must submit the following for review and approval. SD-02, SD-03 and SD-05 submittals received prior to the review and approval of the qualifications will be returned Disapproved Without Review.

1.4.1.1 Shop Drawing

3 copies of the shop drawings, no later than 28 days prior to the start of system installation. Working drawings conforming to the requirements prescribed in NFPA 13 and must be no smaller than ANSI D or the Contract Drawings. Each set of drawings must include the following:

a. A descriptive index with drawings listed in sequence by number. A legend sheet identifying device symbols, nomenclature, and conventions used in the package.

b. Floor plans drawn to a scale not less than 1/8-inch equals 1-foot clearly showing locations of devices, equipment, risers, and other details required to clearly describe the proposed arrangement.

c. Actual center-to-center dimensions between sprinklers on branch lines and between branch lines; from end sprinklers to adjacent walls; from walls to branch lines; from sprinkler feed mains, cross mains and branch lines to finished floor and roof or ceiling. A detail must show the dimension from the sprinkler and sprinkler deflector to the ceiling in finished areas.

d. Longitudinal and transverse building sections showing typical branch line and cross main pipe routing, elevation of each typical sprinkler above finished floor and elevation of "cloud" or false ceilings in relation to the building ceilings.

e. Plan and elevation views which establish that the equipment will fit the allotted spaces with clearance for installation and maintenance.

f. Riser layout drawings drawn to a scale of not less than 1/2-inch equals 1-foot to show details of each system component, clearances between each other and from other equipment and construction in the room.

g. Details of each type of riser assembly, pipe hanger, sway bracing for earthquake protection, and restraint of underground water main at point-of-entry into the building, and electrical devices and interconnecting wiring. The dimension from the edge of vertical piping to the nearest adjacent wall(s) must be indicated on the drawings when vertical piping is located in stairs or other portions of the means of egress.

h. Details of each type of pipe hanger, seismic bracing/restraint and related components.

i. Include fire pump curve with shop drawings and hydraulic calculations.

1.4.1.2 Product Data

3 copies of annotated catalog data to show the specific model, type, and size of each item. Catalog cuts must also indicate the NRTL listing. The data must be highlighted to show model, size, options, and other pertinent information, that are intended for consideration. Data must be adequate to demonstrate compliance with all contract requirements. Product data for all equipment must be combined into a single submittal.

1.4.1.3 Hydraulic Calculations

Calculations must be as outlined in NFPA 13 except that calculations must be performed by computer using software intended specifically for fire protection system design using the design data shown on the drawings. Calculations must include isometric diagram indicating hydraulic nodes and pipe segments. Include fire pump curve with submittal.

1.4.1.4 Operating and Maintenance (O&M) Instructions

Submit in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA as supplemented and modified by this specification section.

Provide six manuals and one pdf version on electronic media. The manuals must include the manufacturer's name, model number, parts list, list of parts and tools that should be kept in stock by the owner for routine maintenance, troubleshooting guide, and recommended service organization (including address and telephone number) for each item of equipment. Each service organization submitted must be capable of providing 4-hour on-site response to a service call on an emergency basis.

Submit spare parts data for each different item of material and equipment specified. The data must include a complete list of parts and supplies, and a list of parts recommended by the manufacturer to be replaced after 1-year and 3 years of service. Include a list of special tools and test equipment required for maintenance and testing of the products supplied.

1.4.2 Qualifications

1.4.2.1 Sprinkler System Designer

The sprinkler system designer must be certified as a Level III Technician by National Institute for Certification in Engineering Technologies (NICET) in the Water-Based Systems Layout subfield of Fire Protection Engineering Technology in accordance with NICET 1014-7.

1.4.2.2 Sprinkler System Installer

The sprinkler system installer must be regularly engaged in the installation of the type and complexity of system specified in the contract documents, and must have served in a similar capacity for at least three systems that have performed in the manner intended for a period of not less than 6 months.

1.4.3 Regulatory Requirements

Equipment and material must be listed or approved. Listed or approved, as used in this Section, means listed, labeled or approved by a Nationally Recognized Testing Laboratory (NRTL) such as UL Fire Prot Dir or FM APP GUIDE. The omission of these terms under the description of an item or equipment described must not be construed as waiving this requirement. All listings or approvals by testing laboratories must be from an existing ANSI or UL published standard. The recommended practices stated in the manufacturer's literature or documentation are mandatory requirements.

1.5 DELIVERY, STORAGE, AND HANDLING

Protect all equipment delivered and placed in storage from the weather, excessive humidity and temperature variations, dirt and dust, or other contaminants. All pipes must be either capped or plugged until installed.

1.6 EXTRA MATERIALS

Spare sprinklers and wrench(es) must be provided as spare parts in accordance with NFPA 13.

PART 2 PRODUCTS

2.1 MATERIALS AND EQUIPMENT

2.1.1 Standard Products

Provide materials, equipment, and devices listed for fire protection service when so required by NFPA 13 or this specification. Select material from one manufacturer, where possible, and not a combination of manufacturers, for a classification of material. Material and equipment must be standard products of a manufacturer regularly engaged in the manufacture of the products for at least 2 years prior to bid.

2.1.2 Nameplates

Major components of equipment must have the manufacturer's name, address, type or style, model or serial number, catalog number, date of installation, installing Contractor's name and address, and the contract number provided on a new name plate permanently affixed to the item or equipment. Nameplates must be etched metal or plastic, permanently attached by screws to control units, panels or adjacent walls.

2.1.3 Identification and Marking

Pipe and fitting markings must include name or identifying symbol of manufacturer and nominal size. Pipe must be marked with ASTM designation. Valves and equipment markings must have name or identifying symbol of manufacturer, specific model number, nominal size, name of device, arrow indicating direction of flow, and position of installation (horizontal or vertical), except if valve can be installed in either position. Markings must be included on the body casting or on an etched or stamped metal nameplate permanently on the valve or cover plate.

2.1.4 Pressure Ratings

Valves, fittings, couplings, alarm switches, and similar devices must be rated for the maximum working pressures that can be experienced in the system, but in no case less than 175 psi.

2.2 UNDERGROUND PIPING COMPONENTS

2.2.1 Pipe

Pipe must comply with NFPA 24. Minimum pipe size is 6 inches. Piping more than 5 feet outside the building walls must comply with Section 33 11 00 WATER UTILITY DISTRIBUTION PIPING. A continuous section of welded stainless steel fire water service piping from a point outside the building perimeter to a flanged fitting at least 1-foot above the finished floor within the building is acceptable.

2.2.2 Fittings and Gaskets

Fittings must be ductile-iron conforming to AWWA C110/A21.10 with cement mortar lining conforming to AWWA C104/A21.4. Gaskets must be suitable in design and size for the pipe with which such gaskets are to be used. Gaskets for ductile-iron pipe joints must conform to AWWA C111/A21.11.

2.2.3 Gate Valve and Indicator Posts

Installation must comply with NFPA 24. Gate valves for use with indicator post must conform to UL 262. Indicator posts must conform to UL 789. Provide each indicator post with one coat of primer and two coats of red enamel paint.

2.2.4 Valve Boxes

Except where indicator posts are provided, for each buried valve, provide a cast-iron valve box of a suitable size. Provide cast-iron cover for valve box with the word "WATER" cast on the cover. The minimum box shaft diameter must be 5.25 inches. Coat cast-iron boxes with bituminous paint applied to a minimum dry-film thickness of 10 mils. Provide a valve box and stem as indicated on the drawings.

2.2.5 Buried Utility Warning and Identification Tape

Provide detectable aluminum foil plastic backed tape or detectable magnetic plastic tape manufactured specifically for warning and identification of buried piping. Tape must be detectable by an electronic detection instrument. Provide tape, 3 inches minimum width, color coded for the utility involved with warning and identification imprinted in bold block letters continuously and repeatedly over the entire tape length. Warning and identification must read "CAUTION BURIED WATER PIPING BELOW" or similar wording. Use permanent code and letter coloring unaffected by moisture and other substances contained in trench backfill material.

2.3 ABOVEGROUND PIPING COMPONENTS

2.3.1 Steel Piping Components

2.3.1.1 Steel Pipe

Except as modified herein, steel pipe must be black as permitted by NFPA 13 and conform to the applicable provisions of ASTM A53/A53M, ASTM A135/A135M or ASTM A153/A153M.

Steel pipe must be minimum Schedule 40 for sizes 2 inches and less; and minimum Schedule 10 for sizes larger than 2 inches. Steel piping with wall thickness less than Schedule 40 must not be threaded.

2.3.1.2 Fittings

Fittings must be welded, threaded, or grooved-end type. Threaded fittings must be cast-iron conforming to ASME B16.4, malleable-iron conforming to ASME B16.3 or ductile-iron conforming to ASTM A536. Plain-end fittings with mechanical couplings, fittings that use steel gripping devices to bite into the pipe, steel press fittings and field welded fittings are not permitted. Fittings, mechanical couplings, and rubber gaskets must be supplied by the same manufacturer. Threaded fittings must use Teflon tape or manufacturer's approved joint compound. Reducing couplings are not permitted except as allowed by NFPA 13.

2.3.1.3 Grooved Mechanical Joints and Fittings

Joints and fittings must be designed for not less than 175 psi service and the product of the same manufacturer. Field welded fittings must not be used. Fitting and coupling housing must be malleable-iron conforming to ASTM A47/A47M, Grade 32510; ductile-iron conforming to ASTM A536, Grade 65-45-12. Rubber gasketed grooved-end pipe and fittings with mechanical couplings are permitted in pipe sizes 2 inches and larger. Gasket must be the flush type that fills the entire cavity between the fitting and the pipe. Nuts and bolts must be heat-treated steel conforming to ASTM A183 and must be cadmium-plated or zinc-electroplated.

2.3.1.4 Flanges

Flanges must conform to NFPA 13 and ASME B16.1. Gaskets must be non-asbestos compressed material in accordance with ASME B16.21, 1/16-inch thick, and full face or self-centering flat ring type.

2.3.2 Flexible Sprinkler Hose

The use of flexible hose is not permitted.

2.3.3 Pipe Hangers and Supports

Provide galvanized pipe hangers, supports and seismic bracing in accordance with NFPA 13. Design and install seismic protection in accordance with the requirements of NFPA 13 section titled "Protection of Piping Against Damage Where Subject to Earthquakes for Seismic Design Category "C".

2.3.4 Valves

Provide valves of types approved for fire service. Valves must open by counterclockwise rotation.

2.3.4.1 Control Valve

Manually operated sprinkler control/gate valve must be outside stem and yoke (OS&Y) type or butterfly type as indicated on the drawings and must be listed.

2.3.4.2 Check Valves

Check valves must comply with UL 312. Check valves 4 inches and larger must be of the swing type, have a clear waterway and meet the requirements of MSS SP-71, for Type 3 or 4. Inspection plate must be provided on valves larger than 6 inches.

2.3.4.3 Hose Valve

Valve must comply with UL 668.

2.3.5 Riser Check Valves

Provide riser check valve, pressure gauges and main drain.

2.4 ALARM INITIATING AND SUPERVISORY DEVICES

2.4.1 Sprinkler Alarm Switch

Vane or pressure-type flow switch(es). Connection of switch must be by the fire alarm installer. Vane type alarm actuating devices must have mechanical diaphragm controlled retard device adjustable from 10 to 60 seconds and must instantly recycle. Flow switches for elevator power shunt must not have a retard feature.

2.4.2 Valve Supervisory (Tamper) Switch

Switch must be integral to the control valve or suitable for mounting to the type of control valve to be supervised open. The switch must be tamper resistant and contain SPDT (Form C) contacts arranged to transfer upon removal of the housing cover or closure of the valve of more than two rotations of the valve stem.

2.5 BACKFLOW PREVENTION ASSEMBLY

Double-check valve assembly backflow preventer complying with ASSE 1013, ASSE 1015, AWWA M14, and as indicated on the drawings. Each check valve must have a drain. Backflow prevention assemblies must have current "Certificate of Approval from the Foundation for Cross-Connection Control and Hydraulic Research, FCCCHR List" and be listed for fire protection use. Listing of the specific make, model, design, and size in the FCCCHR List is acceptable as the required documentation.

2.5.1 Backflow Preventer Test Connection

Test connection must consist of a series of listed hose valves with 2 1/2-inch National Standard male hose threads with cap and chain.

2.6 FIRE DEPARTMENT CONNECTION

Fire department connection must be projecting type with cast-brass body, matching wall escutcheon lettered "Auto Spkr" with a polished-brass or chromium-plated finish. The connection must have individual self-closing clappers, caps with drip drains and chains. Female inlets must have 2 1/2-inch diameter American National Fire Hose Connection Screw Threads (NH) per NFPA 1963. Comply with UL 405.

2.7 SPRINKLERS

Sprinklers must comply with UL 199 and NFPA 13. Sprinklers with internal O-rings are not acceptable. Sprinklers in high heat areas including attic spaces or in close proximity to unit heaters must have temperature classification in accordance with NFPA 13. Extended coverage sprinklers are permitted for loading docks, residential occupancies and high-piled storage applications only.

2.7.1 Pendent Sprinkler

Pendent sprinkler must be recessed quick-response type with nominal K-factor of 5.6 and 8.0. Pendent sprinklers must have a polished chrome finish. Assembly must include an integral escutcheon.

2.7.2 Upright Sprinkler

Upright sprinkler must be chrome-plated quick-response typeand have a nominal K-factor of 8.0.

Amendment 0001 **

2.7.3 **Sidewall Sprinkler**

Sidewall sprinkler must be the quick-response type. Sidewall sprinkler must have a nominal K-factor of 8.0. Sidewall sprinkler must have a polished-chrome finish.

** **Amendment 0001**

2.7.4 Concealed Sprinkler

Concealed sprinkler must be chrome-plated and have a nominal K-factor of 4.2, 5.6, or 8.0. Coverplate must be chrome.

2.7.5 Residential Sprinkler

Residential sprinkler must be recessed pendent type with nominal K-factor of 4.2 or 5.6. Residential sprinkler must have a polished-chrome finish. Sprinkler must comply with UL 1626.

2.8 ACCESSORIES

2.8.1 Sprinkler Cabinet

Provide spare sprinklers in accordance with NFPA 13 and must be placed in a suitable metal or plastic cabinet of sufficient size to accommodate all the spare sprinklers and wrenches in designated locations. Spare sprinklers must be representative of, and in proportion to, the number of each type and temperature rating of the sprinklers installed as required by NFPA 13. At least one wrench of each type required must be provided.

2.8.2 Pendent Sprinkler Escutcheon

Escutcheon must be one-piece metallic type with a depth of less than 3/4-inch and suitable for installation on pendent sprinklers. The escutcheon must have a factory finish that matches the pendent sprinkler.

2.8.3 Pipe Escutcheon

Provide split hinge metal plates for piping entering walls, floors, and ceilings in exposed spaces. Provide polished stainless steel plates or chromium-plated finish on copper alloy plates in finished spaces. Provide paint finish on metal plates in unfinished spaces.

2.8.4 Sprinkler Guard

Listed guard must be a steel wire cage designed to encase the sprinkler and protect it from mechanical damage. Guards must be provided on sprinklers located within 7 feet of the floor.

2.8.5 Relief Valve

Relief valves must be listed and installed at the riser in accordance with NFPA 13.

2.8.6 Air Vent

Air vents must be of the automatic type and piped to drain to the building exterior.

2.8.7 Identification Sign

Valve identification sign must be minimum 6 inches wide by 2 inches high with enamel baked finish on minimum 18 gage steel or 0.024-inch aluminum with red letters on a white background or white letters on red background. Wording of sign must include, but not be limited to "main drain", "auxiliary drain", "inspector's test", "alarm test", "alarm line", and similar wording as required to identify operational components. Where there is more than one sprinkler system, signage must include specific details as to the respective system.

PART 3 EXECUTION

3.1 VERIFYING ACTUAL FIELD CONDITIONS

Before commencing work, examine all adjoining work on which the contractor's work that is dependent for perfect workmanship according to the intent of this specification section, and report to the Contracting Officer's Representative a condition that prevents performance of first class work. No "waiver of responsibility" for incomplete, inadequate or defective adjoining work will be considered unless notice has been filed before submittal of a proposal.

3.2 INSTALLATION

The installation must be in accordance with the applicable provisions of NFPA 13, NFPA 24 and publications referenced therein. Locate sprinklers in a consistent pattern with ceiling grid, lights, and air supply diffusers. Install sprinkler system over and under ducts, piping and platforms when such equipment can negatively affect or disrupt the sprinkler discharge pattern and coverage.

a. Piping offsets, fittings, and other accessories required must be furnished to provide a complete installation and to eliminate interference with other construction.

b. Wherever the contractor's work interconnects with work of other trades the Contractor must coordinate with other Contractors to insure all Contractors have the information necessary so that they may properly install all necessary connections and equipment. Identify all work items needing access (dampers and similar equipment) that are concealed above hung ceilings by permanent color coded pins/tabs in the ceiling directly below the item.

c. Provide required supports and hangers for piping, conduit, and equipment so that loading will not exceed allowable loadings of structure. Submittal of a bid must be a deemed representation that the contractor submitting such bid has ascertained allowable loadings and has included in his estimates the costs associated in furnishing required supports.

3.2.1 Waste Removal

At the conclusion of each day's work, clean up and stockpile on site all waste, debris, and trash which may have accumulated during the day as a result of work by the contractor and of his presence on the job. Sidewalks and streets adjoining the property must be kept broom clean and free of waste, debris, trash and obstructions caused by work of the contractor, which will affect the condition and safety of streets, walks, utilities, and property.

3.3 UNDERGROUND PIPING INSTALLATION

The fire protection water main must be laid, and joints anchored, in accordance with NFPA 24. Minimum depth of cover must be 3 feet or the frost line, whichever is deeper. The supply line must terminate inside the building with a flanged piece, the bottom of which must be set not less than 1-foot above the finished floor. A blind flange must be installed temporarily on top of the flanged piece to prevent the entrance of foreign matter into the supply line. A concrete thrust block must be provided at the elbow where the pipe turns up toward the floor. In addition, joints must be anchored in accordance with NFPA 24. Buried steel components must be provided with a corrosion protective coating in accordance with AWWA C203. Piping more than 5 feet outside the building walls must meet the requirements of Section 33 11 00 WATER UTILITY DISTRIBUTION PIPING.

3.4 ABOVEGROUND PIPING INSTALLATION

The methods of fabrication and installation of the aboveground piping must fully comply with the requirements and recommended practices of NFPA 13 and this specification section.

3.4.1 Protection of Piping Against Earthquake Damage

Seismic restraint is required.

3.4.2 Piping in Exposed Areas

Install exposed piping without diminishing exit access widths, corridors or equipment access. Exposed horizontal piping, including drain piping, must be installed to provide maximum headroom.

3.4.3 Piping in Finished Areas

In areas with suspended or dropped ceilings and in areas with concealed spaces above the ceiling, piping must be concealed above ceilings. Piping must be inspected, hydrostatically tested and approved before being concealed. Risers and similar vertical runs of piping in finished areas must be concealed.

3.4.4 Pendent Sprinklers

a. Drop nipples to pendent sprinklers must consist of minimum 1-inch pipe with a reducing coupling into which the sprinkler must be threaded.

b. Where sprinklers are installed below suspended or dropped ceilings, drop nipples must be cut such that sprinkler ceiling plates or escutcheons are of a uniform depth throughout the finished space. The outlet of the reducing coupling must not extend below the underside of the ceiling.

c. Recessed pendent sprinklers must be installed such that the distance from the sprinkler deflector to the underside of the ceiling must not exceed the manufacturer's listed range and must be of uniform depth throughout the finished area.

d. Pendent sprinklers in suspended ceilings must be located in the center of the tile (plus or minus 2 inches).

e. Where the maximum static or flowing pressure, whichever is greater at the sprinkler, applied other than through the fire department connection, exceeds 100 psi and a branch line above the ceiling supplies sprinklers in a pendent position below the ceiling, the cumulative horizontal length of an unsupported armover to a sprinkler or sprinkler drop must not exceed 12 inches for steel pipe and 6 inches for copper tube.

3.4.5 Upright Sprinklers

Riser nipples or "sprigs" to upright sprinklers must contain no fittings between the branch line tee and the reducing coupling at the sprinkler.

3.4.6 Pipe Joints

Pipe joints must conform to NFPA 13, except as modified herein. Not more than four threads must show after joint is made up. Welded joints will be permitted, only if welding operations are performed as required by NFPA 13 at the Contractor's fabrication shop, not at the project construction site. Flanged joints must be provided where indicated or required by NFPA 13. Grooved pipe and fittings must be prepared in accordance with the manufacturer's latest published specification according to pipe material, wall thickness and size. Grooved couplings, fittings and grooving tools must be products of the same manufacturer. For copper tubing, pipe and groove dimensions must comply with the tolerances specified by the coupling manufacturer. The diameter of grooves made in the field must be measured using a "go/no-go" gauge, vernier or dial caliper, narrow-land micrometer, or other method specifically approved by the coupling manufacturer for the intended application. Groove width and dimension of groove from end of pipe must be measured and recorded for each change in grooving tool setup to verify compliance with coupling manufacturer's tolerances.

3.4.7 Reducers

Reductions in pipe sizes must be made with one-piece tapered reducing fittings. When standard fittings of the required size are not manufactured, single bushings of the face or hex type will be permitted. Where used, face bushings must be installed with the outer face flush with the face of the fitting opening being reduced. Bushings cannot be used in elbow fittings, in more than one outlet of a tee, in more than two outlets of a cross, or where the reduction in size is less than 1/2-inch.

3.4.8 Pipe Penetrations

a. Cutting structural members for passage of pipes or for pipe-hanger fastenings will not be permitted. Pipes that must penetrate concrete or masonry walls or concrete floors must be core-drilled and provided with pipe sleeves. Each sleeve must be Schedule 40 galvanized steel, ductile-iron or cast-iron pipe and extend through its respective wall or floor and be cut flush with each wall surface. Sleeves must provide required clearance between the pipe and the sleeve per NFPA 13. The space between the sleeve and the pipe must be firmly packed with mineral wool insulation.

b. Where pipes and sleeves penetrate fire walls, fire partitions, or floors, pipes/sleeves must be firestopped in accordance with Section 07 84 00 FIRESTOPPING.

c. In penetrations that are not fire-rated or not a floor penetration, the space between the sleeve and the pipe must be sealed at both ends with plastic waterproof cement that will dry to a firm but pliable mass or with a mechanically adjustable segmented elastomer seal.

d. All penetrations through the boundary of rooms/areas identified as secure space area must meet ICS 705-1.

3.4.9 Escutcheons

Escutcheons must be provided for pipe penetration in finished areas of ceilings, floors and walls. Escutcheons must be securely fastened to the pipe at surfaces through which piping passes.

3.4.10 Inspector's Test Connection

Unless otherwise indicated, the test connection must consist of 1-inch pipe connected to the remote branch line, as located on the drawings, and at the riser as a combination test and drain valve; a test valve located approximately 7 feet above the floor; a smooth bore brass outlet equivalent to the smallest orifice sprinkler used in the system; and a painted metal identification sign affixed to the valve with the words "Inspector's Test". All test connection piping must be inside of the building and penetrate the exterior wall at the location of the discharge orifice only. The discharge orifice must be located outside the building wall no more than 2 feet above finished grade, directed so as not to cause damage to adjacent construction or landscaping during full flow discharge, or to the sanitary sewer. Discharge to the exterior must not interfere with exiting from the facility. Water discharge or runoff must not cross the path of egress from the building. Do not discharge to the roof. Discharge to floor drains, janitor sinks or similar fixtures is not permitted.

Provide concrete splash blocks at all drain and inspector's test connection discharge locations if not discharging to a concrete surface. Splash blocks must be large enough to mitigate erosion and not become dislodged during a full flow of the drain. Ensure all discharged water drains away from the facility and does not cause property damage.

3.4.10.1 Access Panels

Access panels must be provided for concealed valves and controls, or any item requiring inspection or maintenance. Access panels must be of sufficient size and located so that the concealed items may be serviced, maintained, or replaced. Access panels must be as specified in Section 08 31 00 ACCESS DOORS AND PANELS.

3.4.11 Backflow Preventer

Locate within the building or in a heated enclosure in locations subject to freezing. For heated enclosures, provide a low temperature supervisory alarm connected to the facility fire alarm system. Heat trace is not permitted to be used.

Install backflow preventers so that the bottom of the assembly is a minimum of 6 inches above the finished floor/grade. Install horizontal backflow preventers so that the bottom of the assembly is no greater than 24 inches above the finished floor/grade. Clearance around control valve handles must be minimum 6 inches above grade/finished floor and away from walls.

3.4.11.1 Test Connection

Provide downstream of the backflow prevention assembly UL 668 hose valves with 2.5-inch National Standard male hose threads with cap and chain. Provide one valve for each 250 gpm of system demand or fraction thereof. Provide a permanent sign in accordance with paragraph entitled "Identification Signs" which reads, "Test Valve". Indicate location of test header. If an exterior connection, provide a control valve inside a heated mechanical room to prevent freezing.

3.4.12 Drains

a. Main drain piping must be provided to discharge at a safe point outside the building, no more than 2 feet above finished grade and at the location indicated on drawings. Provide a concrete splash block at drain outlet. Discharge to the exterior must not interfere with exiting from the facility. Water discharge or runoff must not cross the path of egress from the building.

b. Auxiliary drains must be provided as required by NFPA 13. Auxiliary drains are permitted to discharge to a floor drain if the drain is sized to accommodate full flow (min 40 gpm). Discharge to service sinks or similar plumbing fixtures is not permitted.

3.4.13 Installation of Fire Department Connection

Connection must be mounted on the exterior wall approximately 3 feet above finished grade. The piping between the connection and the check valve must be provided with an automatic drip in accordance with NFPA 13 and piped to drain to the outside or a floor drain within the same room.

3.4.14 Identification Signs

Signs must be affixed to each control valve, inspector test valve, main drain, auxiliary drain, test valve, and similar valves as appropriate or as required by NFPA 13. Main drain test results must be etched into main drain identification sign. Hydraulic design data must be etched into the nameplates and permanently affixed to each sprinkler riser as specified in NFPA 13. Provide labeling on the surfaces of all feed and cross mains to show the pipe function (e.g., "Sprinkler System", "Fire Department Connection", "Standpipe") and normal valve position (e.g. "Normally Open", "Normally Closed"). For pipe sizes 4-inch and larger provide white painted stenciled letters and arrows, a minimum of 2 inches in height and visible from at least two sides when viewed from the floor. For pipe sizes less than 4-inch, provide white painted stenciled letters and arrows, a minimum of 0.75-inch in height and visible from the floor. Provide properly lettered and approved metal sign to elevator flow switch stating the circuits' voltage, and identify the switch as an "Elevator Power Shunt Flow Switch".

3.5 ELECTRICAL

Except as modified herein, electric equipment and wiring must be in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Alarm signal wiring connected to the building fire alarm control system must be by the fire alarm installer.

3.6 PAINTING

Color code mark piping as specified in Section 09 90 00 PAINTS AND COATINGS.

3.7 FIELD QUALITY CONTROL

3.7.1 Test Procedures

Submit detailed test procedures, prepared and signed by the NICET Level III Fire Sprinkler Technician, and the representative of the installing company, and reviewed by the QFPE 60 days prior to performing system tests. Detailed test procedures must list all components of the installed system. Test procedures must include sequence of testing, time estimate for each test, and sample test data forms. The test data forms must be in a check-off format (pass/fail with space to add applicable test data; similar to the forms in NFPA 13). The test procedures and accompanying test data forms must be used for the pre-Government testing and the Government final testing.

a. Provide space to identify the date and time of each test. Provide space to identify the names and signatures of the individuals conducting and witnessing each test.

3.7.2 Pre-Government Testing

3.7.2.1 Verification of Compliant Installation

Conduct inspections and tests to ensure that equipment is functioning properly. Tests must meet the requirements of paragraph entitled "Minimum System Tests" and "System Acceptance" as noted in NFPA 13. The Contractor and QFPE must be in attendance at the pre-Government testing to make necessary adjustments. After inspection and testing is complete, provide a signed Verification of Compliant Installation letter by the QFPE that the installation is complete, compliant with the specification and fully operable. The letter must include the names and titles of the witnesses to the pre-Government tests. Provide all completion documentation as required by NFPA 13 and the test reports noted below.

a. NFPA 13 Aboveground Material and Test Certificate

b. NFPA 13 Underground Material and Test Certificate

3.7.2.2 Request for Government Final Test

When the verification of compliant installation has been completed, submit a formal request for Government final test to the Fort Campbell Fire Prevention Section and Directorate of Public Works Engineer Design Branch (DPW-EDB)Contracting Officers Designated Representative (COR). Government final testing will not be scheduled until the DFPE has received copies of the request for Government final testing and Verification of Compliant Installation letter with all required reports. Government final testing will not be performed until after the connections to the building fire alarm system and installation fire alarm reporting system have been completed and tested to confirm communications are fully functional. Submit request for test at least 15 calendar days prior to the requested test date.

3.7.3 Correction of Deficiencies

If equipment was found to be defective or non-compliant with contract requirements, perform corrective actions and repeat the tests. Tests must be conducted and repeated if necessary until the system has been demonstrated to comply with all contract requirements.

3.7.4 Government Final Tests

The tests must be performed in accordance with the approved test procedures in the presence of the DFPE. Furnish instruments and personnel required for the tests. The following must be provided at the job site for Government Final Testing:

a. The manufacturer's technical representative.

b. The contractor's Qualified Fire Protection Engineer (QFPE).

c. Marked-up red line drawings of the system as actually installed.

Government Final Tests will be witnessed by the Fort Campbell Fire Prevention Section and Directorate of Public Works Engineer Design Branch (DPW-EDB), Contracting Officer, Qualified Fire Protection Engineer (QFPE). At this time, all required tests noted in the paragraph "Minimum System Tests" must be repeated at their discretion.

3.8 MINIMUM SYSTEM TESTS

The system, including the underground water mains, and the aboveground piping and system components, must be tested to ensure that equipment and components function as intended. The underground and aboveground interior piping systems and attached appurtenances subjected to system working pressure must be tested in accordance with NFPA 13 and NFPA 24.

3.8.1 Underground Piping

3.8.1.1 Flushing

Underground piping must be flushed at a minimum of 10 fps in accordance with NFPA 24.

3.8.1.2 Hydrostatic Test

New underground piping must be hydrostatically tested in accordance with NFPA 24.

3.8.2 Aboveground Piping

3.8.2.1 Hydrostatic Test

Aboveground piping must be hydrostatically tested in accordance with NFPA 13. There must be no drop in gauge pressure or visible leakage when the system is subjected to the hydrostatic test. The test pressure must be read from a gauge located at the low elevation point of the system or portion being tested.

3.8.2.2 Backflow Prevention Assembly Forward Flow Test

Each backflow prevention assembly must be tested at system flow demand, including all applicable hose streams, as specified in NFPA 13. The Contractor must provide all equipment and instruments necessary to conduct a complete forward flow test, including 2.5-inch diameter hoses, playpipe nozzles or flow diffusers, calibrated pressure gauges, and pitot tube gauge. The Contractor must provide all necessary supports to safely secure hoses and nozzles during the test. At the system demand flow, the pressure readings and pressure drop (friction loss) across the assembly must be recorded. A metal placard must be provided on the backflow prevention assembly that lists the pressure readings both upstream and downstream of the assembly, total pressure drop, and the system test flow rate determined during the preliminary testing. The pressure drop must be compared to the manufacturer's data and the readings observed during the final inspections and tests.

3.8.3 Main Drain Flow Test

Following flushing of the underground piping, a main drain test must be made to verify the adequacy of the water supply. Static and residual pressures must be recorded on the certificate specified in paragraph SUBMITTALS.

3.9 SYSTEM ACCEPTANCE

Following acceptance of the system, as-built drawings and O&M manuals must be delivered to the Contracting Officer for review and acceptance. Submit six sets of detailed as-built drawings. The drawings must show the system as installed, including deviations from both the project drawings and the approved shop drawings. These drawings must be submitted within two weeks after the final acceptance test of the system. At least one set of as-built (marked-up) drawings must be provided at the time of, or prior to the final acceptance test.

a. Provide one set of full size paper as-built drawings and schematics. The drawings must be prepared electronically and sized no less than the contract drawings. Furnish one set of CDs or DVDs containing software back-up and CAD based drawings in latest version of MicroStationDXF and portable document formats of as-built drawings and schematics.

b. Provide operating and maintenance (O&M) instructions.

3.10 ONSITE TRAINING

Conduct a training course for the responding fire department and operating and maintenance personnel as designated by the Contracting Officer. Training must be performed on two separate days (to accommodate different shifts of Fire Department personnel) for a period of 8 hours of normal working time and must start after the system is functionally complete and after the final acceptance test. The on-site training must cover all of the items contained in the approved Operating and Maintenance Instructions.

-- End of Section --

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INTERIOR FIRE ALARM AND MASS NOTIFICATION SYSTEM, ADDRESSABLE

08/20

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SECTION 28 31 76

INTERIOR FIRE ALARM AND MASS NOTIFICATION SYSTEM, ADDRESSABLE
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PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ACOUSTICAL SOCIETY OF AMERICA (ASA)

ASA S3.2	(2020) American National Standard Method for Measuring the Intelligibility of Speech Over Communication Systems (ASA 85)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME A17.1/CSA B44	(2021) Safety Code for Elevators and Escalators

FM GLOBAL (FM)

FM APP GUIDE	(updated on-line) Approval Guide http://www.approvalguide.com/

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C62.41.1	(2002; R 2008) Guide on the Surges Environment in Low-Voltage (1000 V and Less) AC Power Circuits
IEEE C62.41.2	(2002) Recommended Practice on Characterization of Surges in Low-Voltage (1000 V and Less) AC Power Circuits

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 4	(2018) Standard for Integrated Fire Protection and Life Safety System Testing
NFPA 70	(2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA 20-1; TIA 20-2; TIA 20-3; TIA 20-4) National Electrical Code
NFPA 72	(2022) National Fire Alarm and Signaling Code
NFPA 90A	(2021) Standard for the Installation of Air Conditioning and Ventilating Systems
NFPA 170	(2021) Standard for Fire Safety and

Emergency Symbols

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-601-02	(2010) Operations and Maintenance: Inspection, Testing, and Maintenance of Fire Protection Systems
UFC 4-010-06	(2016; with Change 1, 2017) Cybersecurity of Facility-Related Control Systems

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

47 CFR 15	Radio Frequency Devices
47 CFR 90	Private Land Mobile Radio Services

UNDERWRITERS LABORATORIES (UL)

UL 268	(2016; Reprint Nov 2021) UL Standard for Safety Smoke Detectors for Fire Alarm Systems
UL 268A	(2008; Reprint Oct 2014) Smoke Detectors for Duct Application
UL 464	(2016; Reprint Sep 2017) UL Standard for Safety Audible Signaling Devices for Fire Alarm and Signaling Systems, Including Accessories
UL 497A	(2001; Bul. 2019) UL Standard for Safety Secondary Protectors for Communications Circuits
UL 497B	(2004; Reprint Dec 2012) Protectors for Data Communication Circuits
UL 864	(2014; Reprint May 2020) UL Standard for Safety Control Units and Accessories for Fire Alarm Systems
UL 1283	(2017) UL Standard for Safety Electromagnetic Interference Filters
UL 1449	(2021) UL Standard for Safety Surge Protective Devices
UL 1480	(2016; Reprint Sep 2017) UL Standard for Safety Speakers for Fire Alarm and Signaling Systems, Including Accessories
UL 1638	(2016; Reprint Sep 2017) UL Standard for Safety Visible Signaling Devices for Fire Alarm and Signaling Systems, Including Accessories
UL 1971	(2002; Reprint Oct 2008) Signaling Devices for the Hearing Impaired

UL 2017	(2008; Reprint Dec 2018) UL Standard for Safety General-Purpose Signaling Devices and Systems
UL 2572	(2016; Bul. 2018) UL Standard for Safety Mass Notification Systems
UL Fire Prot Dir	(2012) Fire Protection Equipment Directory

1.2 RELATED SECTIONS

Sections 25 05 11 Cybersecurity for Facility-Related Control Systems, and and Fort Campbell (FTC) Technical Design Guide, Revision 05 (12/15/21), Section 28 31 76 apply to this section, with the additions and modifications specified herein. In addition, refer to the following sections for related work and coordination:

Section 21 13 13 WET PIPE SPRINKLER SYSTEM, FIRE PROTECTION
Section 21 30 00 FIRE PUMPS
Section 23 30 00 HVAC AIR DISTRIBUTION
Section 08 71 00 DOOR HARDWARE for door unlocking and additional work related to finish hardware.
Section 14 21 23 ELECTRIC TRACTION PASSENGER ELEVATORS for additional work related to elevators.
Section 07 84 00 FIRESTOPPING for additional work related to firestopping.

1.3 SUMMARY

1.3.1 Scope

a. This work includes designing and providing a new, complete, fire alarm and mass notification (MNS) system as described herein and on the contract drawings for the entire building. Include system wiring, raceways, pull boxes, terminal cabinets, outlet and mounting boxes, control equipment, initiating devices, notification appliances, supervising station fire alarm transmitters/mass notification transceiver, and other accessories and miscellaneous items required for a complete operational system even though each item is not specifically mentioned or described. Provide system complete and ready for operation. Design and installation must comply with UFGS 25 05 11, UFC 4-010-06 and AFGM 2019-320-02.

b. Provide equipment, materials, installation, workmanship, inspection, and testing in strict accordance with NFPA 72, except as modified herein. The system layout on the drawings show the intent of coverage and suggested locations. Final quantity, system layout, and coordination are the responsibility of the Contractor.

c. Each remote fire alarm control unit must be powered from a wiring riser specifically for that use or from a local emergency power panel located on the same floor as the remote fire alarm control unit. Where remote fire control units are provided, equipment for notification appliances may be located in the remote fire alarm control units.

d. Where a fire pump is provided, the fire alarm and mass notification system must monitor and transmit the fire pump controller signals in accordance with the provisions of NFPA 72.

f. The fire alarm and mass notification system must be independent of the building security, building management, and energy/utility monitoring systems other than for control functions.

1.3.2 Qualified Fire Protection Engineer (QFPE)

Services of the QFPE must include:

a. Reviewing SD-02, SD-03, and SD-05 submittal packages for completeness and compliance with the provisions of this specification. Construction (shop) drawings and calculations must be prepared by, or prepared under the immediate supervision of, the QFPE. The QFPE must affix their professional engineering stamp with signature to the shop drawings, calculations, and material data sheets, indicating approval prior to submitting the shop drawings to the DFPE.

b. Providing a letter documenting that the SD-02, SD-03, and SD-05 submittal package has been reviewed and noting any outstanding comments.

c. Performing in-progress construction surveillance prior to installation of ceilings (rough-in inspection).

d. Witnessing pre-Government and final Government functional performance testing and performing a final installation review.

e. Signing applicable certificates under SD-07.

1.4 DEFINITIONS

Wherever mentioned in this specification or on the drawings, the equipment, devices, and functions must be defined as follows:

1.4.1 Interface Device

An addressable device that interconnects hard wired systems or devices to an analog/addressable system.

1.4.2 Fire Alarm and Mass Notification Control Unit (FMCU)

A master control unit having the features of a fire alarm control unit (FACU) and an autonomous control unit (ACU) where these units are interconnected to function as a combined fire alarm/mass notification system. The FACU and ACU functions may be contained in a single cabinet or in independent, interconnected, and co-located cabinets.

1.4.3 Remote Fire Alarm and Mass Notification Control Unit

A control unit, physically remote from the fire alarm and mass notification control unit, that receives inputs from automatic and manual fire alarm devices; may supply power to detection devices and interface devices; may provide transfer of power to the notification appliances; may provide transfer of condition to relays or devices connected to the control unit; and reports to and receives signals from the fire alarm and mass notification control unit.

1.4.4 Local Operating Console (LOC)

A unit designed to allow emergency responders and/or building occupants to operate the MNS including delivery of recorded messages and/or live voice announcements, initiate visual, textual visual, and audible appliance operation and other relayed functions.

1.4.5 Terminal Cabinet

A steel cabinet with locking, hinge-mounted door where terminal strips are securely mounted inside the cabinet.

1.4.6 Control Module and Relay Module

Terms utilized to describe emergency control function interface devices as defined by NFPA 72.

1.4.7 Designated Fire Protection Engineer (DFPE)

The DoD fire protection engineer that oversees that Area of Responsibility for that project. This is sometimes referred to as the "cognizant" fire protection engineer. Interpret reference to "authority having jurisdiction" and/or AHJ in referenced standards to mean the Designated Fire Protection Engineer (DFPE). The DFPE may be responsible for review of the contractor submittals having a "G" designation, and for witnessing final inspection and testing.

1.4.8 Qualified Fire Protection Engineer (QFPE)

A QFPE is an individual who is a licensed professional engineer (P.E.), who has passed the fire protection engineering written examination administered by the National Council of Examiners for Engineering and Surveying (NCEES) and has relevant fire protection engineering experience.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government.

Shop drawings (SD-02), product data (SD-03) and calculations (SD-05) must be prepared by the fire alarm designer and combined and submitted as one complete package. The QFPE must review the SD-02/SD-03/SD-05 submittal package for completeness and compliance with the Contract provisions prior to submission to the Government. The QFPE must provide a Letter of Confirmation that they have reviewed the submittal package for compliance with the contract provisions. This letter must include their registered professional engineer stamp and signature. Partial submittals and submittals not reviewed by the QFPE will be returned by the Government disapproved without review.

Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualified Fire Protection Engineer (QFPE); G, AE

Fire alarm system designer; G, AE

Supervisor; G, AE

Technician; G, AE

Installer; G, AE

Test Technician; G, AE

SD-02 Shop Drawings

Nameplates; G, AE

Instructions; G, AE

Wiring Diagrams; G, AE

System Layout; G, AE

Notification Appliances; G, AE

Initiating devices; G, AE

Amplifiers; G, AE

Battery Power; G, AE

Voltage Drop Calculations; G, AE

SD-03 Product Data

Fire Alarm and Mass Notification Control Unit (FMCU); G, AE

Local Operating Console (LOC); G, AE

Amplifiers; G, AE

Tone Generators; G, AE

Digitalized voice generators; G, AE

LCD Annunciator; G, AE

Manual Stations; G, AE

Smoke Detectors; G, AE

Duct Smoke Detectors; G, AE

Heat Detectors; G, AE

Addressable Interface Devices; G, AE

Addressable Control Modules; G, AE

Notification Appliances; G, AE

Textual Display Sign Control Panel; G, AE

Textual Display Signs; G, AE

Batteries; G, AE

Battery Chargers; G, AE

Surge Protective Devices; G, AE

Alarm Wiring; G, AE

Back Boxes and Conduit; G, AE

Ceiling Bridges for Ceiling-Mounted Appliances; G, AE

Terminal Cabinets; G, AE

Automatic Fire Alarm Transmitters (including housing); G, AE

Radio Transmitter and Interface Panels; G, AE

Mass Notification Transceiver; G, AE

Electromagnetic Door Holders; G, AE

Environmental Enclosures or Guards; G, AE

Firefighter Telephone; G, AE

Printer; G, AE

Document Storage Cabinet; G, AE

SD-05 Design Data

SD-06 Test Reports

Test Procedures; G, AE,FTC

SD-07 Certificates

Verification of Compliant Installation; G, AE,FTC

Request for Government Final Test; G, AE,FTC

SD-10 Operation and Maintenance Data

Operation and Maintenance (O&M) Instructions; G,AE,FTC

Instruction of Government Employees; G, AE,FTC

SD-11 Closeout Submittals

As-Built Drawings

Spare Parts

1.6 SYSTEM OPERATION

Fire alarm system/mass notification system including textual display sign control panel(s), components requiring power, except for the FMCU(s) power supply, must operate on 24 volts DC unless noted otherwise in this section.

The interior fire alarm and mass notification system must be a complete, supervised, noncoded, analog/addressable fire alarm and mass notification system conforming to NFPA 72, UL 864, and UL 2572. Systems meeting UL 2017 only are not acceptable. The system must be activated into the alarm mode by actuation of an alarm initiating device. The system must remain in the alarm mode until the initiating device is reset and the control unit is reset and restored to normal. The system may be placed in the alarm mode as indicated in FAMNS sequence of operation found in 'FA' series sheets in plan drawings.

1.6.1 Alarm Initiating Devices and Notification Appliances (Visual, Voice, Textual)

a. Connect alarm initiating devices to initiating device circuits (IDC) Class "A", or to signaling line circuits (SLC) Class "A" and installed in accordance with NFPA 72.

b. Connect notification appliances to notification appliance circuits (NAC) Class "A". A looped conduit system shall be provided so that if the conduit and all conductors within are severed or exposed to fire at any point, all IDC, NAC and SLC will remain functional. Should the design or building layout preclude separation, then a fire-rated separation shall be provided in accordance with NFPA 72. The conduit loop requirement is not applicable to the signal transmission link from the local panels (at the protected premises) to the supervising station (fire station, fire alarm central communication center).

1.6.2 Functions and Operating Features

The system must provide the following functions and operating features:

a. Power, annunciation, supervision, and control for the system. Addressable systems must be microcomputer (microprocessor or microcontroller) based with a minimum word size of eight bits with sufficient memory to perform as specified.

b. Visual alarm notification appliances must be synchronized as required by NFPA 72.

c. Electrical supervision of the primary power (AC) supply, presence of the battery, battery voltage, and placement of system modules within the control unit.

d. An audible and visual trouble signal to activate upon a single break or open condition, or ground fault. The trouble signal must also operate upon loss of primary power (AC) supply, absence of a battery supply, low battery voltage, or removal of alarm or supervisory control unit modules. After the system returns to normal operating conditions, the trouble signal must again sound until the trouble is acknowledged. A smoke detector in the process of being verified for the actual presence of smoke must not initiate a trouble condition.

e. A trouble signal silence feature that must silence the audible trouble signal, without affecting the visual indicator.

f. Program capability via switches in a locked portion of the FMCU to bypass the automatic notification appliance circuits, air handler shutdown and elevator recall features. Operation of this programmed action must indicate on the FMCU display and printer output as a supervisory or trouble condition.

g. Alarm functions must override trouble or supervisory functions. Supervisory functions must override trouble functions.

h. The system must be capable of being programmed from the control unit keyboard. Programmed information must be stored in non-volatile memory.

i. The system must be capable of operating, supervising, and/or monitoring non-addressable alarm and supervisory devices.

j. There must be no limit, other than maximum system capacity, as to the number of addressable devices that may be in alarm simultaneously.

k. Where the fire alarm/mass notification system is responsible for initiating an action in another emergency control device or system, such as HVAC, or an elevator recall, the addressable fire alarm relay must be located within 3ft of the emergency control device.

l. An alarm signal must automatically initiate the following functions:

(1) Transmission of an alarm signal to the fire department via FTC IRACS (Kingfisher Industrial Radio Alarm Control System).

(2) Visual indication of the device operated on the FMCU, LED textual display signs, and on the remote annunciators.

(3) Record the event on the system printer.

(4) Actuation of alarm notification appliances.

(5) Recording of the event electronically in the history log of the FMCU.

(6) Release of power to electric locks (delayed egress locks) on doors that are part of the means of egress.

(7) Elevator recall as described in this section.

(8) Operation of a sprinkler waterflow switch serving an elevator machinery room or elevator shaft must operate shunt trip circuit breaker(s) to shut down power to the elevators in accordance with ASME A17.1/CSA B44 and as indicated in UFC 3-490-06 - Chapter 6.

m. A supervisory signal must automatically initiate the following functions:

(1) Visual indication of the device operated on the FMCU, and on the remote annunciator.

(2) Record the event on the system printer.

(3) Transmission of a supervisory signal to the fire department via FTC IRACS.

(4) Operation of a duct smoke detector must shut down the appropriate air handler in accordance with NFPA 90A in addition to other requirements of this paragraph and as allowed by NFPA 72.

(5) Recording of the event electronically in the history log of the FMCU.

n. A trouble condition must automatically initiate the following functions:

(1) Visual indication of the device operated on the FMCU, and on the remote annunciator.(2) Record the event on the system printer.

(3) Transmission of a trouble signal to the fire department via FTC IRACS.

(4) Recording of the event electronically in the history log of the FMCU.

o. System control equipment must be programmed to provide a 60-minute to 180-minute delay in transmission of trouble signals resulting from primary power failure.

p. Activation of a LOC pushbutton must activate the audible and visual alarms in the facility. The audible message must be the one associated with the pushbutton activated.

q. In addition to requirements in this section, reference 'FA' series sheets in plan drawings for details and further requirements for mass notification, alarm, supervisory, and trouble condition FAMNS sequence of operation.

1.6.3 Elevator Recall

Provide elevator recall in accordance with ASME A17.1/CSA B44, Section 14 21 23 ELECTRIC TRACTION PASSENGER ELEVATORS, and as specified herein. Activation of any smoke detector in an elevator shaft, machine room, or lobby (except at designated recall level) must cause all elevators associated with that shaft, machine room, or lobby to return nonstop to the designated level. Activation of a smoke detector in the lobby or machine room at the designated level must cause all elevators associated with that lobby to return nonstop to the assigned alternate level. Activation of a detector in an elevator shaft, machine room, or lobby must also cause illumination of elevator cab warning signal (fire hat) and complete operation of fire alarm system as specified in paragraph titled "Functions and Operating Features".

1.7 TECHNICAL DATA AND SITE-SPECIFIC SOFTWARE

Technical data and site-specific software (meaning technical data that relates to computer software) that are specifically identified in this project, and may be required in other specifications, must be delivered, strictly in accordance with the CONTRACT CLAUSES. The fire alarm system manufacturer must submit written confirmation of this contract provision as "Fire Alarm System Site-Specific Software Acknowledgement". Identify data delivered by reference to the specification paragraph against which it is furnished. Data to be submitted must include complete system, equipment, and software descriptions. Descriptions must show how the equipment will operate as a system to meet the performance requirements of this contract. The site-specific software data package must also include the following:

a. Items identified in NFPA 72, titled "Site-Specific Software".

b. Identification of programmable portions of the system equipment and capabilities.

c. Description of system revision and expansion capabilities and methods of implementation detailing both equipment and software requirements.

d. Provision of operational software data on all modes of programmable portions for fire alarm and mass notification.

e. Description of Fire Alarm and Mass Notification Control Unit equipment operation.

f. Description of auxiliary and remote equipment operations.

g. Library of application software.

h. Operation and maintenance manuals.

1.8 QUALITY ASSURANCE

1.8.1 Submittal Documents

1.8.1.1 Preconstruction Submittals

Within 36 days of contract award but not less than 14 days prior to commencing any work on site, the Contractor must submit the following for review and approval. SD-02, SD-03 and SD-05 submittals received prior to the review and approval of the qualifications of the fire alarm subcontractor and QFPE must be returned disapproved without review. All resultant delays must be the sole responsibility of the Contractor.

1.8.1.2 Shop Drawings

Shop drawings must not be smaller than ANSI D. Drawings must comply with the requirements of NFPA 72 and NFPA 170. Minimum scale for floor plans must be 1/8"=1'.

1.8.1.3 Nameplates

Nameplate illustrations and data to obtain approval by the Contracting Officer before installation.

1.8.1.4 Wiring Diagrams

Six copies of point-to-point wiring diagrams showing the points of connection and terminals used for electrical field connections in the system, including interconnections between the equipment or systems that are supervised or controlled by the system. Diagrams must show connections from field devices to the FMCU and remote FMCU, initiating circuits, switches, relays and terminals, including pathway diagrams between the control unit and shared communications equipment within the protected premises. Point-to-point wiring diagrams must be job specific and must not indicate connections or circuits not being utilized. Provide complete riser diagrams indicating the wiring sequence of all devices and their connections to the control equipment. Include a color-code schedule for the wiring.

1.8.1.5 System Layout

Six copies of plan view drawing showing device locations, terminal cabinet locations, junction boxes, other related equipment, conduit routing, conduit sizes, wire counts, conduit fill calculations, wire color-coding, circuit identification in each conduit, and circuit layouts for all floors. Indicate candela rating of each visual notification appliance. Indicate the wattage of each speaker. Clearly identify the locations of isolation modules. Indicate the addresses of all devices, modules, relays, and similar. Show/identify all acoustically similar spaces. Indicate if the environment for the FMCU is within its environmental listing (e.g. temperature/humidity).

Provide a complete description of the system operation in matrix format similar to the "Typical Input/Output Matrix" included in the Annex of NFPA 72.

1.8.1.6 Notification Appliances

Calculations and supporting data on each circuit to indicate that there is at least 25 percent spare capacity for notification appliances. Annotate data for each circuit on the drawings.

1.8.1.7 Initiating Devices

Calculations and supporting data on each circuit to indicate that there is at least 25 percent spare capacity for initiating devices. Annotate data for each circuit on the drawings.

1.8.1.8 Amplifiers

Calculations and supporting data to indicate that amplifiers have sufficient capacity to simultaneously drive all notification speakers at tapped settings plus 25 percent spare capacity. Annotate data for each circuit on the drawings.

1.8.1.9 Battery Power

Calculations and supporting data as required in paragraph Battery Power Calculations for alarm, alert, and supervisory power requirements. Calculations including ampere-hour requirements for each system component and each control unit component, and the battery recharging period, must be included on the drawings.

1.8.1.10 Voltage Drop Calculations

Voltage drop calculations for each notification circuit indicating that sufficient voltage is available for proper operation of the system and all components, at a minimum rated voltage of the system operating on batteries. Include the calculations on the system layout drawings.

1.8.1.11 Product Data

Six copies of annotated descriptive data to show the specific model, type, and size of each item. Catalog cuts must also indicate the NRTL listing. The data must be highlighted to show model, size, and options that are intended for consideration. Data must be adequate to demonstrate compliance with all contract requirements. Product data for all equipment must be combined into a single submittal.

Provide an equipment list identifying the type, quantity, make, and model number of each piece of equipment to be provided under this submittal. The equipment list must include the type, quantity, make and model of spare equipment. Types and quantities of equipment submitted must coincide with the types and quantities of equipment used in the battery calculations and those shown on the shop drawings.

1.8.1.12 Operation and Maintenance (O&M) Instructions

Six copies of the Operation and Maintenance Instructions. The O&M Instructions must be prepared in a single volume or in multiple volumes, with each volume indexed, and may be submitted as a Technical Data Package. Manuals must be approved prior to training. The Interior Fire Alarm And Mass Notification System Operation and Maintenance Instructions must include the following:

a. "Manufacturer Data Package " as specified in Section 01 78 23 OPERATION AND MAINTENANCE DATA.

b. Operating manual outlining step-by-step procedures required for system startup, operation, and shutdown. The manual must include the manufacturer's name, model number, service manual, parts list, and preliminary equipment list complete with description of equipment and their basic operating features.

c. Maintenance manual listing routine maintenance procedures, possible breakdowns and repairs, and troubleshooting guide. The manuals must include conduit layout, equipment layout and simplified wiring, and control diagrams of the system as installed.

d. Complete procedures for system revision and expansion, detailing both equipment and software requirements.

e. Software submitted for this project on CD/DVD media utilized.

f. Printouts of configuration settings for all devices.

g. Routine maintenance checklist. The routine maintenance checklist must be arranged in a columnar format. The first column must list all installed devices, the second column must state the maintenance activity or state no maintenance required, the third column must state the frequency of the maintenance activity, and the fourth column provided for additional comments or reference. All data (devices, testing frequencies, and similar) must comply with UFC 3-601-02.

h. A final Equipment List must be submitted with the Operating and Maintenance (O&M) manual.

1.8.1.13 As-Built Drawings

The drawings must show the system as installed, including deviations from both the project drawings and the approved shop drawings. These drawings must be submitted within two weeks after the final Government test of the system. At least one set of the as-built (marked-up) drawings must be provided at the time of, or prior to the final Government test.

1.8.2 Qualifications

1.8.2.1 Fire Alarm System Designer

The fire alarm system designer must be certified as a Level III (minimum) Technician by National Institute for Certification in Engineering Technologies (NICET) in the Fire Alarm Systems subfield of Fire Protection Engineering Technology or meet the qualifications for a QFPE.

1.8.2.2 Supervisor

A NICET Level III or IV fire alarm technician must supervise the installation of the fire alarm/mass notification system. The fire alarm technicians supervising the installation of equipment must be factory trained in the installation, adjustment, testing, and operation of the equipment specified herein and on the drawings.

1.8.2.3 Technician

Fire alarm technicians with a minimum of four years of experience must be utilized to install and terminate fire alarm/mass notification devices, cabinets and control units. The fire alarm technicians installing the equipment must be factory trained in the installation, adjustment, testing, and operation of the equipment specified herein and on the drawings.

1.8.2.4 Installer

Fire alarm installer with a minimum of two years of experience utilized to assist in the installation of fire alarm/mass notification devices, cabinets and control units . A licensed electrician must be allowed to install wire, cable, conduit and backboxes for the fire alarm system/mass notification system. The fire alarm installer must be factory trained in the installation, adjustment, testing, and operation of the equipment specified herein and on the drawings.

1.8.2.5 Test Technician

Fire alarm technicians with a minimum of eight years of experience and NICET Level III or IV utilized in testing and certification of the installation of the fire alarm/mass notification devices, cabinets and control units. The fire alarm technicians testing the equipment must be factory trained in the installation, adjustment, testing, and operation of the equipment installed as part of this project.

1.8.2.6 Manufacturer

Components must be of current design and must be in regular and recurrent production at the time of installation. Provide design, materials, and devices for a protected premises fire alarm system, complete, conforming to NFPA 72, except as specified herein.

1.8.3 Regulatory Requirements

Equipment and material must be listed or approved. Listed or approved, as used in this section, means listed, labeled or approved by a Nationally Recognized Testing Laboratory (NRTL) such as UL Fire Prot Dir or FM APP GUIDE. The omission of these terms under the description of any item of equipment described must not be construed as waiving this requirement. All listings or approvals by testing laboratories must be from an existing ANSI or UL published standard. The recommended practices stated in the manufacturer's literature or documentation must be considered as mandatory requirements.

1.9 DELIVERY, STORAGE, AND HANDLING

Protect equipment delivered and placed in storage from the weather, humidity, and temperature variation, dirt and dust, and other contaminants.

1.10 MAINTENANCE

1.10.1 Spare Parts

Furnish the following spare parts in the manufacturers original unopened containers:

a. Six complete sets of system keys.

b. Four of each type of fuse required by the system.

c. Ten percent of each initiating and notification device installed, but no less than two.

1.10.2 Special Tools

There shall be no requirement for software locks, special tools and any other proprietary equipment to maintain, add devices to or delete devices from the system, or test the fire alarm system. Fire alarm systems shall be able to be programmed from the control panel and the Government's laptop. Provide any software, cables/interface devices required to manipulate the system without any licensing agreements, signed documents or any requirements upon the Government to relay on any Contractor or manufacturer for maintenance or manipulation of the system.

PART 2 PRODUCTS

2.1 GENERAL PRODUCT REQUIREMENT

All fire alarm and mass notification equipment must be listed for use under the applicable reference standards. Interfacing of UL 864 or similar approved industry listing with Mass Notification equipment listed to UL 2572 must be done in a laboratory listed configuration, if the software programming features cannot provide a listed interface control.

2.2 MATERIALS AND EQUIPMENT

2.2.1 Standard Products

Provide materials, equipment, and devices that have been tested by a nationally recognized testing laboratory and listed for fire protection service when so required by NFPA 72 or this specification. Select material from one manufacturer, where possible, and not a combination of manufacturers, for any particular classification of materials. Material and equipment must be the standard products of a manufacturer regularly engaged in the manufacture of the products for at least 2 years prior to bid opening and must be compatible with standard FTC FAMNS equipment.

2.2.2 Nameplates

Major components of equipment must have the manufacturer's name, address, type or style, model or serial number, catalog number, date of installation, installing Contractor's name and address, and the contract number provided on a new name plate permanently affixed to the item or equipment. Major components include, but are not limited to, the following:

a. FMCU

b. Automatic transmitter/transceiver
c. Terminal Cabinet

Nameplates must be etched metal or plastic, permanently attached by screws to control units or adjacent walls.

2.2.3 Keys

Keys and locks for equipment, control units and devices must be identical. Master all keys and locks to a single key as required by the Installation Fire Department or as required by the Contracting Officer.

2.2.4 Instructions

Provide a typeset printed or typewritten instruction card mounted behind a Lexan plastic or glass cover in a stainless steel or aluminum frame. Install the frame in a conspicuous location observable from the FMCU. The card must show those steps to be taken by an operator when a signal is received as well as the functional operation of the system under all conditions, normal, alarm, supervisory, and trouble. The instructions must also include procedures for operating live voice microphones. The instructions and their mounting location must be approved by the Contracting Officer before being posted.

2.3 FIRE ALARM AND MASS NOTIFICATION CONTROL UNIT

Provide a complete fire alarm and mass notification control unit (FMCU) fully enclosed in a lockable steel cabinet as specified herein. Operations required for testing or for normal care, maintenance, and use of the system must be performed from the front of the enclosure. Ft. Campbell will provide consolidated FACP and ACU (Potter AFC-1000V)unit and Kingfisher equipment (transmitter, antenna, antenna mount, battery, and coax cable) as "standard FTC FAMNS equipment". Contact DPW Engineering

Design Branch, Jake Ramacher at jacob.d.ramacher.civ@army.mil to obtain these items. FACP/ACU are a consolidated unit, referred in this specification as "FMCU". Both FMCU and Transmitter are GFCI.

2.3.1 Silencing Switches

2.3.1.1 Alarm Silencing Switch

Provide an alarm silencing switch at the FMCU that must silence the audible and visual signal, but not affect the visual alarm indicator. Subsequent activation of initiating devices must cause the notification appliances to re-activate.

2.3.1.2 Supervisory/Trouble Silencing Switch

Provide supervisory and trouble silencing switch(es) that must silence the audible trouble and supervisory signal(s), but not extinguish the visual indicator. This switch must be overridden upon activation of a subsequent supervisory or trouble condition. Audible trouble indication must resound automatically every 24 hours after the silencing feature has been operated if the supervisory or trouble condition still exists.

2.3.2 Non-Interfering

Power and supervise each circuit such that a signal from one device does not prevent the receipt of signals from any other device. Initiating devices must be manually reset by switch from the FMCU after the initiating device or devices have been restored to normal.

2.3.3 Audible Notification System

The Audible Notification System must comply with the requirements of NFPA 72 for Emergency Voice/Alarm Communications System requirements, except as specified herein. The system must be a one-way, multi-channel voice notification system incorporating user selectability of a minimum eight distinct sounds for tone signaling, and the incorporation of a voice module for delivery of recorded messages. Audible appliances must produce a three-pulse temporal pattern for three cycles followed by a voice message that is repeated until the control unit is reset or silenced. Automatic messages must be broadcast through speakers throughout the building/facility but not in stairs or elevator cabs. A live voice message must override the automatic audible output through use of a microphone input at the control unit or the LOC.

a. When using the microphone, live messages must be broadcast throughout a selected floor or floors, or all call. The system must be capable of operating all speakers at the same time. The Audible Notification System must support Public Address (PA) paging for the facility. Activation of the public address microphone must not initiate activation of visual notification appliances or LED text displays.

b. The microprocessor must actively interrogate circuitry, field wiring, and digital coding necessary for the immediate and accurate rebroadcasting of the stored voice data into the appropriate amplifier input. Loss of operating power, supervisory power, or any other malfunction that could render the digitalized voice module inoperative must automatically cause the three-pulse temporal pattern to take over all functions assigned to the failed unit in the event an alarm is activated.

2.3.3.1 Outputs and Operational Modules

All outputs and operational modules must be fully supervised with on-board diagnostics and trouble reporting circuits. Provide form "C" contacts for system alarm and trouble conditions. Provide circuits for operation of auxiliary appliance during trouble conditions. During a Mass Notification event, the control unit must not generate nor cause any trouble alarms to be generated with the Fire Alarm system.

2.3.3.2 Mass Notification

a. The system must have the capability of utilizing an LOC with redundant controls of the FMCU. Notification Appliance Circuits (NAC) must be provided for the activation of strobe appliances. Audio output must be selectable for line level. A hand-held microphone must be provided and, upon activation, must take priority over any tone signal, recorded message or PA microphone operation in progress, while maintaining the strobe NAC circuit activation.

b. The Mass Notification functions must override the manual or automatic fire alarm notification, and public address (PA) functions. Other fire alarm functions including transmission of a signal(s) to the fire department must remain operational. When a mass notification announcement is disengaged and a fire alarm condition still exists, the audible and visual notification appliances must resume activation for alarm conditions. The fire alarm message must be of lower priority that all other messages (except any "test" messages) and must not override any other messages.

c. The mass notification messages shall be programmed in male/female voice as specified by UFC 3-600-01 for message type and occupancy. Messages recorded from the system microphone must not be accepted. The Mass Notification messages will be generated by the text-to-speech system Speechify, Nuance, 2005 as used by NOAA for weather information and Emergency Alert System messaging. The voices used will be "Tom" and "Donna" as identified for each message type in UFC 3-600-01. The current Nuance text-to-speech product is Vocalizer 5.0, Nuance 2013.

   Update all messages:

   (1) Fire Emergency: “MAY I HAVE YOUR ATTENTION PLEASE. MAY I HAVE YOUR ATTENTION PLEASE. A FIRE EMERGENCY HAS BEEN REPORTED IN THE BUILDING. PLEASE LEAVE THE BUILDING BY THE NEAREST EXIT OR EXIT STAIRWAY. DO NOT USE THE ELEVATORS.” For single-story facilities, delete “OR EXIT STAIRWAY. DO NOT USE THE ELEVATORS”.

   (2) Shelter In Place: “A SHELTER IN PLACE EMERGENCY HAS BEEN DECLARED; PLEASE TAKE SHELTER IN A DESIGNATED SAFE AREA IMMEDIATELY”.

   (3) Weather: “A WEATHER EMERGENCY HAS BEEN DECLARED; PLEASE TAKE SHELTER IN A DESIGNATED SAFE AREA IMMEDIATELY; DO NOT USE THE ELEVATORS.” For single-story facilities, delete “DO NOT USE THE ELEVATORS”.

   (4) Evacuate: A FORCE PROTECTION EMERGENCY HAS BEEN DECLARED; PLEASE LEAVE THE BUILDING BY THE NEAREST EXIT”.

   (5) Test: “TEST, TEST, TEST, THIS IS AN EMERGENCY NOTIFICATION AUDIO

SYSTEM TEST, YOU MAY CONTINUE NORMAL OPERATIONS, TEST, TEST, TEST".

(6) All Clear: "THE EMERGENCY HAS BEEN RESOLVED; RETURN TO NORMAL OPERATIONS."

d. Auxiliary Input Module must be designed to be an outboard expansion module to either expand the number of optional LOC's, or allow a telephone interface.

2.3.3.3 Installation-Wide Control

If an installation-wide control system for mass notification exists on the Base, the autonomous control unit must communicate with the central control unit of the Installation-wide system. The autonomous control unit must receive commands/messages from the central control unit and provide status information.

2.3.4 Memory

Provide each control unit with non-volatile memory and logic for all functions. The use of long life batteries, capacitors, or other age-dependent devices must not be considered as equal to non-volatile processors, PROMS, or EPROMS.

2.3.5 Field Programmability

Provide control units and control units that are fully field programmable for both input and output of control, initiation, notification, supervisory, and trouble functions. The system program configuration must be menu driven. System changes must be password protected. Any proprietary equipment and proprietary software needed by qualified technicians to implement future changes to the fire alarm system must be provided as part of this contract.

2.3.6 Input/Output Modifications

The FMCU must contain features that allow the bypassing of input devices from the system or the modification of system outputs. These control features must consist of a control unit mounted keypad. Any bypass or modification to the system must indicate a trouble condition on the FMCU.

2.3.7 Resetting

Provide the necessary controls to prevent the resetting of any alarm, supervisory, or trouble signal while the alarm, supervisory or trouble condition on the system still exists.

2.3.8 Walk Test

The FMCU must have a walk test feature. When using this feature, operation of initiating devices must result in limited system outputs, so that the notification appliances operate for only a few seconds and the event is indicated in the history log and on the system printer, but no other outputs occur.

2.3.9 History Logging

The control unit must have the ability to store a minimum of 400 events in a log. These events must be stored in a battery-protected memory and must remain in the memory until the memory is downloaded or cleared manually. Resetting of the control unit must not clear the memory.

2.3.10 Manual Access

An operator at the control unit, having a proper access level, must have the capability to manually access the following information for each initiating device.

a. Primary status.

b. Device type.

c. Present average value.

d. Present sensitivity selected.

e. Detector range (normal, dirty).

2.3.11 Heat Detector Self-Test Routines

Automatic self-test routines must be performed on each detector that will functionally check detector sensitivity electronics and ensure the accuracy of the value being transmitted. Any detector that fails this test must indicate a trouble condition with the detector location at the control unit.

2.4 LOCAL OPERATING CONSOLES (LOC)

2.4.1 General

The LOC must consist of a remote microphone station incorporating a push-to-talk (PTT) hand-held microphone and system status indicators. The LOC must have the capability of being utilized to activate prerecorded messages. The unit must incorporate microphone override of any tone generation or recorded messages. The unit must be fully supervised from the FMCU. The housing for the LOC must not be lockable. The LOC must have public address capability with the provision of a separate microphone. The PA paging function must not override any alarm or notification functions. The PA microphone must be hand-held style. Hand-held microphones must be housed in a separate protective cabinet. The cabinet must be accessible without the use of a key. The location of the microphones must be approved by the Designated Fire Protection Engineer (DFPE). Activation of the PA microphone must not initiate activation of visual notification appliances or LED text displays. The PA paging function must not override any alarm or notification functions.

2.4.2 Multiple LOCs

When an installation has more than one LOC, the LOCs must be programmed to allow only one LOC to be available for paging or messaging at a time. Once one LOC becomes active, all other LOC's will have an indication that the system is busy (Amber Busy Light) and cannot be used at that time. This is to avoid two messages being given at the same time. It must be possible to override or lockout the LOC's from the FMCU.

2.5 AMPLIFIERS, PREAMPLIFIERS, TONE GENERATORS

Any amplifiers, preamplifiers, tone generators, digitalized voice generators, and other hardware necessary for a complete, operational, textual audible circuit conforming to NFPA 72 must be housed in a remote FMCU, terminal cabinet, or in the FMCU. Individual amplifiers must be 100 watts maximum.

2.5.1 Operation

The system must automatically operate and control all building speakers except those installed in the stairs. The speakers in the stairs must operate only when the microphone is used to deliver live messages.

2.5.2 Construction

Amplifiers must utilize computer grade solid state components and must be provided with output protection devices sufficient to protect the amplifier against any transient up to 10 times the highest rated voltage in the system.

2.5.3 Inputs

Equip each system with separate inputs for the tone generator, digitalized voice driver and control unit mounted microphone . Microphone inputs must be of the low impedance, balanced line type. Both microphone and tone generator input must be operational on any amplifier.

2.5.4 Tone Generator

The tone generator must produce a three-pulse temporal pattern and must be constantly repeated until interrupted by either the digitalized voice message, the microphone input, or the alarm silence mode as specified. The tone generator must be single channel with an automatic backup generator per channel such that failure of the primary tone generator causes the backup generator to automatically take over the functions of the failed unit and also causes transfer of the common trouble relay. The tone generator must be provided with securely attached labels to identify the component as a tone generator and to identify the specific tone it produces.

2.5.5 Protection Circuits

Each amplifier must be constantly supervised for any condition that could render the amplifier inoperable at its maximum output. Failure of any component must cause illumination of a visual "amplifier trouble" indicator on the control unit, appropriate logging of the condition in the history log and on the system printer, and other actions for trouble conditions as specified.

2.6 REMOTE ANNUNCIATOR

2.6.1 LCD Annunciator

Located where indicated in plan drawings, provide a flush mounted annunciator that includes an LCD display. The display must indicate the device in trouble/alarm or any supervisory device. Display the device name, address, and actual building location. The remote annunciator must duplicate functions of the FMCU for message display, fire alarm, supervisory alarm, and trouble conditions, visual and audible notification, and system reset functions. Remote annunciator must require the use of a key for accessing the reset, control and other functions.

A building floor plan must be provided and mounted (behind Plexiglass or similar protective material) at the annunciator location. The floor plan must indicate all rooms by name and number including the locations of stairs and elevators. The floor plan must show all devices and their programmed address to facilitate identification of their physical location from the LCD display information.

2.6.2 Printer

a. Provide a system printer to record alarm, supervisory, and trouble conditions without loss of any signal or signals. Printout must be by circuit, device, and function as provided in the FMCU. Printer must operate on a 120 VAC, 60 Hz power supply.

b. The printer must have at least 80 characters per line and have a 96 ASCII character set. The printer must have a microprocessor-controlled, bi-directional, logic seeking head capable of printing 120 characters per second utilizing a 9 by 7 dot matrix print head. Printer must not contain internal software which is essential for proper operation.

c. When the FMCU receives a signal, the alarm, supervisory, and trouble condition must be printed. The printout must include the type of signal, the circuit or device reporting, the date, and the time of the occurrence. The printer must differentiate alarm signals from other printed indications. When the system is reset, this condition must also be printed including the same information concerning device, location, date, and time. Provide a means to automatically print a list of existing alarm, supervisory, and trouble conditions in the system. If a printer is off-line when an alarm is received, the system must have a buffer to retain the data and it must be printed when the printer is restored to service. The printer must have an indicator to alert the operator that the paper has run out.

2.7 MANUAL STATIONS

Provide metal or plastic, semi-flush mounted, double-action, addressable manual stations, that are not subject to operation by jarring or vibration. Stations must be equipped with screw terminals for each conductor. Stations that require the replacement of any portion of the device after activation are not permitted. Stations must be finished in red with molded raised lettering operating instructions of contrasting color. The use of a key must be required to reset the station.

2.8 SMOKE DETECTORS

2.8.1 Spot Type Detectors

Provide addressable photoelectric smoke detectors as follows:

a. Provide analog/addressable photoelectric smoke detectors utilizing the photoelectric light scattering principle for operation in accordance with UL 268. Smoke detectors must be listed for use with the FMCU.

b. Provide self-restoring type detectors that do not require any readjustment after actuation at the FMCU to restore them to normal operation. The detector must have a visual indicator to show actuation.

c. Vibration must have no effect on the detector's operation. Protect the detection chamber with a fine mesh metallic screen that prevents the entrance of insects or airborne materials. The screen must not inhibit the movement of smoke particles into the chamber.

d. Provide twist lock bases with sounder that produces a minimum of 90 dBA at 10 feet with screw terminals for each conductor. The detectors must maintain contact with their bases without the use of springs.

e. The detector address must identify the particular unit, its location within the system, and its sensitivity setting. Detectors must be of the low voltage type rated for use on a 24 VDC system.

2.8.2 Duct Smoke Detectors

Duct-mounted addressable photoelectric smoke detectors must consist of a smoke detector, as specified in paragraph Spot Type Detectors, mounted in a special housing fitted with duct sampling tubes. Detector circuitry must be mounted in a metallic or plastic enclosure exterior to the duct. It is not permitted to cut the duct insulation to install the duct detector directly on the duct. Detectors must be listed for operation over the complete range of air velocities, temperature and humidity expected at the detector when the air-handling system is operating. Detectors must be powered from the FMCU.

a. Sampling tubes must run the full width of the duct. The duct detector package must conform to the requirements of NFPA 90A, UL 268A, and must be listed for use in air-handling systems. The control functions, operation, reset, and bypass must be controlled from the FMCU.

b. Lights to indicate the operation and alarm condition must be visible and accessible with the unit installed and the cover in place. Remote indicators must be provided where required by NFPA 72. Remote indicators as well as the affected fan units must be properly identified in etched plastic placards.

c. Detectors must provide for control of auxiliary contacts that provide control, interlock, and shutdown functions specified in Section 23 09 00 to INSTRUMENTATION AND CONTROL FOR HVAC. Auxiliary contacts provide for this function must be located within 3 feet of the controlled circuit or appliance. The auxiliary contacts must be supplied by the fire alarm system manufacturer to ensure complete system compatibility.

2.9 ADDRESSABLE INTERFACE DEVICES

The initiating device being monitored must be configured as a Class "A" initiating device circuits. The module must be listed as compatible with the control unit. The module must provide address setting means compatible with the control unit's SLC supervision and store an internal identifying code. Monitor module must contain an integral LED that flashes each time the monitor module is polled and is visible through the device cover plate. Pull stations with a monitor module in a common backbox are not required to have an LED. Modules must be listed for the environmental conditions in which they will be installed.

2.10 ADDRESSABLE CONTROL MODULES

The control module must be capable of operating as a relay (dry contact form C) for interfacing the control unit with other systems, and to control door holders or initiate elevator fire service. The module must be listed as compatible with the control unit. The indicating device or the external load being controlled must be configured as Class A notification appliance circuits. The system must be capable of supervising, audible, visual and dry contact circuits. The control module must have both an input and output address. The supervision must detect a short on the supervised circuit and must prevent power from being applied to the circuit. The control module must provide address setting means compatible with the control unit's SLC supervision and store an internal identifying code. The control module must contain an integral LED that flashes each time the control module is polled and is visible through the device cover plate. Control Modules must be listed for the environmental conditions in which they will be installed.

2.11 NOTIFICATION APPLIANCES

2.11.1 Audible Notification Appliances

Audible appliances must conform to the applicable requirements of UL 464. Appliances must be connected into notification appliance circuits. Surface mounted audible appliances must be painted red or white. Recessed audible appliances must be installed with a grill that is painted red or white. Audible appliances shall generate a unique audible sound from other devices provided in the building and surrounding area.

2.11.1.1 Speakers

a. Speakers must conform to the applicable requirements of UL 1480. Speakers must have six different sound output levels and operate with audio line input levels of 70.7 VRMs and 25 VRMs, by means of selectable tap settings. Interior speaker tap settings must include taps of 1/4, 1/2, 1, and 2 watt, at a minimum. Exterior speakers must also be multi-tapped with no more than 15 watt maximum setting. Speakers must incorporate a high efficiency speaker for maximum output at minimum power across a frequency range of 400 Hz to 4,000 Hz, and must have a sealed back construction. Speakers must be capable of installation on standard 4-inch square electrical boxes. Where speakers and strobes are provided in the same location, they may be combined into a single unit. All inputs must be polarized for compatibility with standard reverse polarity supervision of circuit wiring via the FMCU.

b. Provide speaker mounting plates constructed of cold rolled steel having a minimum thickness of 16 gage or molded high impact plastic and equipped with mounting holes and other openings as needed for a complete installation. Fabrication marks and holes must be ground and finished to provide a smooth and neat appearance for each plate. Each plate must be primed and painted.

c. Speakers must utilize screw terminals for termination of all field wiring.

2.11.2 Visual Notification Appliances

Visual notification appliances must conform to the applicable requirements of UL 1638, UL 1971 and conform to the Architectural Barriers Act (ABA). Visual Notification Appliances must have clear high intensity optic lens, xenon flash tubes, or light emitting diode (LED) and be marked "Alert" in letters of contrasting color. The light pattern must be dispersed so that it is visible above and below the strobe and from a 90 degree angle on both sides of the strobe. Strobe flash rate must be 1 flash per second and a minimum of 15 candela based on the UL 1971 test. Strobe must be semi-flush mounted.

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2.11.3 **Textual Display Signs**

As required by ECB 2018-17 - Textual display signs must be LED and must not exceed a height necessary to meet the requirements of NFPA 72. The text display must spell out the word "EVACUATE" or "ANNOUNCEMENT" in conjunction with fire alarm activation or during time when voice messages are broadcast over mass notification system (respectively). The design of text display must be such that it cannot be read when not illuminated.

LCD or LED text displays must meet the following requirements at a minimum:

a. Two lines of information for high priority messaging.

b. Display must be wall or ceiling mounted.

c. Mounting brackets for a convenient wall/cubicle mount.

d. The system must interface with the textual display sign control panel to activate the proper message.

e. Per FTC requirement, scrolling text displays are not permitted.

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2.12 ELECTRIC POWER

2.12.1 Primary Power

Power must be 120 VAC 60 Hz service for the FMCU from the AC service to the building in accordance with NFPA 72.

2.13 SECONDARY POWER SUPPLY

Provide for system operation in the event of primary power source failure. Transfer from normal to auxiliary (secondary) power or restoration from auxiliary to normal power must be automatic and must not cause transmission of a false alarm.

2.13.1 Batteries

Provide sealed, maintenance-free, sealed lead acid batteries as the source for emergency power to the FMCU. Batteries must contain suspended electrolyte. The battery system must be maintained in a fully charged condition by means of a solid state battery charger. Provide an automatic transfer switch to transfer the load to the batteries in the event of the failure of primary power.

2.13.1.1 Capacity

Battery size must be the greater of the following two capacities. This capacity applies to every control unit associated with this system, including supplemental notification appliance circuit panels, auxiliary power supply panels, fire alarm transmitters, and Base-wide mass notification transceivers. When determining the required capacity under alarm condition, visual notification appliances must include both textual and non-textual type appliances.

a. Sufficient capacity to operate the fire alarm system under supervisory and trouble conditions, including audible trouble signal devices for 48 hours and audible and visual signal devices under alarm conditions for an additional 15 minutes.

b. Sufficient capacity to operate the mass notification for 60 minutes after loss of AC power.

2.13.1.2 Battery Power Calculations

a. Verify that battery capacity exceeds supervisory and alarm power requirements for the criteria noted in the paragraph "Capacity" above.

(1) Substantiate the battery calculations for alarm and supervisory power requirements. Include ampere-hour requirements for each system component and each control unit component, and compliance with UL 864.

(2) Provide complete battery calculations for both the alarm and supervisory power requirements. Submit ampere-hour requirements for each system component with the calculations.

(3) Provide voltage drop calculations to indicate that sufficient voltage is available for proper operation of the system and all components. Calculations must be performed using the minimum rated voltage of each component.

b. For battery calculations assume a starting voltage of 24 VDC for starting the calculations to size the batteries. Calculate the required Amp-Hours for the specified standby time, and then calculate the required Amp-Hours for the specified alarm time. Using 20.4 VDC as starting voltage, perform a voltage drop calculation for circuits containing device and/or appliances remote from the power sources.

2.13.2 Battery Chargers

Provide a solid state, fully automatic, variable charging rate battery charger. The charger must be capable of providing 120 percent of the connected system load and must maintain the batteries at full charge. In the event the batteries are fully discharged (20.4 Volts dc), the charger must recharge the batteries back to 95 percent of full charge within 48 hours after a single discharge cycle as described in paragraph CAPACITY above. Provide pilot light to indicate when batteries are manually placed on a high rate of charge as part of the unit assembly if a high rate switch is provided.

2.14 SURGE PROTECTIVE DEVICES

Surge protective devices must be provided to suppress all voltage transients which might damage fire alarm control unit components. Systems having circuits located outdoors, communications equipment must be protected against surges induced on any signaling line circuit. Cables and conductors, that serve as communications links, must have surge protection circuits installed at each end. The surge protective device must wire in series to the power supply of the protected equipment with screw terminations. Line voltage surge arrestor must be installed directly adjacent to the power panel where the FMCU breaker is located.

a. Surge protective devices for nominal 120 VAC must be UL 1449 listed with a maximum 500 volt suppression level and have a maximum response time of 5 nanoseconds. The surge protective device must also meet IEEE C62.41.1 and IEEE C62.41.2 category B tests for surge capacity. The surge protective device must feature multi-stage construction and be provided with a long-life indicator lamp (either light emitting diode or neon) which extinguishes upon failure of protected components. Any unit fusing must be externally accessible.

b. Surge protective devices for nominal 24 VAC, fire alarm telephone dialer, or ethernet connection must be UL 497B listed, meet IEEE C62.41.1 and have a maximum response time of 1-nanosecond. The surge protective device must feature multi-stage construction and be self-resetting. The surge protective device must be a base and plug style. The base assembly must have screw terminals for fire alarm wiring. The base assembly must accept "plug-in" surge protective module.

c. All surge protective devices (SPD) must be the standard product of a single manufacturer and be equal or better than the following:

   (1) For 120 VAC nominal line voltage: UL 1449 and UL 1283 listed, series connected 120 VAC, 20A rated, surge protective device in a NEMA 4x enclosure. Minimum 50,000 amp surge current rating with EMI/RFI filtering and a dry contact circuit for remote monitoring of surge protection status.

   (2) For 24-volt nominal line voltage: UL 497B listed, series connected low voltage, 24-volt, 5A rated, loop circuit protector, base and replaceable module.

   (3) For alarm telephone dialers: UL 497A listed, series connected, 130-volt, 150 mA rated with self-resetting fuse, dialer circuit protector with modular plug and play.

   (4) For IP-DACTS: UL 497B listed, series connected, 6.4-volt, 1.5A rated with 20 kA/pair surge current, data network protector with modular plug and play.

2.15 WIRING

Provide wiring materials under this section as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM with the additions and modifications specified herein.

2.15.1 Alarm Wiring

IDC and SLC wiring must be solid copper cable in accordance with the manufacturers requirements. Copper signaling line circuits and initiating device circuit field wiring must be No. 18 AWG size conductors at a minimum. Visual notification appliance circuit conductors, that contain audible alarm appliances, must be copper No. 14 AWG size conductors at a minimum. Speaker circuits must be copper No. 16 AWG size twisted and shielded conductors at a minimum. Wiring for textual notification appliance circuits must be in accordance with manufacturer's requirements but must be supervised by the FMCU. Wire size must be sufficient to prevent voltage drop problems. Circuits operating at 24 VDC must not operate at less than the listed voltages for the detectors and/or appliances. Power wiring, operating at 120 VAC minimum, must be a minimum No. 12 AWG solid copper having similar insulation. Acceptable power-limited cables are FPL, FPLR or FPLP as appropriate with red colored covering. Nonpower-limited cables must comply with NFPA 70.

2.16 INTERFACE TO THE BASE-WIDE MASS NOTIFICATION NETWORK

There is no connection to base-wide mass notification network.

2.17 AUTOMATIC FIRE ALARM TRANSMITTERS

2.17.1 Radio Transmitter and Interface Panels

Transmitters must be compatible with proprietary supervising station receiving equipment. Each radio alarm transmitter must be the manufacturer's recognized commercial product, completely assembled, wired, factory tested, and delivered ready for installation and operation. Transmitters must be provided in accordance with applicable portions of NFPA 72, Federal Communications Commission (FCC) 47 CFR 90 and Federal Communications Commission (FCC) 47 CFR 15. Transmitter electronics module must be contained within the physical housing as an integral, removable assembly and is to be installed in first floor electrical room, per plan drawings. The proprietary transmitting equipment is a King-Fisher radio transmitter(IRACS), furnished by the Government and installed by the Contractor in accordance with the manufacturer's instructions.

2.17.1.1 Operation

Operate each transmitter from 120-volt ac power. In the event of 120-volt ac power loss, the transmitter must automatically switch to battery operation. Switchover must be accomplished with no interruption of protective service, and must automatically transmit a trouble message. Upon restoration of ac power, transfer back to normal ac power supply must also be automatic.

2.17.1.2 Battery Power

Transmitter standby battery capacity must provide sufficient power to operate the transmitter in a normal standby status for a minimum of 72 hours and be capable of transmitting alarms during that period.

2.17.1.3 Transmitter Housing

Use NEMA Type 1 for housing. The housing must contain a lock that is keyed identical to the fire alarm system for the building. Radio alarm transmitter housing must be factory painted with a suitable priming coat and not less than two coats of a hard, durable weatherproof enamel.

2.17.1.4 Antenna

Antenna is specified by FTC Fire Department and is to be mounted as indicated in plan drawings. Do not mount antennas to any portion of the building roofing system. Protect the antenna from physical damage.

2.17.2 Signals to Be Transmitted to the Base Receiving Station

Reference 'FA' series sheets in plan drawings for FAMNS sequence of operation that indiates all communicated events which are transmitted to the Base Receiving Station.

2.18 SYSTEM MONITORING

2.18.1 Valves

Each valve affecting the proper operation of a fire protection system, including automatic sprinkler control valves, sprinkler service entrance valve, valves at fire pumps, isolating valves for pressure type waterflow or supervision switches, and valves at backflow preventers, whether supplied under this contract or existing, must be electrically monitored to ensure its proper position. Provide each tamper switch with a separate address.

2.19 ENVIRONMENTAL ENCLOSURES OR GUARDS

Environmental enclosures must be provided to permit fire alarm/mass notification components to be used in areas that exceed the environmental limits of the listing. The enclosure must be listed for the device or appliance as either a manufactured part number or as a listed compatible accessory for the component is currently listed. Guards required to deter mechanical damage must be either a listed manufactured part or a listed accessory for the category of the initiating device or notification appliance.

2.20 FIREFIGHTER TELEPHONE COMMUNICATION SYSTEM

2.20.1 General

Provide a firefighter telephone communication system with complete, common talk, closed circuits. The system must include, but not be limited to, a master control station mounted in the fire alarm control unit, a power supply and standby battery system, and remote telephone stations.

2.20.2 Features

The system must include the following features:

a. A master control station which must provide power, supervision, and control for wiring, components, and circuits. The act of lifting any remote telephone hand set from its cradle must cause both a visual and audible signal to annunciate at the master control station. Removing the hand set at the master control station and depressing a button at the remote telephone hand set must cause the automatic silencing of the audible signal.

b. Communication between the master control station hand set and any/or all remote hand sets must require the depressing of a push-to-talk switch located on any/all remote hand sets. During the time that the master control hand set is removed from its cradle it must be possible to communicate between five remote hand sets and the master control station.

c. Hand sets must be able to monitor any conversation in progress and join the conversation by pressing the push-to-talk button. It must not be possible to communicate between two or more remote hand sets with the master control station hand set in its cradle.

d. The master control station hand set must be red in color and equipped with a 5-foot long strain-relieved coiled cord.

e. The master control station must monitor wire and connections for any opens, shorts, or grounds which would render the system inoperable or unintelligible.

f. The master control station must be equipped with a silencing switch and ring-back feature such that any audible trouble signal can be silenced and must be so indicated by the lighting of an amber LED. Once any trouble condition has been corrected, the amber LED must be extinguished.

g. The master control station must be equipped with a separate, LED annunciated switch for each telephone circuit. In addition, LEDs must provide for the annunciation of operating and supervisory power.

h. The loss of operating or supervisory power must cause an audible and visual indication at the master control station and must also cause the fire alarm trouble signal to sound on the FMCU.

i. Switches, LEDs, and controls must be fully labeled.

2.20.3 Handsets

Handsets must have the following features:

a. Provide flush mounted remote telephone stations.

b. Each station must be equipped with a hinged door that is magnetically locked.

c. Each hand set must be permanently wired in place with a coiled cord.

d. Each hand set must be red high-impact cycolac and must be equipped with a push-to-talk switch which, when operated, must signal the master control station and a switch-equipped, storage cradle.

e. Provide operating and supervising power from the same supply circuit(s) utilized for the FMCU.

PART 3 EXECUTION

3.1 VERIFYING ACTUAL FIELD CONDITIONS

Before commencing work, examine all adjoining work on which the contractor's work is in any way dependent for perfect workmanship according to the intent of this specification section, and report to the Contracting Officer's Representative any condition which prevents performance of first class work. No "waiver of responsibility" for incomplete, inadequate or defective adjoining work will be considered unless notice has been filed before submittal of a proposal.

3.2 INSTALLATION

3.2.1 Fire Alarm and Mass Notification Control Unit (FMCU)

Locate the FMCU where indicated on the drawings. Surface mount the enclosure with the top of the cabinet 6 feet above the finished floor or center the cabinet at 5 feet, whichever is lower. Conductor terminations must be labeled and a drawing containing conductors, their labels, their circuits, and their interconnection must be permanently mounted in the FMCU. Locate the document storage cabinet adjacent to the FMCU unless the Contracting Officer directs otherwise.

3.2.2 Battery Cabinets

When batteries will not fit in the FMCU, locate battery cabinets below or adjacent to the FMCU. Battery cabinets must be installed at an accessible location when standing at floor level. Battery cabinets must not be installed lower than 12 inches above finished floor, measured to the bottom of the cabinet, nor higher than 36 inches above the floor, measured to the top of the cabinet. Installing batteries above drop ceilings or in inaccessible locations is prohibited. Battery cabinets must be large enough to accommodate batteries and also to allow ample gutter space for interconnection of control units as well as field wiring. The cabinet must be provided in a sturdy steel housing, complete with back box, hinged steel door with cylinder lock, and surface mounting provisions.The cabinet must be identified by an engraved phenolic resin nameplate. Lettering on the nameplate must indicate the control unit(s) the batteries power and must not be less than 1-inch high.

3.2.3 Manual Stations

Locate manual stations as required by NFPA 72 and as indicated on the drawings. Mount stations so they are located no farther than 5 feet from the exit door they serve, measured horizontally. Manual stations must be mounted at 44 inches measured to the operating handle.

3.2.4 Notification Appliances

a. Locate notification appliance devices as required by NFPA 72 and to meet the intelligibility requirements. Where two or more visual notification appliances are located in the same room or corridor or field of view, provide synchronized operation. Devices must use screw terminals for all field wiring. Audible and visual notification appliances mounted in acoustical ceiling tiles must be centered in the tiles plus or minus 2 inches.

b. Audible and visual notification appliances mounted on the exterior of the building, within unconditioned spaces, or in the vicinity of showers must be listed weatherproof appliances installed on weatherproof backboxes.

c. Speakers must not be located in close proximity to the FMCU or LOC so as to cause feedback when the microphone is in use.

3.2.5 Smoke and Heat Detectors

Locate detectors as required by NFPA 72 and their listing on a 4-inch mounting box. Install heat detectors not less than 4 inches from a side wall to the near edge. Heat detectors located on the wall must have the top of the detector at least 4 inches below the ceiling, but not more than 12 inches below the ceiling. Smoke detectors are permitted to be on the wall no lower than 12 inches from the ceiling with no minimum distance from the ceiling. Install smoke detectors no closer than 3 feet from air handling supply diffusers. Detectors installed in acoustical ceiling tiles must be centered in the tiles plus or minus 2 inches.

3.2.6 Graphic Annunciator (Wall-Mounted Textual Display Signs)

Locate the graphic annunciator as shown on the drawings. Mount the annunciator, with the lower edge 80-96 inches inches above the finished floor, but no closer than 6 inches to the ceiling.

3.2.7 LCD REMOTE Annunciator

Locate the LCD annunciators as shown on the drawings. Mount the annunciator, with the top 6 feet above the finished floor or center the annunciator at 5 feet, whichever is lower.

3.2.8 Firefighter Telephones

FTC only requires the single firefigher telephone within the elevator cab.

3.2.9 Local Operating Console (LOC)

Locate the LOC(s) as required by NFPA 72 and as indicated. Mount the console so that the top message button and microphone is no higher than 4 feet above the floor and the bottom (lowest) message button and microphone is at least 3 feet above the finished floor.

3.2.10 Ceiling Bridges

Provide ceiling bridges for ceiling-mounted appliances. Ceiling bridges must be as recommended/required by the manufacturer of the ceiling-mounted notification appliance.

3.3 SYSTEM FIELD WIRING

3.3.1 Wiring within Cabinets, Enclosures, and Boxes

Provide wiring installed in a neat and workmanlike manner and installed parallel with or at right angles to the sides and back of any box, enclosure, or cabinet. Conductors that are terminated, spliced, or otherwise interrupted in any enclosure, cabinet, mounting, or junction box must be connected to screw-type terminal blocks. Mark each terminal in accordance with the wiring diagrams of the system. The use of wire nuts or similar devices is prohibited. Wiring to conform with NFPA 70.

Indicate the following in the wiring diagrams:

a. Point-to-point wiring diagrams showing the points of connection and terminals used for electrical field connections in the system, including interconnections between the equipment or systems that are supervised or controlled by the system. Diagrams must show connections from field devices to the FMCU and remote fire alarm/mass notification control units, initiating circuits, switches, relays and terminals.

b. Complete riser diagrams indicating the wiring sequence of devices and their connections to the control equipment. Include a color code schedule for the wiring. Include floor plans showing the locations of devices and equipment.

3.3.2 Terminal Cabinets

Provide a terminal cabinet at the base of any circuit riser, on each floor at each riser, and where indicated on the drawings. Terminal size must be appropriate for the size of the wiring to be connected. Conductor terminations must be labeled and a drawing containing conductors, their labels, their circuits, and their interconnection must be permanently mounted in the terminal cabinet. Minimum size is 8 inches by 8 inches. Only screw-type terminals are permitted. Provide an identification label, that displays "FIRE ALARM TERMINAL CABINET" with 2-inch lettering, on the front of the terminal cabinet.

3.3.3 Alarm Wiring

a. Voltages must not be mixed in any junction box, housing or device, except those containing power supplies and control relays.

b. Utilize shielded wiring where recommended by the manufacturer. For shielded wiring, ground the shield at only one point, in or adjacent to the FMCU.

c. Pigtail or T-tap connections to signal line circuits, initiating device circuits, supervisory alarm circuits, and notification appliance circuits are prohibited.

d. Color coding is required for circuits and must be maintained throughout the circuit. Conductors used for the same functions must be similarly color coded. Conform wiring to NFPA 70.

e. Pull all conductors splice free. The use of wire nuts, crimped connectors, or twisting of conductors is prohibited. Where splices are unavoidable, the location of the junction box or pull box where they occur must be identified on the as-built drawings. The number and location of splices must be subject to approval by the Designated Fire Protection Engineer (DFPE).

3.3.4 Back Boxes and Conduit

In addition to the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM, provide all wiring in rigid metal conduit or intermediate metal conduit unless specifically indicated otherwise. Minimum conduit size must be 3/4-inch in diameter. Do not use electrical non-metallic tubing (ENT) or flexible non-metallic tubing and associated fittings.

a. Galvanized rigid steel (GRS) conduit must be utilized where exposed to weather, where subject to physical damage, and where exposed on exterior of buildings. Intermediate metal conduit (IMC) may be used in lieu of GRS as allowed by NFPA 70.

b. Electrical metallic tubing (EMT) is permitted above suspended ceilings or exposed where not subject to physical damage. Do not use EMT underground, encased in concrete, mortar, or grout, in hazardous locations, where exposed to physical damage, outdoors or in fire pump rooms. Use die-cast compression connectors.

c. For rigid metallic conduit (RMC), only threaded type fitting are permitted for wet or damp locations.

d. Flexible metal conduit is permitted for initiating device circuits 6 feet in length or less. Flexible metal conduit is prohibited for notification appliance circuits and signaling line circuits. Use liquid tight flexible metal conduit in damp and wet locations.

e. Schedule 40 (minimum) polyvinyl chloride (PVC) is permitted where conduit is routed underground or underground below floor slabs. Convert non-metallic conduit, other than PVC Schedule 40 or 80, to plastic-coated rigid, or IMC, steel conduit before turning up through floor slab.

f. Exterior wall penetrations must be weathertight. Conduit must be sealed to prevent the infiltration of moisture.

g. For Class "A" or "X" circuits with conductor lengths of 10 feet or less, the conductors must be permitted to be installed in the same raceway in accordance with NFPA 72.

3.3.5 Conductor Terminations

Labeling of conductors at terminal blocks in terminal cabinets, FMCU and the LOC must be provided at each conductor connection. Each conductor or cable must have a shrink-wrap label to provide a unique and specific designation. Each terminal cabinet, FMCU, and remote FMCU must contain a laminated drawing that indicates each conductor, its label, circuit, and terminal. The laminated drawing must be neat, using 12 point lettering minimum size, and mounted within each cabinet, control unit, or unit so that it does not interfere with the wiring or terminals. Maintain existing color code scheme where connecting to existing equipment.

3.4 FIRESTOPPING

Provide firestopping for holes at conduit penetrations through floor slabs, fire-rated walls, partitions with fire-rated doors, corridor walls, and vertical service shafts in accordance with Section 07 84 00 FIRESTOPPING.

3.5 PAINTING

a. In unfinished areas (including areas above drop ceilings), paint all exposed electrical conduit (serving fire alarm equipment), fire alarm conduit, surface metal raceway, junction boxes and covers red. In lieu of painting conduit, the contractor may utilize red conduit with a factory applied finish.

b. In finished areas, paint exposed electrical conduit (serving fire alarm equipment), fire alarm conduit, surface metal raceways, junction boxes, and electrical boxes to match adjacent finishes. The inside cover of the junction box must be identified as "Fire Alarm" and the conduit must have painted red bands 3/4-inch wide at 10-foot centers and at each side of a floor, wall, or ceiling penetration.

c. Painting must comply with Section 09 90 00 PAINTS AND COATINGS.

3.6 FIELD QUALITY CONTROL

3.6.1 Test Procedures

Submit detailed test procedures, prepared and signed by the NICET Level III Fire Alarm Technician, and the representative of the installing company, and reviewed by the QFPE 60 days prior to performing system tests. Detailed test procedures must list all components of the installed system such as initiating devices and circuits, notification appliances and circuits, signaling line devices and circuits, control devices/equipment, batteries, transmitting and receiving equipment, power sources/supply, annunciators, special hazard equipment, emergency communication equipment, interface equipment, and surge protective devices. Test procedures must include sequence of testing, time estimate for each test, and sample test data forms. The test data forms must be in a check-off format (pass/fail with space to add applicable test data; similar to the forms in NFPA 72 and NFPA 4.) The test procedures and accompanying test data forms must be used for the pre-Government testing and the Government testing. The test data forms must record the test results and must:

a. Identify the NFPA Class of all Initiating Device Circuits (IDC), and Notification Appliance Circuits (NAC), Voice Notification System Circuits (NAC Audio), and Signaling Line Circuits (SLC).

b. Identify each test required by NFPA 72 Test Methods and required test herein to be performed on each component, and describe how these tests must be performed.

c. Identify each component and circuit as to type, location within the facility, and unique identity within the installed system. Provide necessary floor plan sheets showing each component location, test location, and alphanumeric identity.

d. Identify all test equipment and personnel required to perform each test (including equipment necessary for smoke detector testing. The use of magnets is not permitted.

e. Provide space to identify the date and time of each test. Provide space to identify the names and signatures of the individuals conducting and witnessing each test.

3.6.2 Pre-Government Testing

3.6.2.1 Verification of Compliant Installation

Conduct inspections and tests to ensure that devices and circuits are functioning properly. Tests must meet the requirements of paragraph entitled "Minimum System Tests" as required by NFPA 72. The contractor and an authorized representative from each supplier of equipment must be in attendance at the pre-Government testing to make necessary adjustments. After inspection and testing is complete, provide a signed Verification of Compliant Installation letter by the QFPE that the installation is complete, compliant with the specification and fully operable. The letter must include the names and titles of the witnesses to the pre-Government tests. Provide all completion documentation as required by NFPA 72 including all referenced annex sections and the test reports noted below.

a. NFPA 72 Record of Completion.

b. NFPA 72 Record of Inspection and Testing.

c. Fire Alarm and Emergency Communication System Inspection and Testing Form.

d. Audibility test results with marked-up test floor plans.

e. Intelligibility test results with marked-up floor plans.

f. Documentation that all tests identified in the paragraph "Minimum System Tests" are complete.

3.6.2.2 Request for Government Final Test

When the verification of compliant installation has been completed, submit a formal request for Government final test to the Contracting Officer's Representative (COR). Government final testing will not be scheduled until the DFPE has received copies of the request for Government final testing and Verification of Compliant Installation letter with all required reports. Government final testing will not be performed until after the connections to the installation-wide fire reporting system and the installation-wide mass notification system have been completed and tested to confirm communications are fully functional. Submit request for test at least 15 calendar days prior to the requested test date.

3.6.3 Correction of Deficiencies

If equipment was found to be defective or non-compliant with contract requirements, perform corrective actions and repeat the tests. Tests must be conducted and repeated if necessary until the system has been demonstrated to comply with all contract requirements.

3.6.4 Government Final Tests

The tests must be performed in accordance with the approved test procedures in the presence of the DFPE. Furnish instruments and personnel required for the tests. The following must be provided at the job site for Government Final Testing:

a. The manufacturer's technical representative.

c. Marked-up red line drawings of the system as actually installed.

d. Loop resistance test results.

e. Complete program printout including input/output addresses.

f. Copy of pre-Government Test Certificate, test procedures and completed test data forms.

g. Audibility test results with marked-up floor plans.

h. Intelligibility test results with marked-up floor plans.

Government Final Tests will be witnessed by the , Contracting Officer's Representative (COR). At this time, any and all required tests noted in the paragraph "Minimum System Tests" must be repeated at their discretion.

3.7 MINIMUM SYSTEM TESTS

3.7.1 System Tests

Test the system in accordance with the procedures outlined in NFPA 72. The required tests are as follows:

a. Loop Resistance Tests: Measure and record the resistance of each circuit with each pair of conductors in the circuit short-circuited at the farthest point from the circuit origin. The tests must be witnessed by the Contracting Officer and test results recorded for use at the final Government test.

b. Verify the absence of unwanted voltages between circuit conductors and ground. The tests must be accomplished at the pre-Government test with results available at the final system test.

c. Verify that the control unit is in the normal condition as detailed in the manufacturer's O&M manual.

d. Test each initiating device and notification appliance and circuit for proper operation and response at the control unit. Smoke detectors must be tested in accordance with manufacturer's recommended calibrated test method. Use of magnets is prohibited. Testing of duct smoke detectors must comply with the requirements of NFPA 72 except disconnect at least 20 percent of devices. If there is a failure at these devices, then supervision must be tested at each device.

f. Test the system for specified functions in accordance with the contract drawings and specifications and the manufacturer's O&M manual.

g. Test both primary power and secondary power. Verify, by test, the secondary power system is capable of operating the system for the time period and in the manner specified.

h. Determine that the system is operable under trouble conditions as specified.

i. Visually inspect wiring.

j. Test the battery charger and batteries.

k. Verify that software control and data files have been entered or programmed into the FMCU. Hard copy records of the software must be provided to the Contracting Officer.

l. Verify that red-line drawings are accurate.

m. Measure the current in circuits to ensure there is the calculated spare capacity for the circuits.

n. Measure voltage readings for circuits to ensure that voltage drop is not excessive.

o. Disconnect the verification feature for smoke detectors during tests to minimize the amount of smoke needed to activate the sensor. Testing of smoke detectors must be conducted using real smoke.

p. Measure the voltage drop at the most remote appliance (based on wire length) on each notification appliance circuit.

q. Verify the documentation cabinet is installed and contains all as-built shop drawings, product data sheets, design calculations, site-specific software data package, and all documentation required by paragraph titled "Test Reports".

r. Open the circuit at not less than 10 percent of alarm initiating devices and notification appliances to test the wiring supervisory feature.

3.7.2 Audibility Tests

Sound pressure levels from audible notification appliances must be a minimum of 15 dBa over ambient with a maximum of 110 dBa in any occupiable area. The provisions for audible notification (audibility and intelligibility) must be met with doors, fire shutters, movable partitions, and similar devices closed.

3.7.3 Intelligibility Tests

Intelligibility testing of the System must be accomplished in accordance with NFPA 72 for Voice Evacuation Systems, and ASA S3.2. Following are the specific requirements for intelligibility tests:

a. Intelligibility Requirements: Verify intelligibility by measurement after installation.

b. Ensure that a CIS value greater than the required minimum value is provided in each area where building occupants typically could be found. The minimum required value for CIS is .8. Rounding of values is permitted.

c. Areas of the building provided with hard wall and ceiling surfaces (such as metal or concrete) that are found to cause excessive sound reflections may be permitted to have a CIS score less than the minimum required value if approved by the DFPE, and if building occupants in these areas can determine that a voice signal is being broadcast and they must walk no more than 33 feet to find a location with at least the minimum required CIS value within the same area.

d. Areas of the building where occupants are not expected to be normally present are permitted to have a CIS score less than the minimum required value if personnel can determine that a voice signal is being broadcast and they must walk no more than 50 feet to a location with at least the minimum required CIS value within the same area.

e. Take measurements near the head level applicable for most personnel in the space under normal conditions (e.g., standing, sitting, sleeping, as appropriate).

f. The distance the occupant must walk to the location meeting the minimum required CIS value must be measured on the floor or other walking surface as follows:

   (1) Along the centerline of the natural path of travel, starting from any point subject to occupancy with less than the minimum required CIS value.

(2) Curving around any corners or obstructions, with a 12 inches clearance there from.

(3) Terminating directly below the location where the minimum required CIS value has been obtained.

Use commercially available test instrumentation to measure intelligibility as specified by NFPA 72 as applicable. Use the mean value of at least three readings to compute the intelligibility score at each test location.

3.8 SYSTEM ACCEPTANCE

Following acceptance of the system, as-built drawings and O&M manuals must be delivered to the Contracting Officer for review and acceptance. The drawings must show the system as installed, including deviations from both the project drawings and the approved shop drawings. These drawings must be submitted within two weeks after the final Government test of the system. At least one set of as-built (marked-up) drawings must be provided at the time of, or prior to the Final Government Test.

a. The drawings must be prepared electronically and sized no less than the contract drawings. Furnish one set of CDs or DVDs containing software back-up (if required) and CAD based drawings in latest version of AutoCAD, DXF and portable document formats of as-built drawings and schematics.

b. Include complete wiring diagrams showing connections between devices and equipment, both factory and field wired.

c. Include a riser diagram and drawings showing the as-built location of devices and equipment.

d. Provide Operation and Maintenance (O&M) Instructions.

3.9 INSTRUCTION OF GOVERNMENT EMPLOYEES

3.9.1 Instructor

Provide the services of an instructor, who has received specific training from the manufacturer for the training of other persons regarding the operation, inspection, testing, and maintenance of the system provided. The instructor must train the Government employees designated by the Contracting Officer, in the care, adjustment, maintenance, and operation of the fire alarm system. The instructor must be thoroughly familiar with all parts of this installation. The instructor must be trained in operating theory as well as in practical O&M work. Submit the instructors information and qualifications including the training history.

3.9.2 Required Instruction Time

Provide 8 hours of instruction after final acceptance of the system. The instruction must be given during regular working hours on such dates and times selected by the Contracting Officer. The instruction may be divided into two or more periods at the discretion of the Contracting Officer. The training must allow for rescheduling for unforeseen maintenance and/or fire department responses.

3.9.2.1 Technical Training

Equipment manufacturer or a factory representative must provide 1 day of on site training. Training must allow for classroom instruction as well as individual hands on programming, troubleshooting and diagnostics exercises.

3.9.3 Technical Training Manual

Provide, in manual format, lesson plans, operating instructions, maintenance procedures, and training data for the training courses. The operations training must familiarize designated government personnel with proper operation of the installed system. The maintenance training course must provide the designated government personnel adequate knowledge required to diagnose, repair, maintain, and expand functions inherent to the system.

3.10 EXTRA MATERIALS

3.10.1 Repair Service/Replacement Parts

Repair services and replacement parts for the system must be available for a period of 10 years after the date of final acceptance of this work by the Contracting Officer. During the warranty period, the service technician must be on-site within 24 hours after notification. All repairs must be completed within 24 hours of arrival on-site.

During the warranty period, the installing fire alarm contractor is responsible for conducting all required testing and maintenance in accordance with the requirements and recommended practices of NFPA 72 and the system manufacturer(s). Installing fire alarm contractor is NOT responsible for any damage resulting from abuse, misuse, or neglect of equipment by the end user.

3.10.2 Spare Parts

Spare parts furnished must be directly interchangeable with the corresponding components of the installed system(s). Spare parts must be suitably packaged and identified by nameplate, tagging, or stamping. Spare parts must be delivered to the Contracting Officer at the time of the Government testing and must be accompanied by an inventory list.

3.10.3 Document Storage Cabinet

Upon completion of the project, but prior to project close-out, place in the document storage cabinet copies of the following record documentation:

a. As-built shop drawings

b. Product data sheets

c. Design calculations

d. Site-specific software data package

e. All documentation required by SD-06.

-- End of Section --